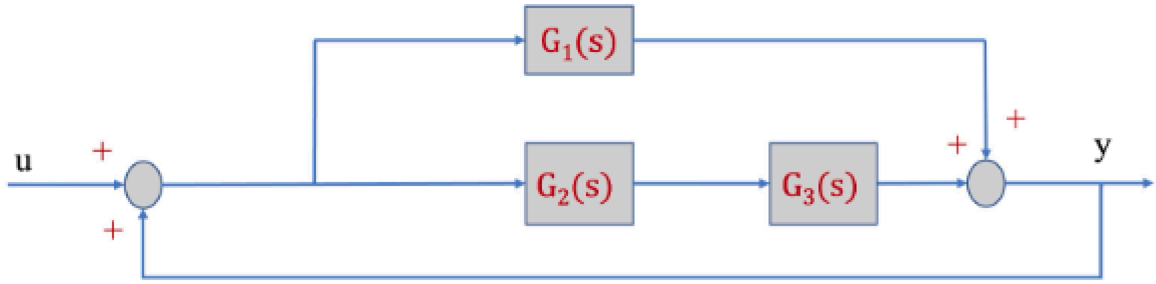
Question 1 Consider the system with input $u(t)$ and output $y(t)$ in the diagram below where the first-order linear systems have the following transfer functions $G_1(s) = \frac{4}{s+10}, G_2(s) = \frac{1}{s+2}, G_3(s) = \frac{s+2}{s+10}$



Calculate the $y(0)$ and the $\lim_{t\to\infty} y(t)$ when $u(t) = e^{-2t} + \theta(t)$, where $\theta(t)$ is the unit step function and $t >= 0$.

- [x] $y(0) = 0, \lim_{t\to\infty} y(t) = 1$
- [ ] $y(0) = 0, \lim_{t\to\infty} y(t) = 0.33$
- [ ] $y(0) = 1, \lim_{t\to\infty} y(t) = 1$
- [ ] $y(0) = 1, \lim_{t\to\infty} y(t) = \infty$

```
clear; clc;

s = tf('s');

G1 = 4/(s+10);
G2 = 1/(s+2);
G3 = (s+2)/(s+10);

% Parallel forward path
H = G1 + G2*G3;

% Positiv feedback: x = u + y
T = feedback(H,1,+1);
T = minreal(T)
```

```
T =

    5
  -----
  s + 5
```

```
Continuous-time transfer function.
Model Properties
```

```
% Slutværdi for u(t) = exp(-2t) + step(t)
% Når t -> inf, går exp(-2t) mod 0, og step(t) går mod 1
y_inf = dcgain(T)
```

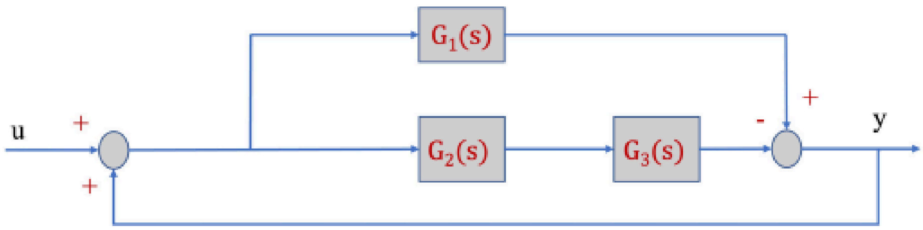
```
y_inf =
1.0000
```

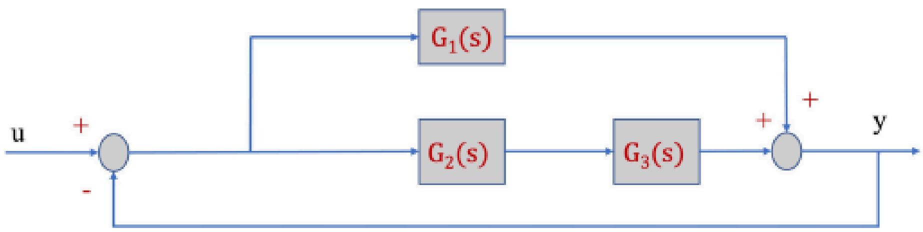
```
% Da systemet er strictly proper og har nul begyndelsesbetingelser:
y_0 = 0
```

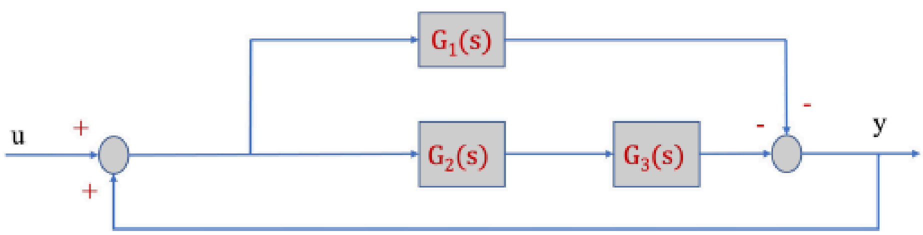
```
y_0 =
0
```

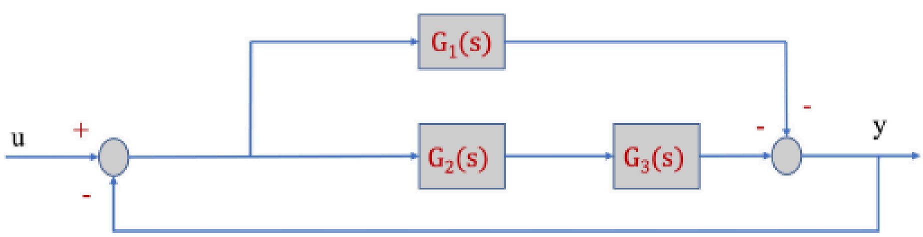
Question 2 Select the correct block diagram that corresponds to the I/O Transfer Function $H(s) = \frac{Y(s)}{U(s)} = \frac{3}{s+7}$.

$G_1(s) = \frac{4}{s+10}, G_2(s) = \frac{1}{s+2}, G_3(s) = \frac{s+2}{s+10}$










```
clear; clc;

s = tf('s');

G1 = 4/(s+10);
G2 = 1/(s+2);
G3 = (s+2)/(s+10);

G23 = minreal(G2*G3);

%% Diagram 1
% Venstre summer: positiv feedback
% Højre summer: G1 positiv, G2G3 negativ
```

```
H1 = G1 - G23;
T1 = feedback(H1, 1, +1);
T1 = minreal(T1)
```

```
T1 =

    3
  -----
  s + 7

Continuous-time transfer function.
Model Properties
```

```
%% Diagram 2
% Venstre summer: negativ feedback
% Højre summer: G1 positiv, G2G3 positiv

H2 = G1 + G23;
T2 = feedback(H2, 1, -1);
T2 = minreal(T2)
```

```
T2 =

    5
  ------
  s + 15

Continuous-time transfer function.
Model Properties
```

```
%% Diagram 3
% Venstre summer: positiv feedback
% Højre summer: G1 negativ, G2G3 positiv

H3 = -G1 - G23;
T3 = feedback(H3, 1, +1);
T3 = minreal(T3)
```

```
T3 =

    -5
  ------
  s + 15

Continuous-time transfer function.
Model Properties
```

```
%% Diagram 4
% Venstre summer: negativ feedback
% Højre summer: G1 negativ, G2G3 negativ

H4 = -G1 - G23;
T4 = feedback(H4, 1, -1);
```

```
T4 = minreal(T4)
```

```
T4 =

   -5
  -----
  s + 5

Continuous-time transfer function.
Model Properties
```

Question 3 Consider the transfer function

$$F(s) = \frac{Y(s)}{U(s)} = \frac{3(s+2)(s+4)}{(s+3)(s^2+6s+6k+8)}$$

For which range of parameter k the system with input $u(t)$ and output $y(t)$ is asymptotically stable?

- ○ $k \in \mathbb{R}$
- ✔ $k > -\frac{4}{3}$
- ○ $k \leq -1$
- ○ $k < 0$
- ○ $k < -\frac{2}{3}$

Løst i python

```python
s, k = symbols("s k", real=True)

p = (s + 3) * (s**2 + 6*s + 6*k + 8)

find_stable_gain_ranges(p, k, s)
```

✓ 0.1s Python

```
{'input_expression': (s + 3)*(6*k + s**2 + 6*s + 8),
 'input_was_characteristic': True,
 'characteristic': 6*k*s + 18*k + s**3 + 9*s**2 + 26*s + 24,
 'discarded_denominator': 1,
 'boundary_expression': 6*I*k*omega + 18*k - I*omega**3 - 9*omega**2 + 26*I*omega + 24,
 'boundary_equations': [18*k - 9*omega**2 + 24, omega*(6*k - omega**2 + 26)],
 'boundary_points': [{'gain': -4/3, 'omega': 0}],
 'boundary_gains': [-4/3],
 'tested_intervals': [{'interval': Interval.open(-oo, -4/3),
   'sample_gain': -7/3,
   'poles': [(-6.872983346207417+0j), (-3+0j), (0.8729833462074169+0j)],
   'stable': False},
  {'interval': Interval.open(-4/3, oo),
   'sample_gain': -1/3,
   'poles': [(-4.732050807568878+0j), (-3+0j), (-1.2679491924311228+0j)],
   'stable': True}],
 'stable_gain_intervals': [Interval.open(-4/3, oo)],
 'positive_stable_gain_intervals': [Interval.open(0, oo)]}
```

Question 4 A unity-feedback closed-loop control system has open-loop transfer function

$$G(s) = \frac{K}{s(s+3)}$$

Which one of the following statements is correct?

- ○ The system has finite steady-state error for a step input and infinite error for ramp and parabolic inputs.
- ○ The system has infinite steady-state error for all standard inputs.
- ✔ The system has zero steady-state error for a step input.
- ○ The system has zero steady-state error for step and ramp inputs and finite error for a parabolic input.

```
clear; clc;

syms s K positive

G = K/(s*(s+3));

% Error transfer function: E(s)/R(s)
E_over_R = simplify(1/(1 + G));

%% Step input: R(s) = 1/s
R_step = 1/s;
E_step = E_over_R * R_step;
ess_step = limit(s*E_step, s, 0)
```

ess_step = 0

```
%% Ramp input: R(s) = 1/s^2
R_ramp = 1/s^2;
E_ramp = E_over_R * R_ramp;
ess_ramp = limit(s*E_ramp, s, 0)
```

ess_ramp =

$$\frac{3}{K}$$

```
%% Parabolic input: R(s) = 1/s^3
R_parabolic = 1/s^3;
E_parabolic = E_over_R * R_parabolic;
ess_parabolic = limit(s*E_parabolic, s, 0)
```

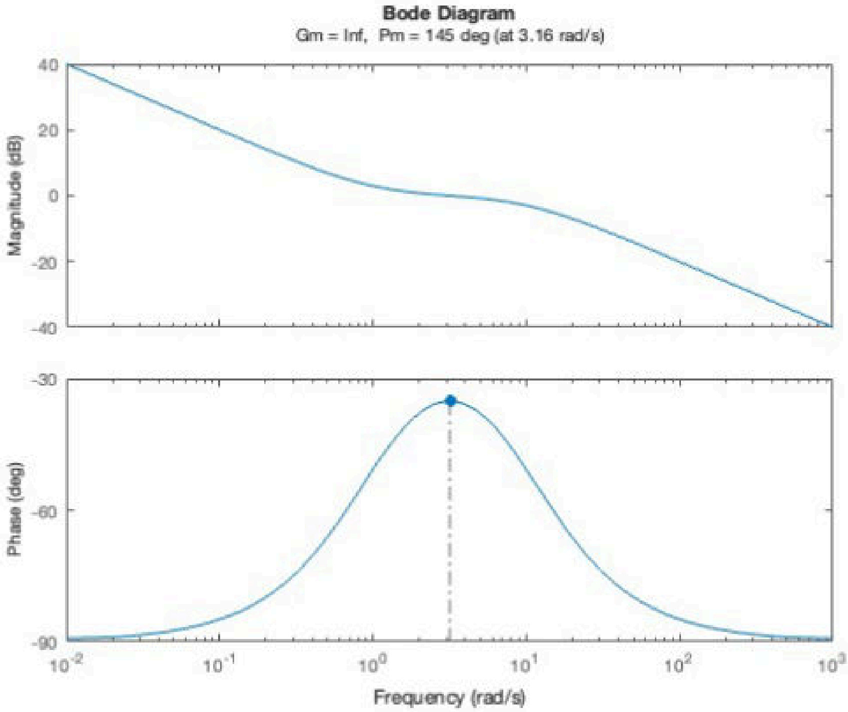
ess_parabolic = NaN

```python
s, K = symbols("s K", positive=True)

G = K / (s * (s + 3))

analyze_transfer_function(G, s)
```
✓ 0.0s

```
{'G(s)': K/(s*(s + 3)),
 'numerator': K,
 'denominator': s**2 + 3*s,
 'zeros_exact': {},
 'poles_exact': {-3: 1, 0: 1},
 'zeros_numeric': None,
 'poles_numeric': None,
 'dc_gain': oo,
 'order': 2,
 'numerator_order': 0,
 'system_type': 1,
 'stable': None,
 'stability_text': 'afhaenger af parametre',
 'y(0+)': 0,
 'y(inf)': oo,
 'omega_n': 0,
 'zeta': zoo,
 'overshoot_fraction_formula': nan,
 'overshoot_percent_formula': nan}
```

System type	Step input	Ramp input	Parabolic input
Type 0	finite error	infinite error	infinite error
Type 1	**zero error**	**finite error**	infinite error
Type 2	zero error	zero error	finite error

Question 5 The Bode diagram shown corresponds to a transfer function:



Which of the following statements about the poles and zeros of the system is most consistent with the diagram?

- [x] The system has a pole at the origin, a zero at $\omega = 1$, and a pole at $\omega = 10$
- [ ] The system has a zero at the origin, a pole at $\omega = 1$, and another pole at $\omega = 10$
- [ ] The system has a single pole at $\omega = 10$, and no zeros
- [ ] The system has two poles at the origin and a zero at $\omega = 1$

```python
s, K = symbols("s K", positive=True)

G = K * (s + 1) / (s * (s + 10))

analyze_transfer_function(G, s)
```

✓ 0.0s


```
{'G(s)': K*(s + 1)/(s*(s + 10)),
 'numerator': K*s + K,
 'denominator': s**2 + 10*s,
 'zeros_exact': {-1: 1},
 'poles_exact': {-10: 1, 0: 1},
 'zeros_numeric': None,
 'poles_numeric': None,
 'dc_gain': oo,
 'order': 2,
```

```
Lav frekvens: slope ≈ -20 dB/dec  → pole at origin
Ved ω = 1: slope stiger +20 dB/dec → zero at ω = 1
Ved ω = 10: slope falder -20 dB/dec → pole at ω = 10
```

Question 6 The system has the following transfer function:

$$G(s) = 8\frac{(s+2)}{(s+0.2)(s^2+10s+25)}$$

For a unit step input, approximately how long does it take for the output to settle within 1% of its final value?

- ( ) 0.2
- ( ) 5
- (x) 25
- ( ) 12.5
- ( ) 2.5

```python
s = symbols("s", real=True)

G = 8*(s + 2) / ((s + 0.2)*(s**2 + 10*s + 25))

analyze_transfer_function(G, s)
```

✓ 0.0s

```
{'G(s)': 0.64*(0.5*s + 1.0)/((0.2*s + 1.0)**2*(1.0*s + 0.2)),
 'numerator': 0.32*s + 0.64,
 'denominator': 0.04*s**3 + 0.408*s**2 + 1.08*s + 0.2,
 'zeros_exact': {-2.00000000000000: 1},
 'poles_exact': {-0.200000000000000: 1,
  -5.0 - 5.50977865084345e-8*I: 1,
  -5.0 + 5.50977865184743e-8*I: 1},
 'zeros_numeric': [(-2+0j)],
 'poles_numeric': [(-0.2+0j),
  (-4.999999999999999-5.5097786508434504e-08j),
  (-4.999999999999999+5.509778651847427e-08j)],
 'dc_gain': 16/5,
 'order': 3,
 'numerator_order': 1,
 'system_type': 0,
 'stable': True,
 'stability_text': 'asymptotisk stabil',
 'y(0+)': 0,
 'y(inf)': 16/5,
 'gain_crossover_frequency': 0.6294734449147297,
 'phase_at_gain_crossover_deg': -69.2538658145092,
 'phase_margin_deg': 110.7461341854908,
 'phase_margin_text': '110.746 grader ved omega = 0.629 rad/s'}
```

```matlab
clear; clc;

s = tf('s');
```

```
G = 8*(s+2)/((s+0.2)*(s^2 + 10*s + 25));

info = stepinfo(G, 'SettlingTimeThreshold', 0.01);

Ts_1percent = info.SettlingTime
```

```
Ts_1percent =
22.9073
```

Den **dominerende pol** er den pol, der ligger tættest på imaginæraksen, altså:

$$s = -0.2$$

Det er den langsomste del af systemet.

For 1% settling time bruger du:

$$T_s \approx \frac{\ln(100)}{|\text{dominerende pol}|}$$

$$T_s \approx \frac{\ln(100/X)}{|\text{dominerende pol}|}$$

Eksempler:

Tolerance	Formel	Tid
1%	$\ln(100)/0.2$	ca. 23.0
2%	$\ln(50)/0.2$	ca. 19.6
5%	$\ln(20)/0.2$	ca. 15.0
33%	$\ln(100/33)/0.2$	ca. 5.5
48%	$\ln(100/48)/0.2$	ca. 3.7
50%	$\ln(2)/0.2$	ca. 3.47

Question 7 Linearize the given nonlinear rotational mass-spring-damper system described by the following differential equation around the working point ($\theta_0 = 2\mathrm{rad}, \dot{\theta}_0 = 4\mathrm{rad/s}$):

$$J\ddot{\theta}(t) + b\theta(t)^3 + d\sqrt{\dot{\theta}(t)} = T(t)$$

where the input is torque $T(t)$ and the output is angular velocity $\dot{\theta}(t)$. The constants are: $J = 5\mathrm{kg{\cdot}m^2}, b = 3\mathrm{N{\cdot}m/rad^3}, d = 2\mathrm{N{\cdot}m{\cdot}s^{1/2}/rad^{1/2}}$, $\theta_0 = 2\mathrm{rad}$, and $\dot{\theta}_0 = 4\mathrm{rad/s}$.

- [x] $\Delta T = 5\Delta\ddot{\theta} + 36\Delta\theta + 0.5\Delta\dot{\theta}$
- [ ] $\Delta T = 5\Delta\ddot{\theta} + 24\Delta\theta + \Delta\dot{\theta}$
- [ ] $\Delta T = 5\Delta\ddot{\theta} + 36\Delta\theta + \Delta\dot{\theta}$
- [ ] $\Delta T = 5\Delta\ddot{\theta} + 24\Delta\theta + 0.5\Delta\dot{\theta}$

```python
theta, theta_dot, theta_ddot = symbols("theta theta_dot theta_ddot", positive=True)

J = 5
b = 3
d = 2

theta0 = 2
theta_dot0 = 4

T_expr = J*theta_ddot + b*theta**3 + d*sqrt(theta_dot)

a_acc = diff(T_expr, theta_ddot)
a_theta = diff(T_expr, theta).subs(theta, theta0)
a_dot = diff(T_expr, theta_dot).subs(theta_dot, theta_dot0)

a_acc, a_theta, a_dot
```

✓ 0.0s

```
(5, 36, 1/2)
```

```matlab
clear; clc;

syms theta theta_dot theta_ddot real
```

```
% Konstanter
J = 5;
b = 3;
d = 2;

% Arbejdspunkt
theta0 = 2;
theta_dot0 = 4;

% Venstresiden af differentialligningen
f = J*theta_ddot + b*theta^3 + d*sqrt(theta_dot);

% Partialafledte
df_dtheta     = diff(f, theta);
df_dtheta_dot = diff(f, theta_dot);
df_dtheta_ddot = diff(f, theta_ddot);

% Indsæt arbejdspunkt
a_theta = subs(df_dtheta, theta, theta0);
a_theta_dot = subs(df_dtheta_dot, theta_dot, theta_dot0);
a_theta_ddot = df_dtheta_ddot;

% Vis koefficienter
a_theta_ddot
```

a_theta_ddot = 5

```
a_theta
```

a_theta = 36

```
a_theta_dot
```

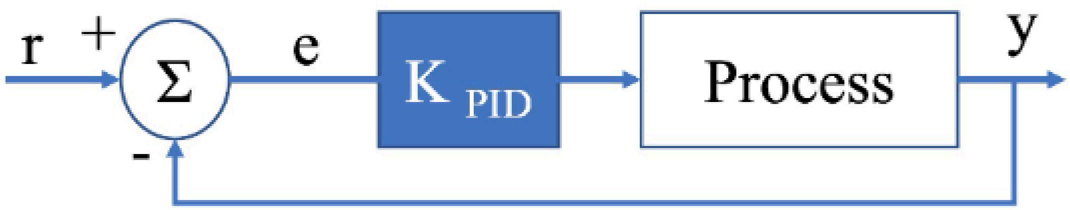
a_theta_dot =

$$\frac{1}{2}$$

Question 8 Consider the PID-controlled system in the figure. Apply Ziegler-Nichols tuning rule for the determination of the values of parameters K_p, T_i and T_d. The closed-loop transfer function is $\frac{Y(s)}{R(s)} = \frac{K_p}{s(s+1)(s+5)+K_p}$

The period of the sustained oscillation is $P_u = \frac{2\pi}{\omega}$ where ω is the frequency of the sustained oscillation when the critical gain $K_u = 30$.

Select the correct transfer function of the PID controller.



- [x] $K_{PID}(s) = 18(1 + \frac{1}{1.405s} + 0.35124s)$
- [ ] $K_{PID}(s) = 15(1 + \frac{1}{1.405s} + 0.35124s)$
- [ ] $K_{PID}(s) = 30(1 + \frac{1}{1.405s} + 0.35124s)$
- [ ] $K_{PID}(s) = 18(1 + \frac{1}{2.809s} + 0.35124s)$

Python

```python
'boundary_points': [
    {'gain': 0, 'omega': 0},
    {'gain': 30, 'omega': sqrt(5)}
]
```

Du skal bruge punktet med **ikke-nul frekvens**:

Python

```python
{'gain': 30, 'omega': sqrt(5)}
```

Det betyder:

$$K_u = 30$$

og

$$\omega = \sqrt{5}$$

Her er K_u den **kritiske gain**, altså den gain hvor systemet begynder at oscillere vedvarende.

Nu bruger du Ziegler-Nichols PID-reglerne:

$$K_p = 0.6K_u$$

```
clear; clc;

Ku = 30;

% Karakteristisk polynomium:
% s(s+1)(s+5) + Ku = s^3 + 6s^2 + 5s + Ku

% Ved marginal stabilitet sættes s = j*w
% Realdel: -6*w^2 + Ku = 0
% Imaginærdel: -w^3 + 5*w = 0

w = sqrt(5);

Pu = 2*pi/w;

Kp = 0.6*Ku;
Ti = Pu/2;
Td = Pu/8;

Kp
```

```
Kp =
18
```

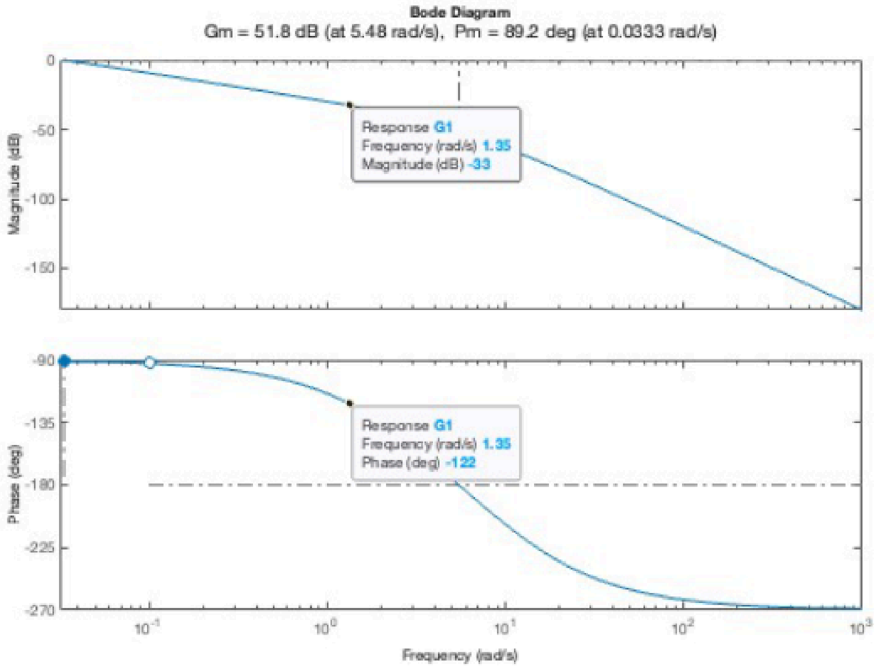
```
Ti
```

```
Ti =
1.4050
```

```
Td
```

```
Td =
0.3512
```

Question 9 The Bode diagram of the open-loop transfer function $G_1(s)$ shows the following stability margin values:
Gain margin: $G_m = 51.8\,\text{dB}$
Phase margin: $P_m = 89.2^\circ$



You wish to adjust the proportional gain K_p so that the phase margin is approximately 60°. What will be the new gain margin after this adjustment?

- ○ The new gain margin is approximately 33 dB
- ○ The gain margin remains unchanged if the phase at -180° does not move
- ○ The new gain margin is greater than 51.8 dB
- ○ The new gain margin is equal to 51.8 dB
- ✔ The new gain margin is less than 51.8 dB

$$P_m \approx 60^\circ$$

Phase margin defineres som:

$$P_m = 180^\circ + \angle G(j\omega_{gc})$$

Så for $P_m = 60^\circ$ skal den nye gain crossover ligge dér, hvor fasen er cirka:

$$\angle G(j\omega) = -120^\circ$$

På plottet står der cirka:

```
ved ω = 1.35 rad/s:
```

```
% Hvis magnitude ved ønsket fase-margin-frekvens er -33 dB,
% så skal proportional gain øges med +33 dB.

old_GM_dB = 51.8;
gain_increase_dB = 33;

new_GM_dB = old_GM_dB - gain_increase_dB
```

```
new_GM_dB =
18.8000
```

Så her skal du ikke kigge efter `Kpid` eller poler. Du skal bruge Bode-reglen:

```
Øg Kp → magnitude-plottet rykkes op
Phase-plottet ændres ikke
Gain margin bliver mindre med samme antal dB
```

```
GM_old_dB = 51.8

# Aflæst fra Bode-plottet ved phase ≈ -120 deg
mag_at_new_crossover_dB = -33

gain_increase_dB = -mag_at_new_crossover_dB
GM_new_dB = GM_old_dB - gain_increase_dB

GM_new_dB
```

✓ 0.0s

```
18.799999999999997
```

Question 10 Which of the following statements about Bode diagrams is true?

- ○ A pole in the transfer function introduces a phase lead of $+90^{\circ}$
- ● The magnitude plot of a Bode diagram for a first-order system with a pole at $\omega = a$ will change slope from 0 to $-20dB/decade$ at $\omega = a$
- ○ The phase shift caused by a pole or zero begins exactly at the break frequency
- ○ A zero in the transfer function contributes a decrease of 20 dB/decade to the magnitude plot starting at its break frequency.

Et førsteordensled med én pol ved

$$\omega = a$$

kan fx skrives som:

$$G(s) = \frac{1}{s+a}$$

eller ofte normaliseret som:

$$G(s) = \frac{1}{1+\frac{s}{a}}$$

I Bode-magnitude gælder:

- før knækfrekvensen $\omega = a$: hældning cirka 0 dB/decade
- efter knækfrekvensen $\omega = a$: hældning cirka -20 dB/decade

Derfor er udsagnet sandt:

$$\boxed{\text{The magnitude plot of a Bode diagram for a first-order system with a pole at } \omega = a \text{ changes slope from 0 to } -20 \text{ dB/decade a}}$$

De andre udsagn er forkerte, fordi:

En **pol** giver ikke $+90^\circ$ phase lead. Den giver typisk phase lag, altså op mod:

$$-90^\circ$$

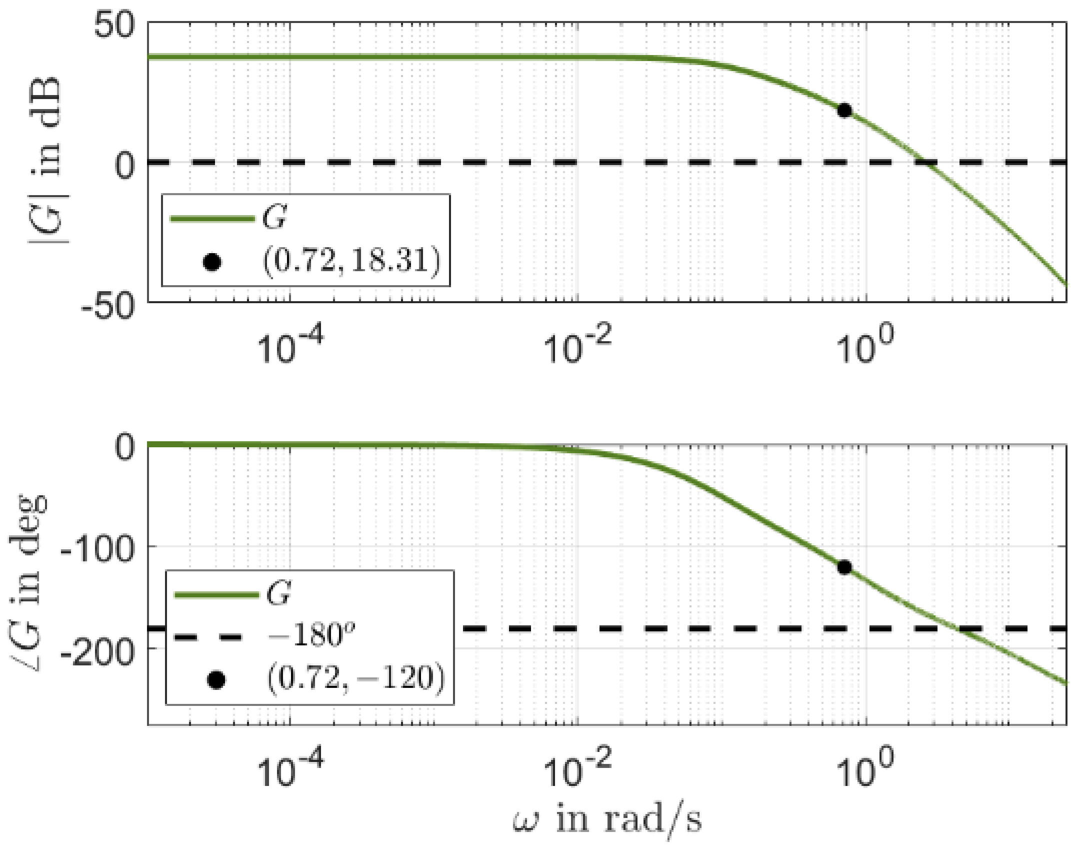
Faseskiftet begynder heller ikke præcis ved knækfrekvensen. I praksis begynder det cirka én dekade før og slutter cirka én dekade efter.

Et **nulpunkt** giver ikke et fald på -20 dB/decade. Det giver en stigning på:

$$+20 \text{ dB/decade}$$

Så den rigtige svarmulighed er den markerede.

Question 11. Consider a system described by a transfer function $G(s)$ with open-loop Bode plot as shown in the following figure.



A P-controller needs to be designed so that a phase margin $\gamma_M = 60^o$ is achieved. The proportional gain K_P is approximately:

- ○ $K_P = 18.31$
- ○ $K_P = 0.9$
- ○ $K_P = -18.31$
- ○ $K_P = 2.7$
- ✔ $K_P = 0.12$

```
clear; clc;

mag_dB = 18.31;

Kp = 10^(-mag_dB/20)
```

```
Kp =
0.1215
```

Question 12. Which of the following statements is correct?

- [x] When identifying the transfer function of an input rate/amplitude-limited system it is better to use small step inputs to produce input/output data.
- [ ] An input amplitude-limited (saturated) system is always linear.
- [ ] Increasing the value of K_a in a feedback-based anti-windup strategy for an input amplitude-limited system decreases the undershoot in the closed-loop step response.
- [ ] Increasing the proportional gain K_P in an input amplitude-limited (saturated) system which is controlled by a PI-Lead controller alleviates (reduces) the effect of integrator wind-up.
- [ ] When designing controllers for an input rate-limited system, the crossover frequency should be chosen as far as possible from the dominant pole of the system.

$$\boxed{\text{Use small step inputs when identifying a transfer function of a rate/amplitude-limited system.}}$$

Grunden er, at **amplitude saturation** og **rate limits** er ikke-lineariteter. Hvis du bruger et stort step-input, kan systemet ramme mætning eller rate limit, og så måler du ikke længere den lineære dynamik.

Når man vil identificere en transferfunktion, ønsker man typisk at finde en **lineær model** omkring et arbejdspunkt. Derfor bruger man små inputs, så systemet bliver i det næsten lineære område.

Altså:

$$u(t) \text{ lille nok } \Rightarrow \text{ ingen saturation/rate limit}$$

og så kan man approksimere systemet med en transferfunktion.

De andre svar er forkerte fordi:

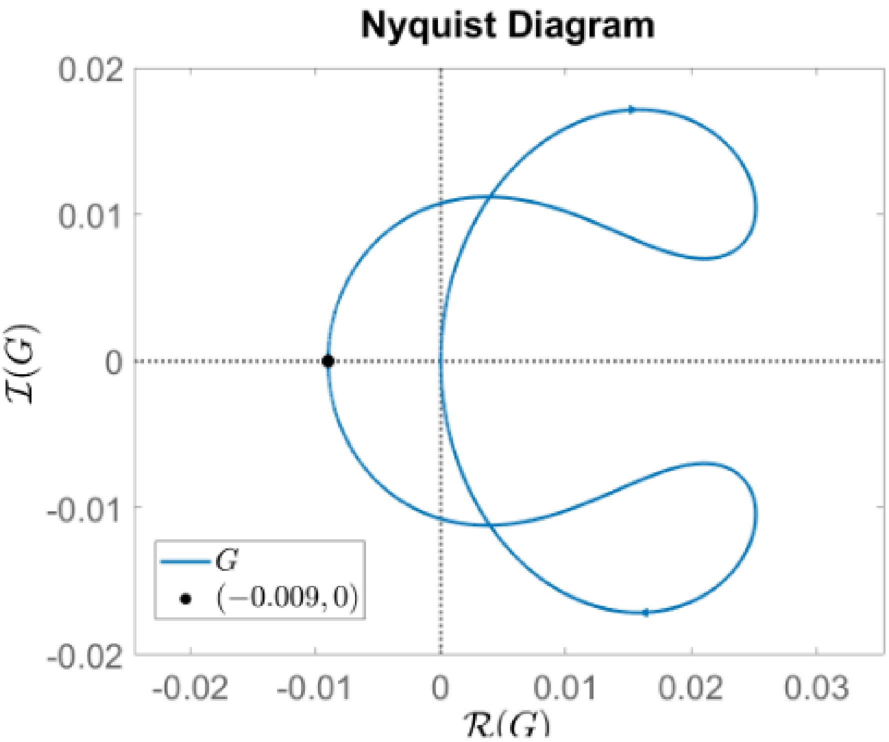
Udsagn	**Hvorfor forkert?**
Et amplitude-limited system er altid lineært	Nej, saturation er en ikke-linearitet
Større K_a i anti-windup mindsker altid undershoot	Ikke altid; for stor K_a kan give dårligere dynamik
Større K_P reducerer integrator wind-up	Typisk nej; større K_P kan gøre saturation værre
Crossover frequency skal vælges så langt som muligt fra dominant pole	Ikke en generel korrekt regel

Så det vigtigste er:

$$\boxed{\text{Til systemidentifikation: brug små nok step-inputs til at undgå saturation/rate limits.}}$$

Men inputtet må heller ikke være så lille, at støj dominerer målingen.

Question 13. Consider a system described by a transfer function $G(s)$ with one pole on the right half complex plane. The associated open-loop Nyquist plot is shown in the following figure.



The Nyquist plot intersects the real axis along the counter-clockwise direction at the point $(-0.009, 0)$. A P-controller with $K_P = 115$ is applied to the system and the loop is closed with unit feedback. Which of the following statements is the correct one?

- [x] The closed-loop system is stable
- [ ] The closed-loop system is unstable.
- [ ] The closed-loop system is marginally stable.
- [ ] The system retains the same stability properties in closed-loop (as in open-loop).
- [ ] The open-loop system is stable.

Når du sætter en P-controller på med:

$$K_P = 115$$

så skaleres hele Nyquist-plottet med 115:

$$K_P G(j\omega)$$

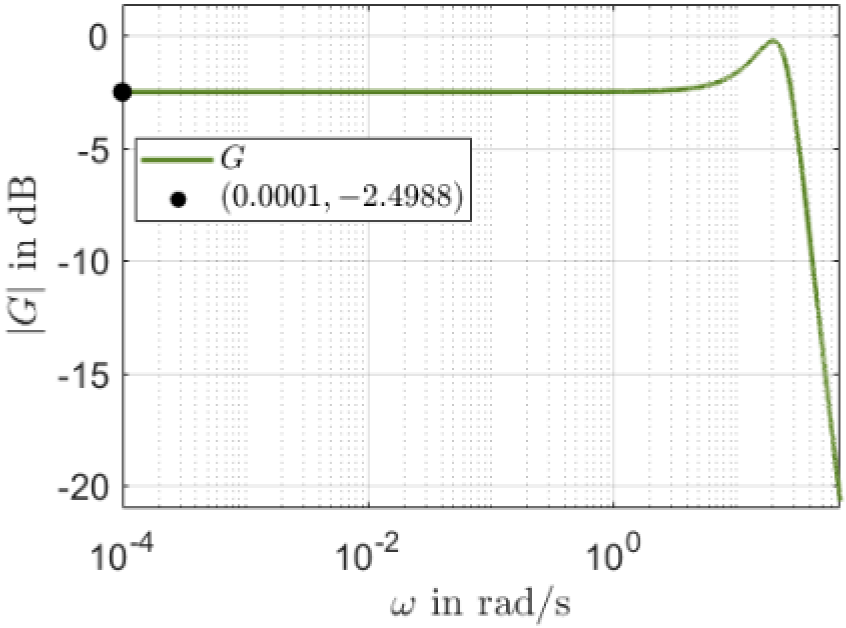
Realakse-krydset bliver derfor:

$$115 \cdot (-0.009) = -1.035$$

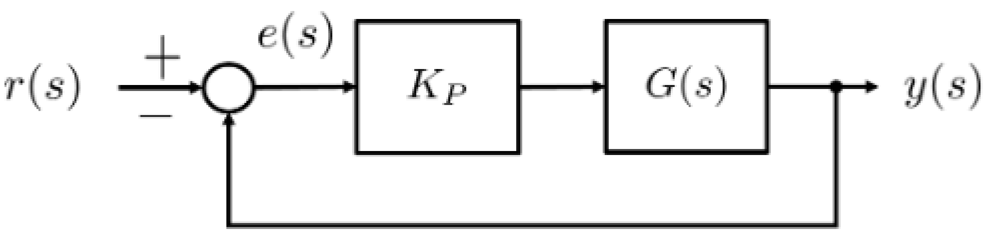
Altså krydser kurven realaksen **til venstre for** -1.

```
Open-loop har 1 ustabil pol
Kp = 115 flytter realakse-krydset til -1.035
Det er venstre for -1
Krydset er counter-clockwise
=> én korrekt encirclement
=> closed-loop stabil
```

Question 14. Consider a system described by a transfer function $G(s)$ with a magnitude Bode plot as shown in the figure below:



A P-controller with gain K_P is applied to the system and the loop is closed with unit feedback as in the following figure.



If the steady state error $e_{SS} = \lim_{s \to 0} e(s)$ due to a unit step change in the reference signal is $e_{SS} = 0.1$, the proportional gain K_P is approximately:

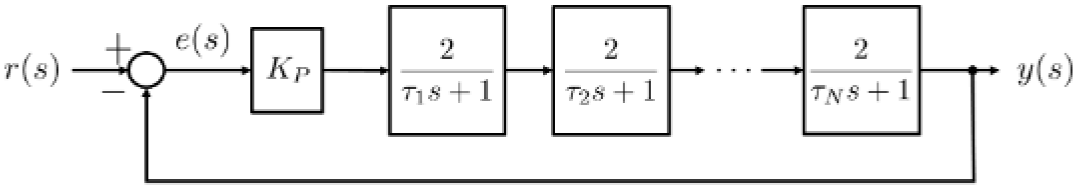
- ○ $K_P = 0.4$
- ○ $K_P = 1$
- ○ $K_P = 2.5$
- ✔ $K_P = 22.5$
- ○ $K_P = 7.95$

```
clear; clc;
```

```
ess = 0.1;

G0_dB = -2.4988;          % aflæst fra Bode-plottet
G0 = 10^(G0_dB/20);       % dB til almindelig gain

Kp = (1/ess - 1)/G0
```

```
Kp =
12.0000
```

Question 15. Consider a system described by a transfer function $G(s)$ that consists of $N = 9$ low-pass filters in series as shown in the block diagram below:



If the steady state unit step response of the closed-loop system is $y_{SS} = 0.999$, then K_P is approximately:

- [x] $K_P = 1.95$
- [ ] $K_P = 1$
- [ ] $K_P = 18$
- [ ] $K_P = 25$
- [ ] $K_P = 5.8$

```
clear; clc;

N = 9;
yss = 0.999;

G0 = 2^N;

Kp = yss / (G0*(1 - yss))
```

```
Kp =
1.9512
```

Question 16. A system is described by the following transfer function

$$G(s) = \frac{14}{s^3 + 30s^2 + 257s + 2}$$

A PI-Lead controller

$$C(s) = K_P \underbrace{\frac{\tau_i s + 1}{\tau_i s}}_{C_{PI}(s)} \underbrace{\frac{\tau_d s + 1}{\alpha \tau_d s + 1}}_{C_D(s)}$$

is to be designed so that the open-loop system $C(s)G(s)$ has phase margin $\gamma_M = 75^o$. If ω_c is the open-loop crossover frequency, $N_i = \omega_c \tau_i = 3$ and $\alpha = 0.05$, the controller proportional gain K_P is approximately:

- ○ $K_P = 34$
- ○ $K_P = 235.9$
- ○ $K_P = -34$
- ● $K_P = 50$
- ○ $K_P = 0.02$

```
clear; clc;

s = tf('s');

G = 14/(s^3 + 30*s^2 + 257*s + 2);

PM = 75;
alpha = 0.05;
Ni = 3;

% PI-fase ved crossover
phi_PI = atan(Ni)*180/pi - 90;

% Maksimal fase fra lead-leddet
phi_lead = asind((1-alpha)/(1+alpha));

% Ønsket fase fra planten G
phi_G_required = -180 + PM - phi_PI - phi_lead;

% Funktion til plantens fase
```

```
plant_phase = @(w) angle(evalfr(G,1j*w))*180/pi;

% Find crossover-frekvensen
% Brug smallere interval, så vi undgår fase-wrap ved -180/+180
wc = fzero(@(w) plant_phase(w) - phi_G_required, [5 10]);

% Find tidskonstanter
tau_i = Ni/wc;
tau_d = 1/(wc*sqrt(alpha));

% Controller uden Kp
C_noKp = ((tau_i*s + 1)/(tau_i*s))*((tau_d*s + 1)/(alpha*tau_d*s + 1));

% Vælg Kp så |C(jwc)G(jwc)| = 1
Kp = 1/abs(evalfr(C_noKp*G,1j*wc));

wc
```

```
wc =
9.8135
```

```
tau_i
```

```
tau_i =
0.3057
```

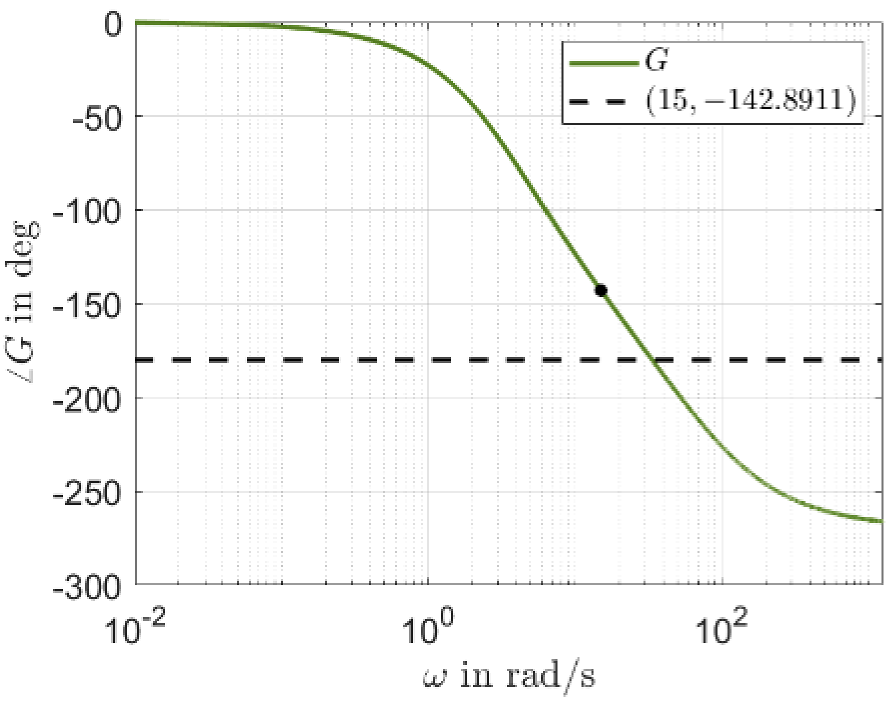
```
tau_d
```

```
tau_d =
0.4557
```

```
Kp
```

```
Kp =
49.8473
```

Question 17. Consider the system described by a transfer function with the following phase Bode plot:



A P-Lead-**Lag** controller

$$C(s) = K_P \frac{\tau_d s + 1}{\alpha \tau_d s + 1} \frac{\tau_i s + 1}{\tau_i s + \frac{1}{\beta}}$$

with $\alpha = 0.2$ and $N_i = 5$ needs to be designed for the system. The new crossover frequency (after applying the controller) is selected as $\omega_c = 15$ rad/s. If a phase margin $\gamma_M = 75^o$ is desired for the open-loop system $C(s)G(s)$, what is the (approximate) value of the parameter β of the Lag-part?

- [x] $\beta = 1.54$
- [ ] $\beta = -2.5$
- [ ] $\beta = 0.17$
- [ ] $\beta = 0.58$
- [ ] $\beta = 0$

```
clear; clc;

PM = 75;
wc = 15;
```

```
phase_G = -142.8911;

alpha = 0.2;
Ni = 5;

% Ønsket total fase fra controlleren
phase_C_required = (-180 + PM) - phase_G;

% Lead-fase
phase_lead = asind((1-alpha)/(1+alpha));

% Krævet lag-fase
phase_lag = phase_C_required - phase_lead;

% Løs for beta
syms beta positive

eqn = atan(Ni)*180/pi - atan(beta*Ni)*180/pi == phase_lag;

beta_sol = double(solve(eqn, beta))
```

```
beta_sol =
1.5419
```

Question 18. Which of the following statements is correct?

- [x] A closed-loop system with 0 dB gain at very low (theoretically 0) frequencies implies zero steady-state error.
- [ ] Pre-filters (reference filters) are used to stabilise open-loop unstable systems.
- [ ] Increasing the proportional gain K_P of a P-controller will shift the phase curve in the open-loop Bode plot upwards.
- [ ] Encirclement of the $(-1, 0)$ point on a Nyquist diagram always implies instability of the unit-feedback closed-loop system.
- [ ] In a PI-controller design, there can be a value N_i for which the I-part gives positive phase contribution at the new crossover frequency.

$$\boxed{\text{A closed-loop system with 0 dB gain at very low frequencies implies zero steady-state error.}}$$

Forklaringen er:

0 dB betyder lineær gain:

$$10^{0/20} = 1$$

Hvis **closed-loop** transferfunktionen har DC-gain:

$$T(0) = 1$$

så vil et unit step input ende med:

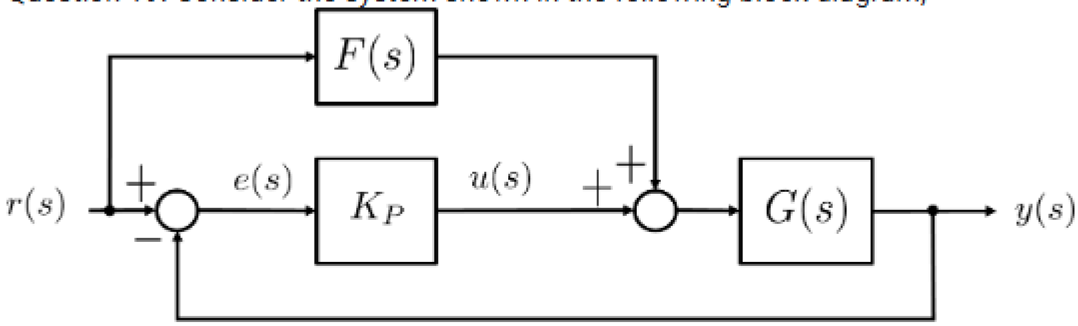
$$y_{ss} = 1$$

og dermed bliver fejlen:

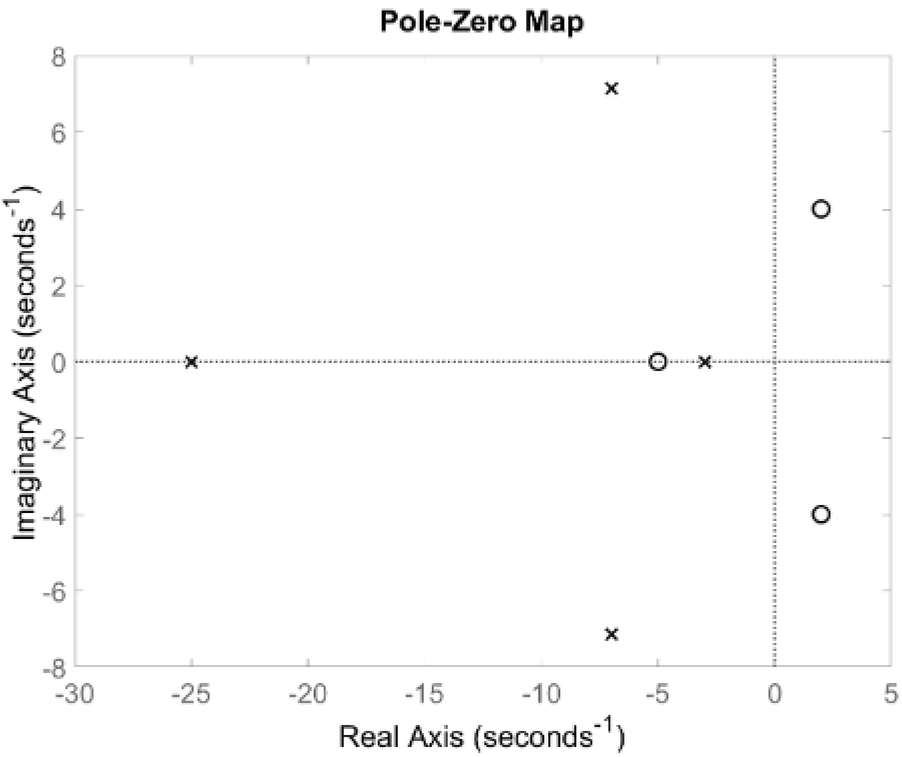
$$e_{ss} = r_{ss} - y_{ss} = 1 - 1 = 0$$

Så 0 dB ved meget lave frekvenser betyder, at systemet følger konstante referencesignaler uden steady-state error.

Question 19. Consider the system shown in the following block diagram,



where $K_P > 0$ and the transfer function $G(s)$ has the following zeros-poles diagram:



A feed-forward controller $F(s)$ needs to be designed such that the steady-state error is zero. Which of the following is an appropriate design for $F(s)$?

- $F(s) = G(0)$
- $F(s) = \dfrac{1}{G(s)}$
- ✔ $F(s) = \dfrac{1}{G(s)(\tau_f s + 1)}, \ \tau_f = 0.01$
- $F(s) = \dfrac{K_P G(s)}{\tau_f s + 1}, \ \tau_f = 0.01$

Fra blokdiagrammet:

$$e(s) = r(s) - y(s)$$

og inputtet til $G(s)$ er:

$$K_P e(s) + F(s)r(s)$$

Derfor:

$$y = G\left(K_P(r-y) + Fr\right)$$

$$y(1 + K_P G) = G(K_P + F)r$$

Fejloverføringen bliver:

$$\frac{E(s)}{R(s)} = \frac{1 - G(s)F(s)}{1 + K_P G(s)}$$

For et unit step er steady-state error:

$$e_{ss} = \lim_{s \to 0} \frac{E(s)}{R(s)} = \frac{1 - G(0)F(0)}{1 + K_P G(0)}$$

For at få:

$$e_{ss} = 0$$

skal:

$$1 - G(0)F(0) = 0$$

altså:

$$F(0) = \frac{1}{G(0)}$$

Derfor er den simple korrekte feed-forward:

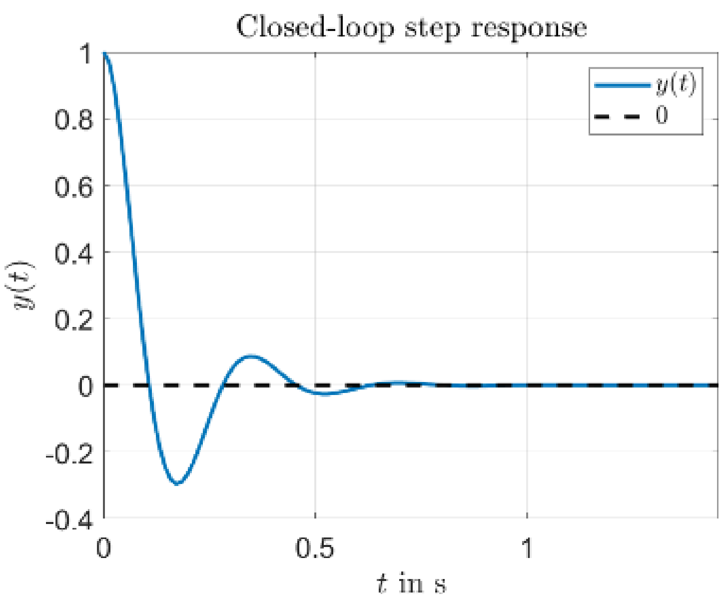
$$\boxed{F(s) = \frac{1}{G(0)}}$$

Question 20. Consider the system described by the following differential equation

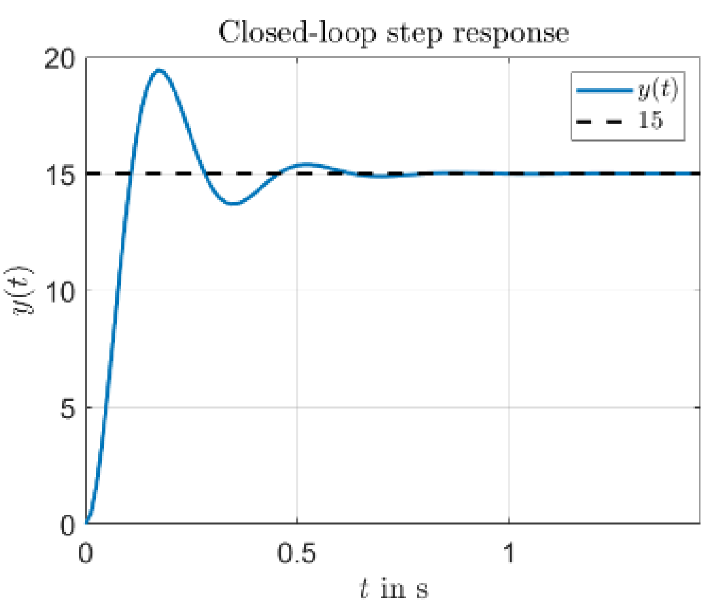
$$\ddot{y}(t) + 14\dot{y}(t) = 25u(t)$$
$$u(t) = K_P(r(t) - y(t))$$
$$r(t) = \begin{cases} 1 \text{ if } t > 0 \\ 0 \text{ elsewhere} \end{cases}$$

where $y(t)$ is the output of the system, $u(t)$ is the control input and $r(t)$ is a unit step reference signal, respectively. With $K_P = 15$ and assuming zero initial conditions, i.e. $y(0) = \dot{y}(0) = \ddot{y}(0) = u(0) = r(0) = 0$, which of the following plots represents the step response of the closed-loop system?

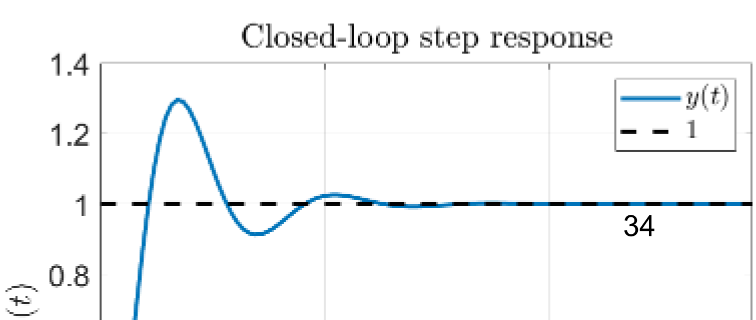
○



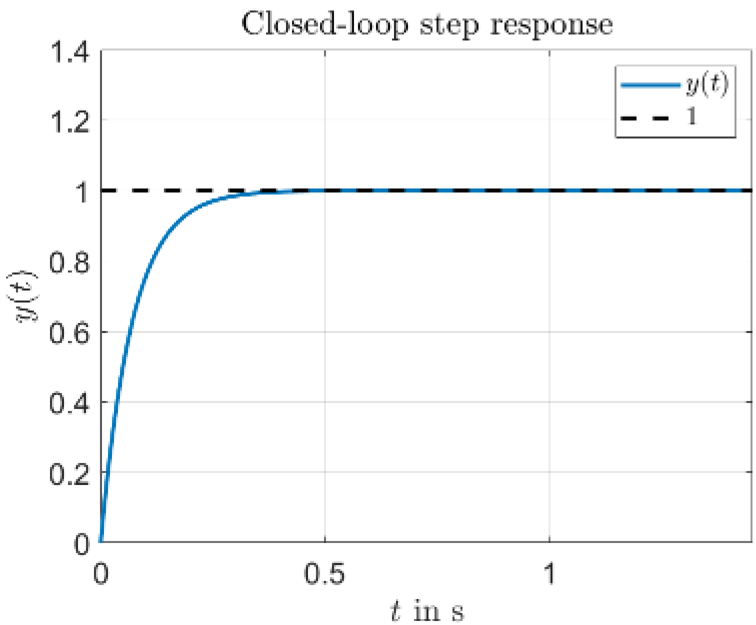
○



✔



○



Closed-loop step response
$y(t)$
1
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
0.5
1
t in s
$y(t)$

Hvis spørgsmålet handler om, at feed-forward'en er valgt så steady-state error er nul, så skal outputtet for et unit step ende i:

$$y_{ss} = 1$$

fordi referenceinputtet er:

$$r(t) = 1$$

og nul steady-state error betyder:

$$e_{ss} = r_{ss} - y_{ss} = 0$$

altså:

$$1 - y_{ss} = 0$$

$$y_{ss} = 1$$

Derfor skal du vælge grafen, hvor den stiplede slutværdi er **1**.

Så den markerede graf her er korrekt:

$$\boxed{\text{step response der ender i } y(t) = 1}$$

Graferne der ender i fx -14 eller 15, kan ikke passe til nul steady-state error for et unit step, fordi de giver:

$$e_{ss} = 1 - (-14) = 15$$

eller

$$e_{ss} = 1 - 15 = -14$$

Altså ikke nul.

Så for det nye spørgsmål: **vælg grafen med slutværdi 1.**

I opgaven har closed-loop systemet stadig dynamik fra $G(s)$, så responsen forventes at have **oversving og dæmpede svingninger.**

Begge grafer ender i:

$$y_{ss} = 1$$

så begge har nul steady-state error.

Men den rigtige skal også matche transienten. Derfor vælges grafen med **oversving**, ikke den glatte monotone graf.

Block diagram
A block diagram of a closed-loop system is shown in Figure 1.

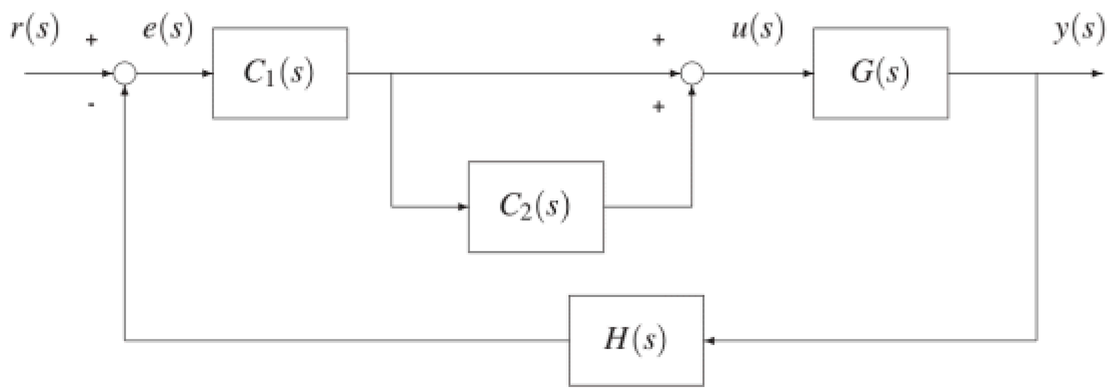


Figure 1.

$C_1(s)$, $C_2(s)$, $G(s)$, and $H(s)$ are dynamic systems. Further, it is assumed that the closed-loop system is stable.

Question 1

The open-loop transfer function $G_{open}(s)$ is given by:

- ○ $G_{open}(s) = (C_1(s) + C_2(s))G(s)H(s)$
- ○ $G_{open}(s) = \frac{C_1(s)}{1+C_2(s)}G(s)H(s)$
- ✔ $G_{open}(s) = C_1(s)(1 + C_2(s))G(s)H(s)$
- ○ $G_{open}(s) = C_1(s)C_2(s)G(s)H(s)$
- ○ $G_{open}(s) = C_1(s)C_2(s) + C_1(s)G(s)H(s)$

```
clear; clc;

syms C1 C2 G H;

G_ol = C1*(1+C2)*G*H
```

G_ol = $C_1\,G\,H\,(C_2+1)$

Question 2

The closed-loop transfer function $G_{cl}(s)$ from input r to output y is given by:

- ○ $G_{cl}(s) = \frac{C_1(s)C_2(s)G(s)}{1+C_1(s)(1+C_2(s))G(s)H(s)}$
- ○ $G_{cl}(s) = \frac{(C_1(s)+C_2(s))G(s)}{1+(C_1(s)+C_2(s))G(s)H(s)}$
- ✔ $G_{cl}(s) = \frac{C_1(s)(1+C_2(s))G(s)}{1+C_1(s)(1+C_2(s))G(s)H(s)}$
- ○ $G_{cl}(s) = \frac{C_1(s)(1+C_2(s))G(s)}{1+C_1(s)C_2(s)G(s)H(s)}$
- ○ $G_{cl}(s) = \frac{C_1(s)(1+C_2(s))G(s)H(s)}{1+C_1(s)(1+C_2(s))G(s)H(s)}$

```
clear; clc;

syms C1 C2 G H;

FB = C1*(1+C2)*G;

G_cl = FB / (1+C1*(1+C2)*G*H)
```

G_cl =

$$\frac{C_1\,G\,(C_2+1)}{C_1\,G\,H\,(C_2+1)+1}$$

Question 3

Let

$C_1(s) = K_1$

$C_2(s) = \frac{K_2}{s}$

$G(s) = \frac{k}{\tau s+1}$

$H(s) = \frac{1}{\tau_h s+1}$

What is the type and order of the open-loop system:

- ○ Type 1 and order 2
- ○ Type 0 and order 2
- ○ Type 3 and order 1
- ● Type 1 and order 3
- ○ Type 0 and order 3

```
clear; clc;

syms K1 K2 s k tau tau_h

C1 = K1;
C2 = K2/s;
G  = k/(tau*s + 1);
H  = 1/(tau_h*s + 1);

% Open-loop transfer function
G_ol = C1*(1 + C2)*G*H;
G_ol = simplify(G_ol)
```

G_ol =

$$\frac{K_1\,k\,\left(\frac{K_2}{s}+1\right)}{(s\,\tau+1)\,(s\,\tau_h+1)}$$

```
[num_ol, den_ol] = numden(G_ol);

num_ol = collect(expand(num_ol), s)
```

num_ol = $(K_1\,k)\,s + K_1\,K_2\,k$

```
den_ol = collect(expand(den_ol), s)
```

den_ol = $(\tau\,\tau_h)\,s^3 + (\tau + \tau_h)\,s^2 + s$

```
% Open-loop poles
poles_ol = solve(den_ol == 0, s)
```

poles_ol =

$$\begin{pmatrix} 0 \\ -\frac{1}{\tau} \\ -\frac{1}{\tau_h} \end{pmatrix}$$

```
% Find multiplicity automatisk
N = double(polynomialDegree(den_ol, s));

mult_ol = zeros(length(poles_ol), 1);

for i = 1:length(poles_ol)

    p = poles_ol(i);
    m = 0;

    for j = 0:N

        test = simplify(subs(diff(den_ol, s, j), s, p));

        if isAlways(test == 0)
            m = m + 1;
        else
            break
        end

    end

    mult_ol(i) = m;

end
```

```
Warning: Unable to prove '-(tau - tau_h)/tau == 0'.
Warning: Unable to prove '(tau - tau_h)/tau_h == 0'.
```

```
% Vis poler og multiplicitet
T_ol = table(poles_ol, mult_ol, ...
```

```
    'VariableNames', {'Pole', 'Multiplicity'})
```

T_ol = 3×2 table

	Pole	Multiplicity
1	0	1
2	-1/tau	1
3	-1/tau_h	1

```
% Forward path uden H
FB = C1*(1 + C2)*G;
FB = simplify(FB)
```

FB =

$$\frac{K_1\, k\, \left(\frac{K_2}{s}+1\right)}{s\,\tau+1}$$

```
% Closed-loop transfer function
G_cl = FB/(1 + G_ol);
G_cl = simplify(G_cl)
```

G_cl =

$$\frac{K_1\, k\, \left(\frac{K_2}{s}+1\right)}{\left(\frac{K_1\, k\, \left(\frac{K_2}{s}+1\right)}{(s\,\tau+1)\,\left(s\,\tau_h+1\right)}+1\right)\,(s\,\tau+1)}$$

```
% Closed-loop numerator og denominator
[num_cl, den_cl] = numden(G_cl);

num_cl = collect(expand(num_cl), s)
```

num_cl = $(K_1\, k\, \tau_h)\, s^2 + (K_1\, k + K_1\, K_2\, k\, \tau_h)\, s + K_1\, K_2\, k$

```
den_cl = collect(expand(den_cl), s)
```

den_cl = $(\tau\, \tau_h)\, s^3 + (\tau + \tau_h)\, s^2 + (K_1\, k + 1)\, s + K_1\, K_2\, k$

```
% Closed-loop poles
poles_cl = solve(den_cl == 0, s)
```

poles_cl =

$$\begin{pmatrix} \mathrm{root}(\sigma_1, z, 1) \\ \mathrm{root}(\sigma_1, z, 2) \\ \mathrm{root}(\sigma_1, z, 3) \end{pmatrix}$$

where

$$\sigma_1 = \tau\,\tau_h\,z^3 + \tau_h\,z^2 + \tau\,z^2 + K_1\,k\,z + z + K_1\,K_2\,k$$

Da vi kun har 1 pol der ligger i origo er det type 1 samt vores closed loop TF er i 3 orden.

Question 4

What is the static loop-gain K_0 for the open-loop system:

- ○ $K_0 = K_1(1 + K_2)k$
- ○ $K_0 = K_1(1 + K_2)$
- ○ $K_0 = (K_1 + K_2)k$
- ✔ $K_0 = K_1 K_2 k$
- ○ $K_0 = \frac{K_1(1+K_2)k}{\tau\tau_h}$

```
clear; clc;

syms K1 K2 s k tau tau_h

C1 = K1;
C2 = K2/s;
G  = k/(tau*s + 1);
H  = 1/(tau_h*s + 1);

G_ol = C1*(1 + C2)*G*H;
G_ol = expand(simplify(G_ol))
```

G_ol =

$$\frac{K_1\,k}{s\,\tau + s\,\tau_h + s^2\,\tau\,\tau_h + 1} + \frac{K_1\,K_2\,k}{s + s^2\,\tau + s^2\,\tau_h + s^3\,\tau\,\tau_h}$$

```
K0 = limit(s*G_ol, s, 0)
```

K0 = $K_1 K_2 k$

Question 5

$C_1(s)$ and $C_2(s)$ is the controller for the system. The controller is:

- [x] PI controller
- [ ] P-lead controller
- [ ] P controller
- [ ] P-lag controller
- [ ] PI-lead controller

Gode regler:

Controller	Typisk form
P	K_P
PI	$K_P + \frac{K_I}{s}$
PD	$K_P + K_D s$
PID	$K_P + \frac{K_I}{s} + K_D s$
Lead	$K\frac{Ts+1}{\alpha Ts+1}$, typisk $0 < \alpha < 1$
Lag	$K\frac{Ts+1}{\beta Ts+1}$, typisk $\beta > 1$

```
clear; clc;

syms K1 K2 s k tau tau_h

C1 = K1;
C2 = K2/s;
G  = k/(tau*s + 1);
H  = 1/(tau_h*s + 1);

% Controller transfer function
```

```
C_ctrl = C1*(1 + C2);
C_ctrl = simplify(C_ctrl)
```

C_ctrl =

$$K_1 \left(\frac{K_2}{s} + 1\right)$$

```
% Numerator og denominator
[num_C, den_C] = numden(C_ctrl);

num_C = collect(expand(num_C), s)
```

num_C = $K_1\,s + K_1\,K_2$

```
den_C = collect(expand(den_C), s)
```

den_C = s

```
% Controller poles og zeros
poles_C = solve(den_C == 0, s)
```

poles_C = 0

```
zeros_C = solve(num_C == 0, s)
```

zeros_C = $-K_2$

```
% Find hvor mange poler i s = 0 controlleren har
m0 = 0;
den_temp = den_C;

while isAlways(subs(den_temp, s, 0) == 0)
    m0 = m0 + 1;
    den_temp = simplify(den_temp/s);
end

% Grader af numerator og denominator
deg_num = double(polynomialDegree(num_C, s));
deg_den = double(polynomialDegree(den_C, s));
deg_den_without_origin = double(polynomialDegree(den_temp, s));

% Afgør controller-type
if deg_den_without_origin == 0

    if m0 == 0 && deg_num == 0
        controller_type = "P controller";

    elseif m0 == 1 && deg_num == 1
```

```
        controller_type = "PI controller";

    elseif m0 == 0 && deg_num == 1
        controller_type = "PD controller";

    elseif m0 == 1 && deg_num == 2
        controller_type = "PID controller";

    else
        controller_type = "Other controller with pure integrator(s)";
    end

else

    if m0 == 0 && deg_num == 1 && deg_den == 1
        controller_type = "P-lead or P-lag controller";

    elseif m0 == 1 && deg_num == 2 && deg_den == 2
        controller_type = "PI-lead or PI-lag controller";

    else
        controller_type = "Other/unknown controller";
    end

end

controller_type
```

```
controller_type =
"PI controller"
```

Dynamic system

Let a dynamic system be described by the following equations:

$\dot{x}_1 = x_2$

$\dot{x}_2 = -ax_1 - bx_2 + cu$

$y = x_1$

where u is the input to the system and y is the output from the system.

Question 6

The type and order of the system is:

- ○ Type 1 and order 3
- ○ Type 0 and order 3
- ○ Type 1 and order 2
- ✔ Type 0 and order 2
- ○ Type 2 and order 0

Systemet er givet ved:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -ax_1 - bx_2 + cu$$

$$y = x_1$$

Fordi $y = x_1$, så er:

$$\dot{y} = \dot{x}_1 = x_2$$

og derfor:

$$\ddot{y} = \dot{x}_2$$

Indsæt $\dot{x}_2$:

$$\ddot{y} = -ay - b\dot{y} + cu$$

Flyt leddene over:

$$\ddot{y} + b\dot{y} + ay = cu$$

Tag Laplace:

$$s^2Y(s) + bsY(s) + aY(s) = cU(s)$$

Så transferfunktionen bliver:

$$\frac{Y(s)}{U(s)} = \frac{c}{s^2 + bs + a}$$

```
clear; clc;

syms a b c s x1 x2 u y

% Laplace-form af differentialligningerne
eq1 = s*x1 == x2;
eq2 = s*x2 == -a*x1 - b*x2 + c*u;
eq3 = y == x1;

% Løs ligningerne for x1, x2 og u
sol = solve([eq1, eq2, eq3], [x1, x2, u]);

% Find u udtrykt ved y
u_expr = simplify(sol.u)
```

u_expr =

$$\frac{y\,(s^2 + b\,s + a)}{c}$$

```
% Transfer function y/u
G = simplify(y/u_expr);
G = collect(G, s)
```

G =

$$\frac{c}{s^2 + b\,s + a}$$

```
% Numerator og denominator
[num, den] = numden(G);

num = collect(expand(num), s)
```

num = c

```
den = collect(expand(den), s)
```

den = $s^2 + b\,s + a$

```
% Poler
poles = solve(den == 0, s)
```

poles =

$$\begin{pmatrix} -\frac{b}{2} - \frac{\sqrt{b^2 - 4\,a}}{2} \\ \frac{\sqrt{b^2 - 4\,a}}{2} - \frac{b}{2} \end{pmatrix}$$

```
% Order
system_order = polynomialDegree(den, s)
```

```
system_order =
2
```

```
% Type: antal rene s-faktorer i nævneren
den_temp = den;
system_type = 0;

while isAlways(subs(den_temp, s, 0) == 0)
    system_type = system_type + 1;
    den_temp = simplify(den_temp/s);
end
```

```
Warning: Unable to prove 'a == 0'.
```

```
system_type
```

```
system_type =
0
```

```
system_order
```

```
system_order =
2
```

Question 7

The open-loop transfer function $G(s)$ from u to y is given by:

- ○ $G(s) = \frac{c}{s(bs+a)}$
- ● $G(s) = \frac{c}{s^2+bs+a}$
- ○ $G(s) = \frac{c}{s(as+b)}$
- ○ $G(s) = \frac{c}{as^2+bs+1}$
- ○ $G(s) = \frac{1}{as^2+bs+c}$

Se fra sidste opgave

Systemet er givet ved:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -ax_1 - bx_2 + cu$$

$$y = x_1$$

Fordi $y = x_1$, så er:

$$\dot{y} = \dot{x}_1 = x_2$$

og derfor:

$$\ddot{y} = \dot{x}_2$$

Indsæt $\dot{x}_2$:

$$\ddot{y} = -ay - b\dot{y} + cu$$

Flyt leddene over:

$$\ddot{y} + b\dot{y} + ay = cu$$

Tag Laplace:

$$s^2Y(s) + bsY(s) + aY(s) = cU(s)$$

Så transferfunktionen bliver:

$$\frac{Y(s)}{U(s)} = \frac{c}{s^2 + bs + a}$$

$$G = \frac{c}{s^2 + bs + a}$$

Bode plot
A Bode plot for an open-loop stable system $G_{open}(s)$ be given shown in Figure 2.

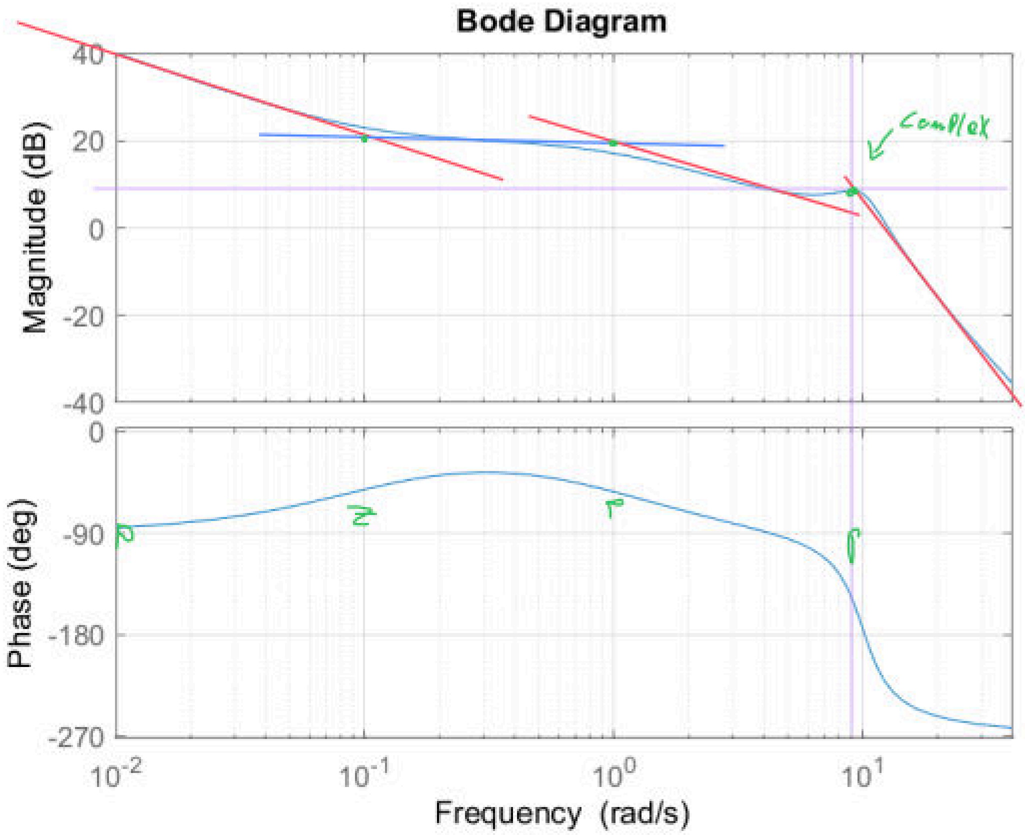
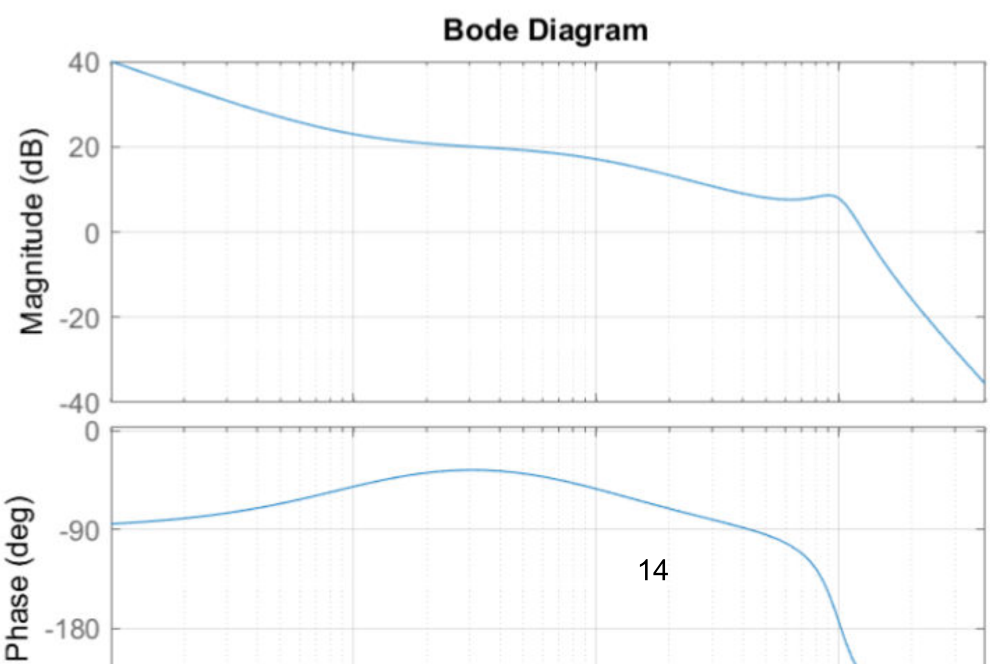


Figure 2.

-90° at origen

Bode plot
A Bode plot for an open-loop stable system $G_{open}(s)$ be given shown in Figure 2.



Question 8

The open-loop system $G_{open}(s)$ includes:

- ○ Four complex poles.
- ✔ One zero and four poles. One pole is in origo, one pole is real and two complex poles.
- ○ Three poles. One pole is in origo and two complex poles.
- ○ One zero and four complex poles.
- ○ One zero and four poles. One pole is in origo and the other poles are real.

I dit plot:

1. **Ved lav frekvens** starter magnitude med ca. -20 dB/dec, og fasen er omkring $-90°$.
   Det betyder:
   $$\boxed{\text{én pol i origo}}$$
2. **Omkring** $\omega \approx 0.1$ bliver hældningen mindre negativ, cirka fra -20 til 0 dB/dec.
   Det betyder et nul, fordi et nul giver $+20$ dB/dec:
   $$\boxed{\text{ét nul}}$$
3. **Omkring** $\omega \approx 1$ begynder hældningen at falde igen med ca. -20 dB/dec.
   Det betyder én reel pol:
   $$\boxed{\text{én reel pol}}$$
4. **Omkring** $\omega \approx 10$ falder magnitude meget kraftigere, cirka yderligere -40 dB/dec, og fasen falder hurtigt mod ca. $-270°$.
   Det svarer til et kompleks konjugeret polpar:
   $$\boxed{\text{to komplekse poler}}$$

Samlet får du:

$$\boxed{\text{ét nul og fire poler}}$$

Element	Magnitude-hældning	Fasebidrag
Pol i origo, $\frac{1}{s}$	-20 dB/dec fra starten	-90
Reelt nul, $(s/z+1)$	$+20$ dB/dec efter knækfrekvens	$+90$
Reel pol, $\frac{1}{s/p+1}$	-20 dB/dec efter knækfrekvens	-90
Kompleks polpar	-40 dB/dec efter knækfrekvens	-180

$$\boxed{+20\ \text{dB/dec} = \text{zero}}$$

$$\boxed{-20\ \text{dB/dec} = \text{reel pol}}$$

$$\boxed{-40\ \text{dB/dec} = \text{to poler / kompleks polpar}}$$

$$\boxed{-20\ \text{dB/dec fra start og fase } -90^\circ = \text{pol i origo}}$$

1. Start med lavfrekvensområdet

På Bode-plottet falder magnituden cirka med

$$-20 \text{ dB/dec}$$

og fasen ligger omkring

$$-90^\circ$$

Det matcher tabellens regel for:

$$\frac{1}{s}$$

Altså har systemet i lavfrekvensområdet noget der opfører sig som:

$$G(s) \approx K\frac{1}{s}$$

For $1/s$-leddet siger tabellen:

$$-20 \text{ dB/dec}$$

og linjen går gennem 0 dB ved

$$\omega = 1$$

Konstanten K flytter hele magnitude-kurven op med:

$$20\log_{10}(K)$$

---

2. Find K med tabellens regel

Aflæs for eksempel ved:

$$\omega = 0.1 \text{ rad/s}$$

Her ligger magnitude cirka ved:

$$26 \text{ dB}$$

For rent $1/s$ ville magnituden ved $\omega = 0.1$ være:

$$20 \text{ dB}$$

fordi $1/s$-linjen går gennem 0 dB ved $\omega = 1$, og én dekade lavere er den 20 dB højere.

Forskellen er derfor:

$$26 - 20 = 6 \text{ dB}$$

Det er bidraget fra K:

$$20\log_{10}(K) = 6$$

$$K = 10^{6/20}$$

$$\boxed{K \approx 2}$$

Så med tabellens regler finder man K ved at sige:

$$\boxed{\text{asymptoten ligger 6 dB over } 1/s\text{-linjen}}$$

3. Find ζ med underdæmpet pol-reglen

Tabellen siger for underdæmpede poler:

$$\frac{1}{\left(\frac{s}{\omega_0}\right)^2 + 2\zeta\left(\frac{s}{\omega_0}\right) + 1}$$

at der kommer en resonanstop omkring:

$$\omega = \omega_0$$

og at toppens højde er:

$$-20\log_{10}(2\zeta)$$

Fra plottet ses resonansen omkring:

$$\omega_0 \approx 10 \text{ rad/s}$$

Nu skal du aflæse, hvor meget toppen ligger **over den asymptotiske linje uden resonanstoppen.**

Hvis toppen for eksempel ligger cirka 3 dB over asymptoten, får du:

$$-20\log_{10}(2\zeta) = 3$$

$$2\zeta = 10^{-3/20}$$

$$\zeta = \frac{10^{-3/20}}{2}$$

$$\boxed{\zeta \approx 0.35}$$

Så med reglen fra tabellen:

$$\boxed{\zeta = \frac{10^{-\Delta M/20}}{2}}$$

hvor ΔM er resonanstoppen i dB over asymptoten.

Typiske værdier:

Resonanstop ΔM	ζ
$2\ \mathrm{dB}$	0.40
$3\ \mathrm{dB}$	0.35
$4\ \mathrm{dB}$	0.32

Så fra figuren vil man typisk sige:

$$\boxed{\zeta \approx 0.3\text{–}0.4}$$

Hvis du aflæser omkring

$$\omega = 0.1\ \mathrm{rad/s}$$

og magnituden er cirka

$$M \approx 26\ \mathrm{dB}$$

så får du

$$|G(j0.1)| = 10^{26/20} \approx 20$$

og dermed

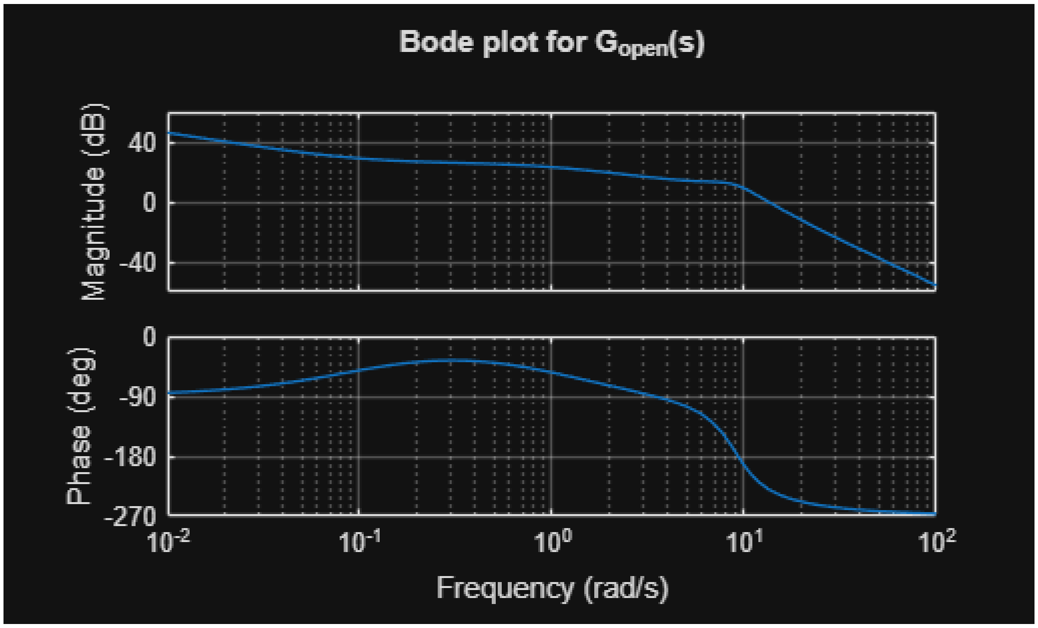
$$K \approx 0.1 \cdot 20 = 2$$

Så:

$$\boxed{K \approx 2}$$

```
clear; clc; close all;

s = tf('s');

% Valgte værdier fra Bode-aflæsning
K = 2;
zeta = 0.3;

% Knækfrekvenser aflæst fra Bode-plottet
wz = 0.1;    % realt nulpunkt
wp = 1;      % reel pol
w0 = 9;      % underdæmpede poler, ca. resonansfrekvens

% Systemets led
P_origin = K/s;                          % K/s, giver -20 dB/dec og -90 grader
Z_real   = s/wz + 1;                     % realt nulpunkt ved wz
P_real   = s/wp + 1;                     % reel pol ved wp
P_2order = (s/w0)^2 + 2*zeta*(s/w0) + 1; % underdæmpede poler

% Samlet open-loop transferfunktion
```

```
G = P_origin * Z_real / (P_real * P_2order);

% Plot Bode-diagram
figure;
bodeplot(G,{0.01,100})
grid on
title('Bode plot for G_{open}(s)')
```




```
% Vis poler og nulpunkter
poles = pole(G)
```

```
poles = 4×1 complex
   0.0000 + 0.0000i
  -2.7000 + 8.5855i
  -2.7000 - 8.5855i
  -1.0000 + 0.0000i
```

```
zeros = zero(G)
```

```
zeros =
-0.1000
```

```
%% Beregn K fra lavfrekvens-asymptoten

% Vælg en lav frekvens hvor systemet ligner K/s
wK = 0.1;

% Find magnitude ved wK
[mag,phase] = bode(G,wK);
mag = squeeze(mag);

% For G ≈ K/s gælder K = w*|G(jw)|
K_calc = wK * mag
```

```
K_calc =
2.8147
```

```matlab
%% Beregn zeta fra resonanstop-reglen

% Ifølge tabellen: peak_dB = -20*log10(2*zeta)
peak_dB = -20*log10(2*zeta)
```

```
peak_dB =
4.4370
```

```matlab
% Hvis man i stedet kender peak_dB, kan zeta beregnes sådan:
zeta_calc = 10^(-peak_dB/20)/2
```

```
zeta_calc =
0.3000
```

Bode plot

A Bode plot for an open-loop stable system $G_{open}(s)$ be given shown in Figure 2.

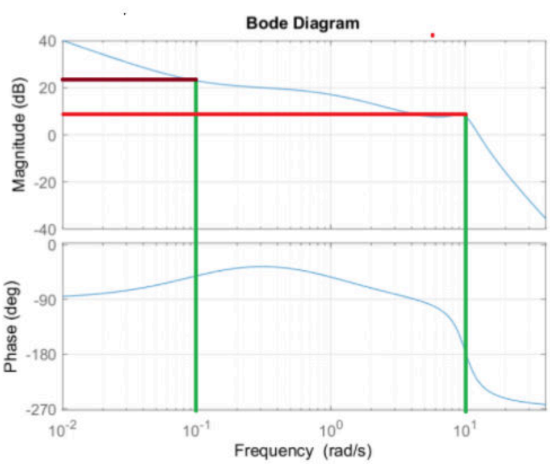


Figure 2.

$\omega_1 = 0.1$

$Mag = 26\,dB$

$10^{\frac{26}{20}} = 20$

$K = \omega_1 \cdot 20 = \underline{\underline{2}}$

$\omega_0 \approx 10 \ \frac{rad}{s}$ (resonans)

$-20 \cdot \log_{10}(2 \cdot \zeta) = 3\,db \Rightarrow$

$2\zeta = 10^{-\frac{3}{20}} \Rightarrow \zeta = \frac{10^{-\frac{3}{20}}}{2} = \underline{\underline{0.35}}$

Question 9.

What is the static loop-gain K_0?

- [x] $K_0 = 1$
- [ ] $K_0 = 100$
- [ ] $K_0 = 40$
- [ ] $K_0 = 2.3$
- [ ] $K_0 = \infty$

```
clear; clc;

s = tf('s');

K = 2;
Zeta = 0.3;

Z = s/0.1 + 1;

P1 = K/s;
P2 = s/1 + 1;
P3 = 1 + 2*Zeta*(s/9) + (s/9)^2;

G = P1*Z/(P3*P2);

dcgain(G)
```

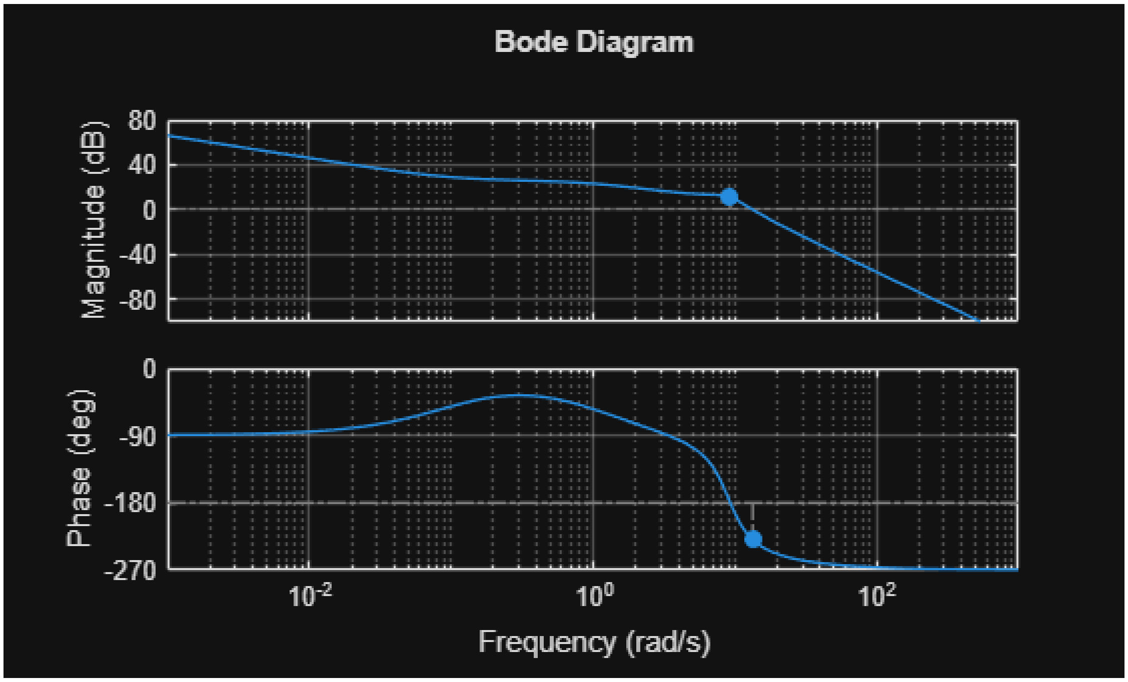
```
ans =
Inf
```

Question 10

What is the phase margin and gain margin:

- ○ Phase margin $\approx 46^0$ and gain margin is infinity
- ☑ Phase margin $\approx -46^0$ and gain margin $\approx -7.6dB$
- ○ Phase margin $\approx -46^0$ and gain margin ≈ 7.6 dB
- ○ Phase margin is $\approx 90^0$ and the gain margin ≈ -7.6 dB
- ○ Phase margin $\approx 46^0$ and gain margin ≈ 7.6 dB

```
clear; clc;

s = tf('s');

K = 2;
Zeta = 0.3;

Z = s/0.1 + 1;

P1 = K/s;
P2 = s/1 + 1;
P3 = 1 + 2*Zeta*(s/9) + (s/9)^2;

G = P1*Z/(P3*P2);

% Plot med margins
margin(G)
grid on
```




```
% Få værdierne ud
[Gm, Pm, Wcg, Wcp] = margin(G);
```

```
Warning: The closed-loop system is unstable.
```

```
% Gain margin i dB
Gm_dB = 20*log10(Gm);

fprintf('Phase margin = %.2f deg\n', Pm);
```

```
Phase margin = -49.59 deg
```

```
fprintf('Gain margin = %.2f dB\n', Gm_dB);
```

```
Gain margin = -10.77 dB
```

```
fprintf('Gain margin frequency = %.2f rad/s\n', Wcg);
```

```
Gain margin frequency = 9.27 rad/s
```

```
fprintf('Phase margin frequency = %.2f rad/s\n', Wcp);
```

```
Phase margin frequency = 13.35 rad/s
```

Question 11

The cross-over frequency ω_c and the π-frequency ω_π are:

- ○ $\omega_c = 10.2\ rad/sec,\ \ \omega_\pi = 12.7\ rad/sec$
- ○ $\omega_c = 0.35\ rad/sec,\ \ \omega_\pi = 10.2\ rad/sec$
- ○ $\omega_c = 12.7\ rad/sec$, and ω_π is not defined
- ✔ $\omega_c = 12.7\ rad/sec,\ \ \omega_\pi = 10.2\ rad/sec$
- ○ $\omega_c = \omega_\pi = 10.2\ rad/sec$

Svaret står i sidste opgave

Question 12

The system is now applied in closed-loop given by:

$G_{cl}(s) = \frac{G_{open}(s)}{1+G_{open}(s)}$

Let the output y be given by:

$y = G_{cl}(s)r$

where r is a unit step input.

We can say the following about the output y:

- [x] $|y(t)| \rightarrow 1 \; as \; t \rightarrow \infty$
- [ ] $|y(t)| \rightarrow \infty \; as \; t \rightarrow \infty$
- [ ] $|y(t)| \rightarrow 0 \; as \; t \rightarrow \infty$
- [ ] $|y(t)| \rightarrow k\sin(\omega t) \; as \; t \rightarrow \infty, \; k \neq 0$
- [ ] $|y(t)| \rightarrow constant \neq 0 \; as \; t \rightarrow \infty$

```
clear; clc;

s = tf('s');

K = 2;
Zeta = 0.3;

Z = s/0.1 + 1;

P1 = K/s;
P2 = s/1 + 1;
P3 = 1 + 2*Zeta*(s/9) + (s/9)^2;

G_ol = P1*Z/(P3*P2)
```

```
G_ol =

              1458 s + 145.8
  ---------------------------------------
  0.9 s^4 + 5.76 s^3 + 77.76 s^2 + 72.9 s

Continuous-time transfer function.
Model Properties
```

```
G_cl = G_ol/(1+G_ol)
```

```
G_cl =

                    1312 s^5 + 8529 s^4 + 1.142e05 s^3 + 1.176e05 s^2 + 1.063e04 s
  -----------------------------------------------------------------------------------------------------
  0.81 s^8 + 10.37 s^7 + 173.1 s^6 + 2339 s^5 + 1.542e04 s^4 + 1.256e05 s^3 + 1.229e05 s^2 + 1.063e04 s

Continuous-time transfer function.
Model Properties
```

```
%Anvender finale value teoremet:
dcgain(G_cl)
```

```
ans =
1
```

Question 13

A P-controller is now designed for $G_{open}(s)$ shown in Figure 2. The P-gain K_p is selected such that a phase margin at 90^0 is obtained.

K_p is:

- ○ $K_p = 2.75$
- ✔ $K_p = 0.36$
- ○ $K_p = 0.10$
- ○ $K_p = 8.80$
- ○ $K_p = 1.00$

```
clear; clc;

s = tf('s');

K = 2;
Zeta = 0.3;

Z = s/0.1 + 1;

P1 = K/s;
P2 = s/1 + 1;
```

```
P3 = 1 + 2*Zeta*(s/9) + (s/9)^2;

G_ol = P1*Z/(P3*P2)
```

```
G_ol =

              1458 s + 145.8
  ---------------------------------------
  0.9 s^4 + 5.76 s^3 + 77.76 s^2 + 72.9 s

Continuous-time transfer function.
Model Properties
```

```


pm = 90
```

```
pm =
90
```

```
phase = pm-180
```

```
phase =
-90
```

```
%på plottet aflæses værdien til ca. 10
1/db2mag(9.5)
```

```
ans =
0.3350
```

Question 14

A section of the Bode plot shown in Figure 2 is shown in Figure 3.

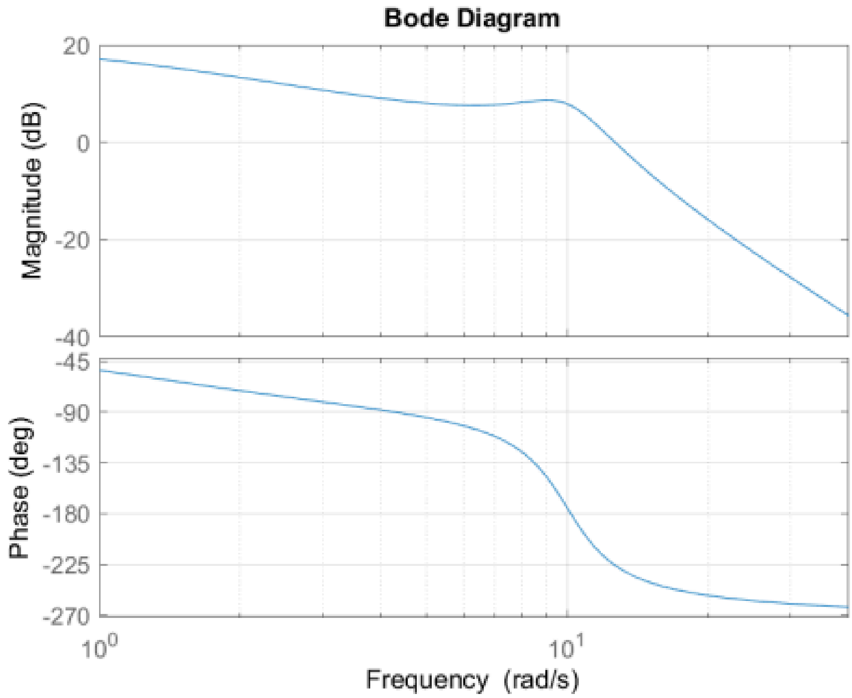


Figure 3.

A P-lead controller is designed for the system with $\alpha = 0.07$, and a phase margin at 60^0. The cross-over frequency ω_c with this design is

- [ ] $\omega_c = 4.2\ rad/sec$
- [ ] $\omega_c = 7.6\ rad/sec$
- [ ] $\omega_c = 6.1\ rad/sec$
- [ ] $\omega_c = 12.6\ rad/sec$
- [x] $\omega_c = 10.2\ rad/sec$

```
clear; clc;

s = tf('s');

K = 2;
Zeta = 0.3;

Z = s/0.1 + 1;
```

```
P1 = K/s;
P2 = s/1 + 1;
P3 = 1 + 2*Zeta*(s/9) + (s/9)^2;

G_ol = P1*Z/(P3*P2)
```

```
G_ol =

              1458 s + 145.8
  ---------------------------------------
  0.9 s^4 + 5.76 s^3 + 77.76 s^2 + 72.9 s

Continuous-time transfer function.
Model Properties
```

```
a = 0.07;
pm = 60;

phi_d = rad2deg(asin((1-a)/(1+a)))
```

```
phi_d =
60.3610
```

```
-180+pm-phi_d %Er ca. phase 180, som kan aflæses på plottet til 10 rad/s
```

```
ans =
-180.3610
```

```
%Svaret er da 10.2rad/s
```

Question 15.

What is the main reason for using a P-lead controller:

- ○ Will reduce the control signal
- ○ Will always give a stable closed-loop system
- ✔ Will reduce the open-loop phase shift at low frequencies.
- ○ Will increase the closed-loop bandwidth.
- ○ Will increase the type of the system

Step response
Two questions about step responses.

- The positive phase shift from Lead part should be exploited at the new crossover frequency of the open-loop transfer function (why?)

Huskeregler:

Controller	Primær effekt
P	ændrer gain
PI	fjerner steady-state error / øger type
PD / lead	gør systemet hurtigere / øger fase-margin
Lag	forbedrer steady-state accuracy, men gør ofte systemet langsommere
Lead	øger bandwidth
Lag	reducerer typisk bandwidth

Så til eksamen:

$$\boxed{\text{Lead} \Rightarrow \text{mere fase, hurtigere respons, større bandwidth}}$$

$$\boxed{\text{PI} \Rightarrow \text{integrator, større type, mindre steady-state error}}$$

$$\boxed{\text{Lag} \Rightarrow \text{bedre lavfrekvensgain, men langsommere system}}$$

Så her ville jeg vælge:

$$\boxed{\text{Will increase the closed-loop bandwidth.}}$$

Question 16.
Figure 4 shows a Bode plot of a stable system $G(s)$.

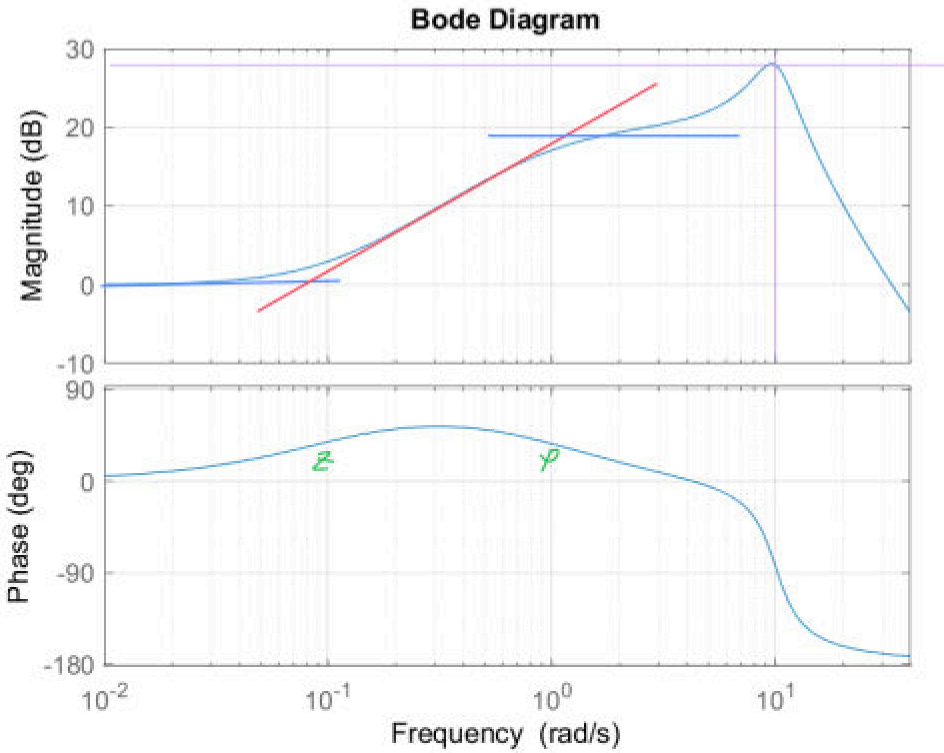
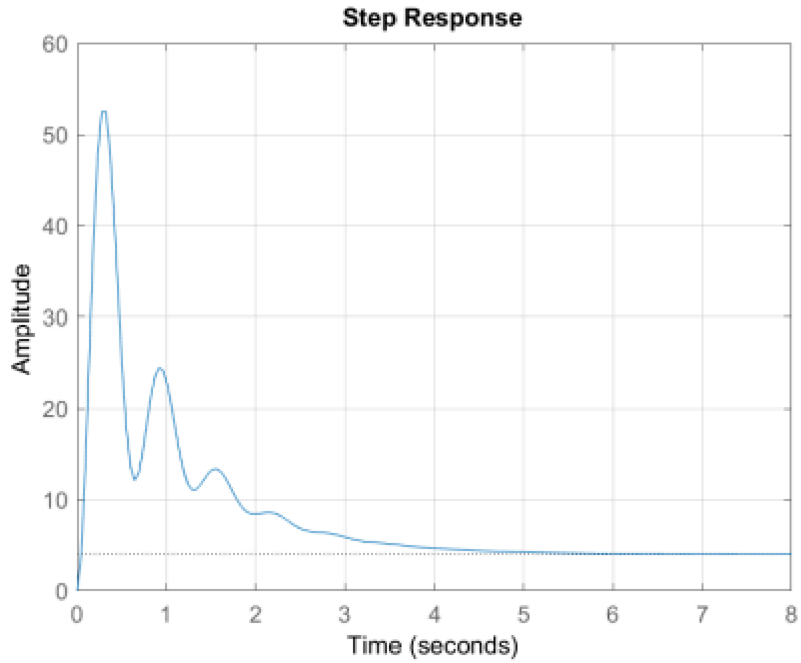
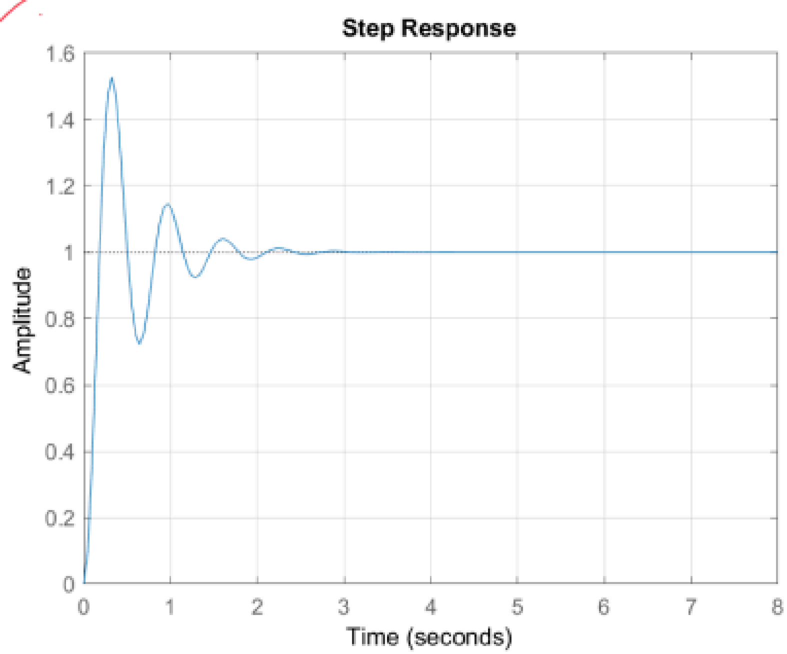
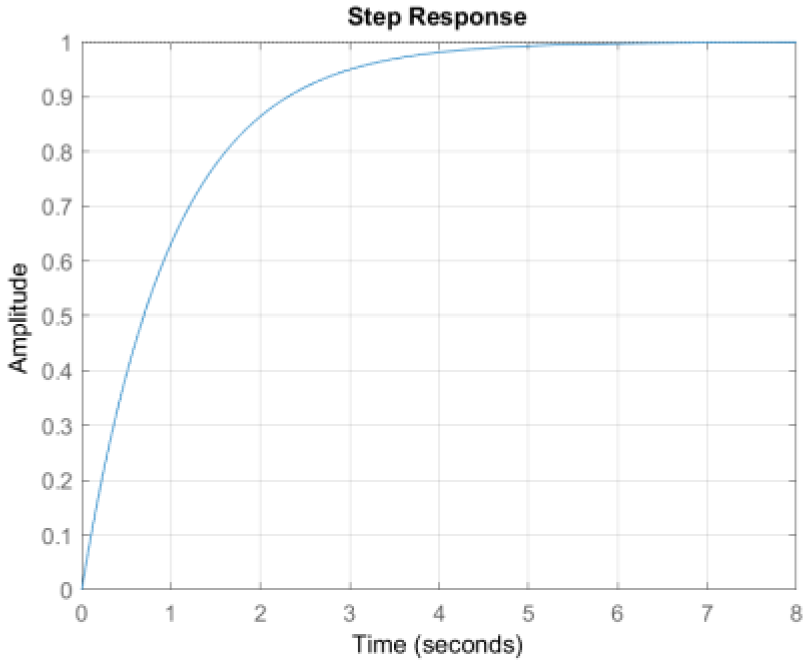


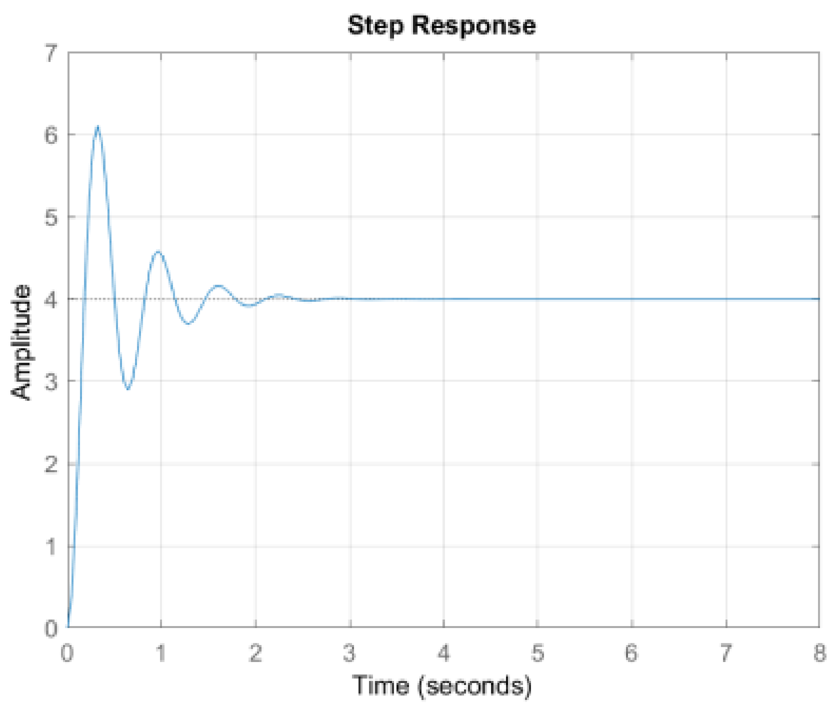
Figure 4.

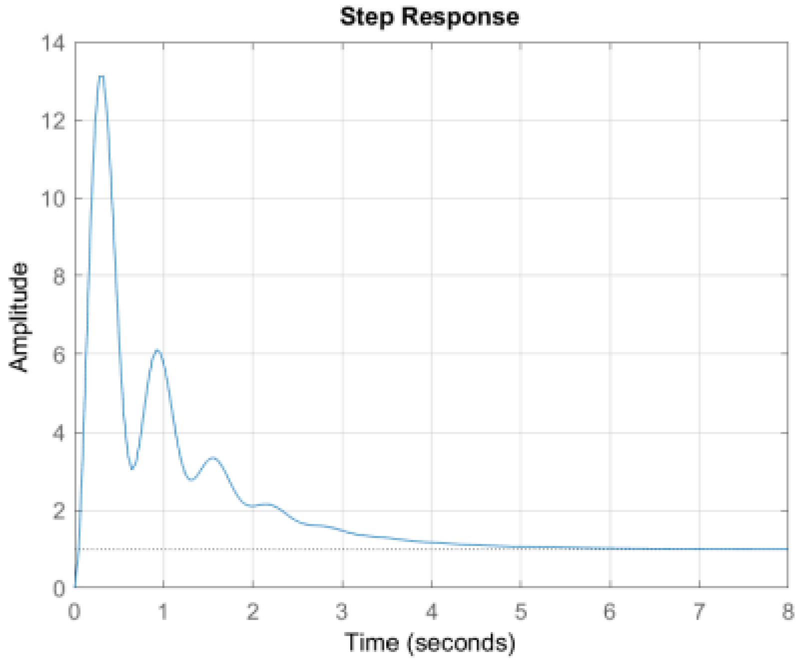
A unit step input is applied on $G(s)$. The response of the unit step input is:


Step Response
Amplitude
Time (seconds)
60
50
40
30
20
10
0
0
1
2
3
4
5
6
7
8

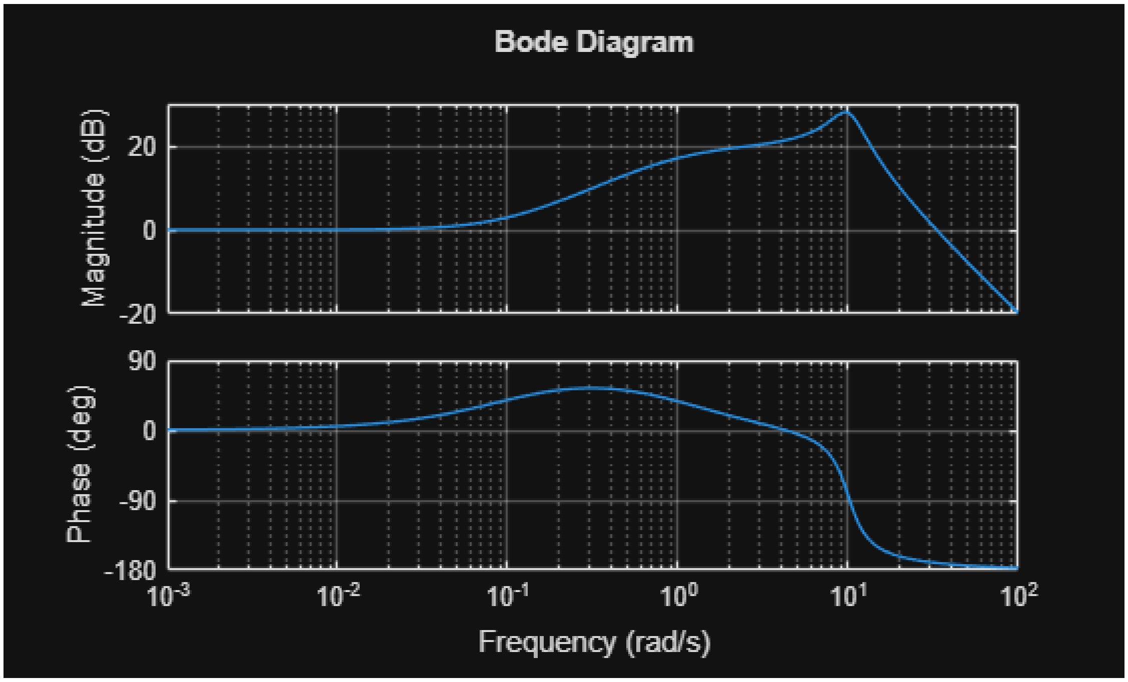

Step Response
Amplitude
Time (seconds)
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
1
2
3
4
5
6
7
8


Step Response
Amplitude
Time (seconds)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
1
2
3
4
5
6
7
8

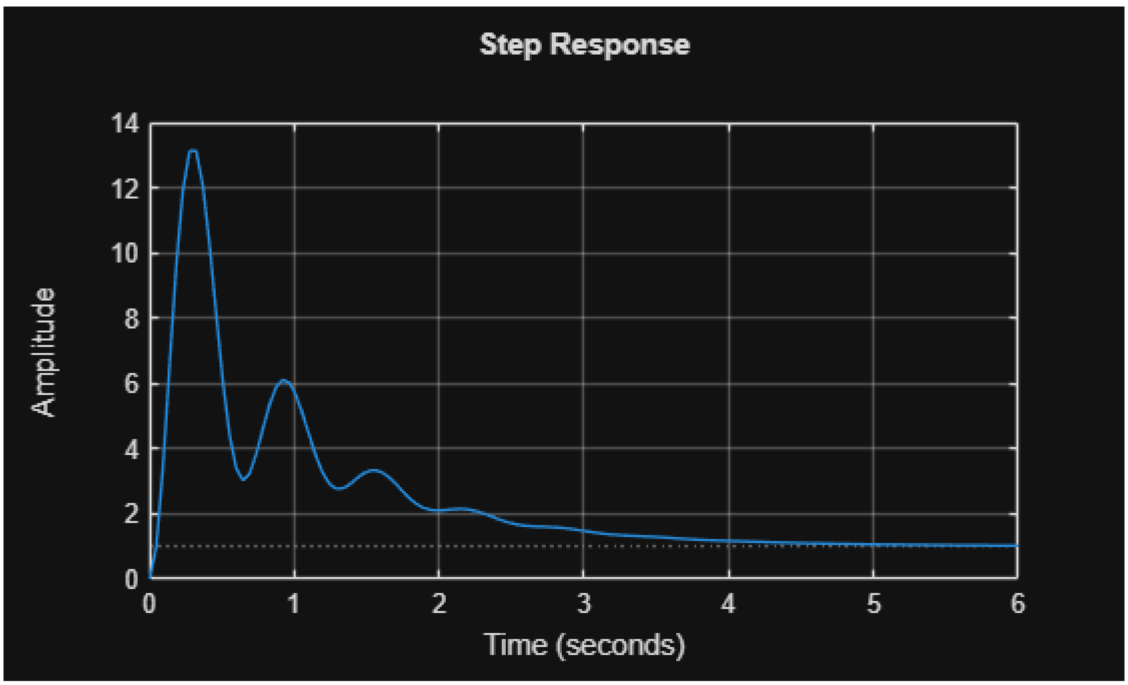

Step Response
Amplitude
Time (seconds)
0
1
2
3
4
5
6
7
0
1
2
3
4
5
6
7
8




```
clear; clc;

s = tf('s');

K = 1;
zeta = 0.2;

Z  = s/0.1 + 1;                     % zero ved 0.1 rad/s
P1 = s/1 + 1;                       % reel pol ved 1 rad/s
P2 = 1 + 2*zeta*(s/10) + (s/10)^2;      % kompleks polpar ved 10 rad/s

G = K*Z/(P1*P2);

bodeplot(G)
grid on
step(G)
```




```
grid on
```




```
pole(G)
```

```
ans = 3×1 complex
  -2.0000 + 9.7980i
  -2.0000 - 9.7980i
  -1.0000 + 0.0000i
```

```
zero(G)
```

```
ans =
```

```
-0.1000
```

Question 17.

Consider the step response for a stable system shown in Figure 10.

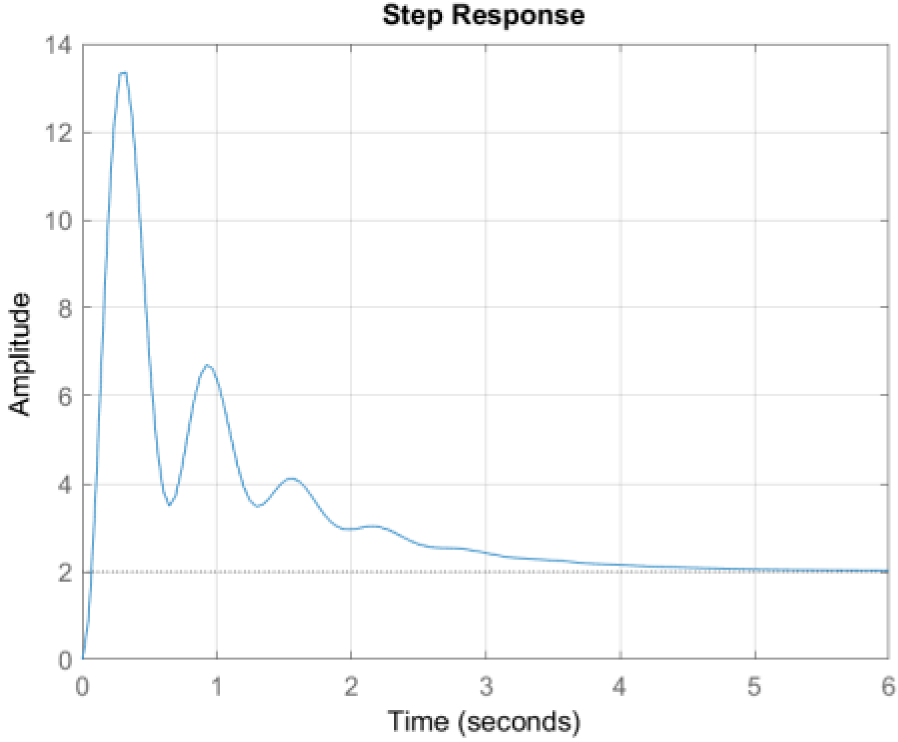


Figure 10.

The system includes a set of complex poles.
What is the damped eigenfrequency ω_d for the system?

- ○ $\omega_d = 0.1\ rad/sec$
- ○ $\omega_d = 0.6\ rad/sec$
- ✔ $\omega_d = 10\ rad/sec$
- ○ $\omega_d = 3.14\ rad/sec$
- ○ $\omega_d = 1.0\ rad/sec$

Steady-state error

Let a stable closed-loop system be given by the block diagram shown in Figure 11.

Du aflæser det fra **perioden mellem to nabotoppe** i stepresponsen.

På figuren ligger toppene cirka ved:

$$t_1 \approx 0.25\ \text{s}$$

og

```
clear; clc;

s = tf('s');

K = 1;
zeta = 0.2;

Z  = s/0.1 + 1;                          % zero ved 0.1 rad/s
P1 = s/1 + 1;                            % reel pol ved 1 rad/s
P2 = 1 + 2*zeta*(s/10) + (s/10)^2;       % kompleks polpar ved 10 rad/s

G = K*Z/(P1*P2);

p = pole(G);
complex_poles = p(abs(imag(p)) > 1e-6);

omega_d = abs(imag(complex_poles(1)))
```

```
omega_d =
9.7980
```

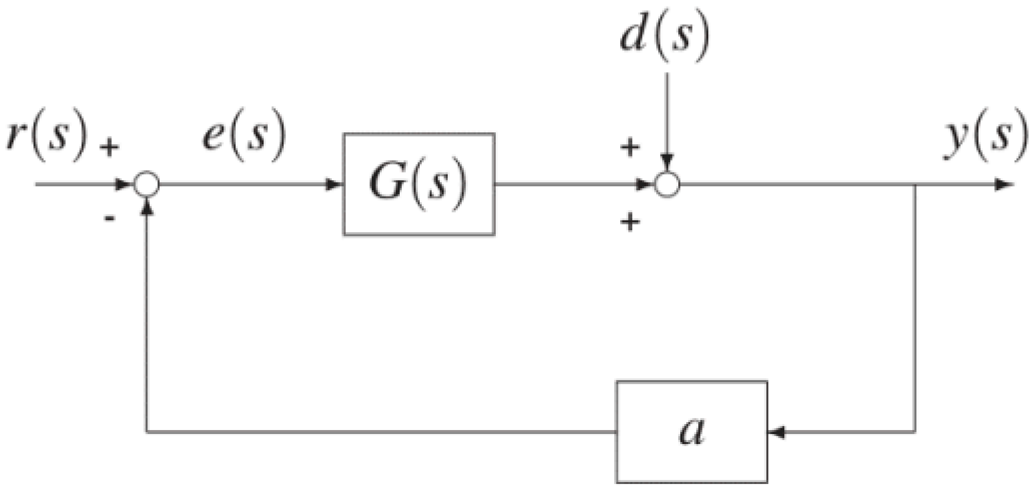


Figure 11.

$G(s)$ is a stable type 0 system with steady-state gain
$G_{ss} = \lim_{s\to 0} G(s) = 1$

Question 18.

The steady-state error $e_{r,ss}$ for a unit step input at r is:

- ○ $e_{r,ss} = \frac{1}{a}$
- ✔ $e_{r,ss} = \frac{1}{1+a}$
- ○ $e_{r,ss} = 0$
- ○ $e_{r,ss} = \frac{a}{1+a}$
- ○ $e_{r,ss} = \frac{-1}{1+a}$

Diagrammet har feedback med gain a, så fejlen er:

$$e(s) = r(s) - a\,y(s)$$

For input r og uden disturbance d, gælder:

$$y(s) = G(s)e(s)$$

Indsæt det i fejlligningen:

$$e(s) = r(s) - aG(s)e(s)$$

Saml $e(s)$:

$$e(s)(1 + aG(s)) = r(s)$$

Derfor:

Huskereglen her er:

Hvis systemet er **type 0**, får du normalt en **ikke-nul steady-state error** for step input.

Hvis feedback gain ikke er 1, men a, så bliver stepfejlen:

$$\boxed{e_{ss} = \frac{1}{1 + aG(0)}}$$

Her er $G(0) = 1$, så:

$$\boxed{e_{ss} = \frac{1}{1 + a}}$$

```
clear; clc;

syms s a;

G = 1;
H = G / (1+G*a)
```

H =

$$\frac{1}{a+1}$$

Question 19.

What is the steady-state value of the output y for a unit step input at d?

- [x] $y_{d,ss} = \frac{1}{1+a}$

$\frac{1}{1+Ga}$

- [ ] $y_{d,ss} = \frac{-1}{1+a}$
- [ ] $y_{d,ss} = \frac{1}{a}$
- [ ] $y_{d,ss} = 0$
- [ ] $y_{d,ss} = \frac{a}{1+a}$

For disturbance-input d sætter vi reference-inputtet til nul:

$$r(s) = 0$$

Fra diagrammet:

$$e(s) = r(s) - ay(s)$$

Når $r(s) = 0$:

$$e(s) = -ay(s)$$

Outputtet er summen af signalet fra $G(s)$ og disturbance $d(s)$:

$$y(s) = G(s)e(s) + d(s)$$

Indsæt $e(s) = -ay(s)$:

$$y(s) = G(s)(-ay(s)) + d(s)$$

$$y(s) = -aG(s)y(s) + d(s)$$

Saml $y(s)$:

$$y(s)(1 + aG(s)) = d(s)$$

Så transferfunktionen fra disturbance til output er:

$$\frac{y(s)}{d(s)} = \frac{1}{1 + aG(s)}$$

For unit step disturbance:

$$d(s) = \frac{1}{s}$$

Huskeregel:
For en disturbance, der lægges direkte på outputtet, får du ofte:

$$\boxed{\frac{y}{d} = \frac{1}{1 + \text{loop gain}}}$$

Her er loop gain ved DC:

$$aG(0) = a$$

så svaret bliver:

$$\boxed{\frac{1}{1+a}}$$

```
clear; clc;

syms s a;

G = 1;
H = G / (1+G*a)
```

H =

$$\frac{1}{a+1}$$

Question 20.

The steady-state error $e_{d,ss}$ is $-\frac{2}{3}$ for a unit step input at d. What is a?

- $a = \frac{5}{2}$
- $a = 5$
- $a = 1$
- $a = 2$
- $a = \frac{1}{2}$

```
clear; clc;

syms a

% Given
G0 = 1;             % G(0) = 1
```

```
d_ss = 1;            % unit step disturbance
e_d_ss_given = sym(-2)/3;

% Steady-state output due to disturbance
y_d_ss = d_ss/(1 + a*G0);

% Error when r = 0:
% e = r - a*y = -a*y
e_d_ss = -a*y_d_ss;

% Solve for a
a_sol = solve(e_d_ss == e_d_ss_given, a)
```

a_sol = 2

1. Q1. Consider the block diagram shown in the following figure Fig.1. **What is the transfer function** $\frac{Y(s)}{R(s)}$ (Select the correct answer)? (1 Point)

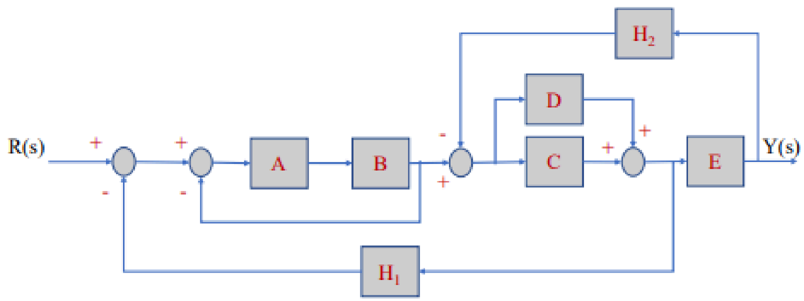


Figure 1: Block Diagram

1. $$\frac{Y(s)}{R(s)} = \frac{ABE^2(C+D)}{(1+AB)[1+(C+D)EH_2]E + ABE(C+D)H_1}$$

   CORRECT

2. $$\frac{Y(s)}{R(s)} = \frac{AB(C+D)}{(1+AB)[E+(C+D)H_2] + AB(C+D)H_1}$$

   wrong reduction of the take-off point

3. $$\frac{Y(S)}{R(s)} = \frac{ABE^2(C+D)}{(1+AB)[1+(C+D)H_2]E - AB(C+D)H_1}$$

   wrong sign and missing multiplication of block E

4. $$\frac{Y(s)}{R(s)} = \frac{AB(C+D)}{(1+AB)[E+(C+D)H_2]E - AB(C+D)H_1}$$

   wrong sign from the last feedback loop reduction and and wrong reduction of the take-off point

Answer:
Q1. [**CORRECT n.1**. The block diagram is reduced using the reduction rules. A and B blocks are connected in series, so they are multiplied. Blocks C and D are in parallel, they are summed. The unit feedback loop containing AB is simplified. The take-off point is shifted after the block E (H1 is divided by E). The block C+D is multiplied to E. The feedback loop containing (C+D)E and H_2 is simplified. The blocks in series are multiplied. Finally, the feedback loop is simplified.]

```
clear; clc;

syms A B C D E H1 H2
```

```
FB = A*B*(C+D)*E;

G_cl = FB / ...
    ((1+A*B)*(1+(C+D)*E*H2) + A*B*(C+D)*H1)
```

G_cl =

$$\frac{A\,B\,\mathrm{E}\,(C+\mathrm{D})}{(\mathrm{E}\,H_2\,(C+\mathrm{D})+1)\,(A\,B+1)+A\,B\,H_1\,(C+\mathrm{D})}$$

2. Q2. Consider the following transfer function of the RC circuit: $G(s) = \frac{V_c(s)}{V_i(s)} = \frac{\frac{1}{RC}}{s+\frac{1}{RC}}$, where V_c is the voltage across the capacitor, V_i is the source voltage, R is the resistance (50Ω), and C is the capacitance (160F). **Which of the following statements is FALSE?**

   1. The time duration after which the capacitor voltage has reached the steady state value is 40ms
      TRUE - 5TAU = 5RC = 40ms

   2. The time constant TAU is the time taken for the capacitor voltage to reach 63.2% of its final value from t=0
      TRUE - after 1tau, it is 63.2 % of its final steady state value.

   3. The initial value of the voltage V_c is 0
      TRUE - apply the initial value theorem

   4. The final value of the unit step response is 1
      TRUE - apply the final value theorem

   5. The time duration after which the capacitor voltage has reached the 63.2% of its final steady state value is 16 ms
      FALSE - it corresponds to 1tau = 1RC = 8 ms

> **Answer:**
> Q2. [**CORRECT n.5**]

```
clear; clc; close all;

syms s t

R = 50;              % ohm
C = 160e-6;          % farad = 160 microfarad

tau = R*C;           % time constant
```

```
G = 1/(R*C*s + 1);   % Transfer function: Vc(s)/Vi(s)

disp('Transfer function G(s):')
```

```
Transfer function G(s):
```

```
pretty(G)
```

$$\frac{1}{\frac{s}{125}+1}$$

```
fprintf('Tau = RC = %.4f s = %.2f ms\n', tau, tau*1000);
```

```
Tau = RC = 0.0080 s = 8.00 ms
```

```
fprintf('5 Tau = %.4f s = %.2f ms\n', 5*tau, 5*tau*1000);
```

```
5 Tau = 0.0400 s = 40.00 ms
```

```
Vc = 1 - exp(-t/tau);

V_initial = limit(Vc, t, 0);
V_final = limit(Vc, t, inf);

V_tau = double(subs(Vc, t, tau));
V_16ms = double(subs(Vc, t, 16e-3));

fprintf('Initial value Vc(0) = %.2f\n', double(V_initial));
```

```
Initial value Vc(0) = 0.00
```

```
fprintf('Final value Vc(inf) = %.2f\n', double(V_final));
```

```
Final value Vc(inf) = 1.00
```

```
fprintf('Vc(tau) = %.4f = %.2f %%\n', V_tau, V_tau*100);
```

```
Vc(tau) = 0.6321 = 63.21 %
```

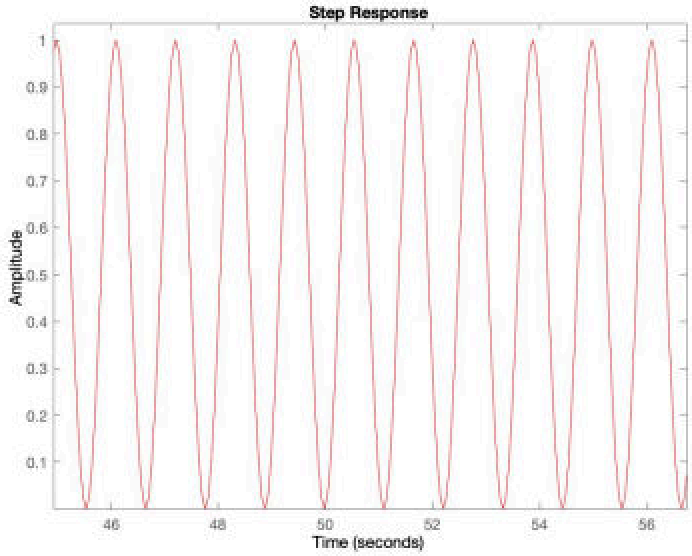
```
fprintf('Vc(16 ms) = %.4f = %.2f %%\n', V_16ms, V_16ms*100);
```

```
Vc(16 ms) = 0.8647 = 86.47 %
```

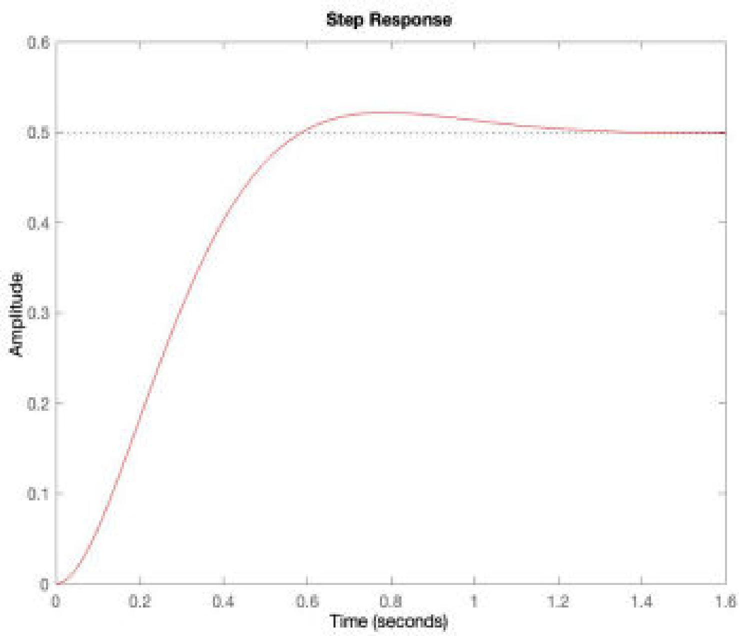
3. Q3. Consider a second order system

$$G(s) = \frac{w_n}{s^2 + 2\zeta w_n s + w_n^2}$$

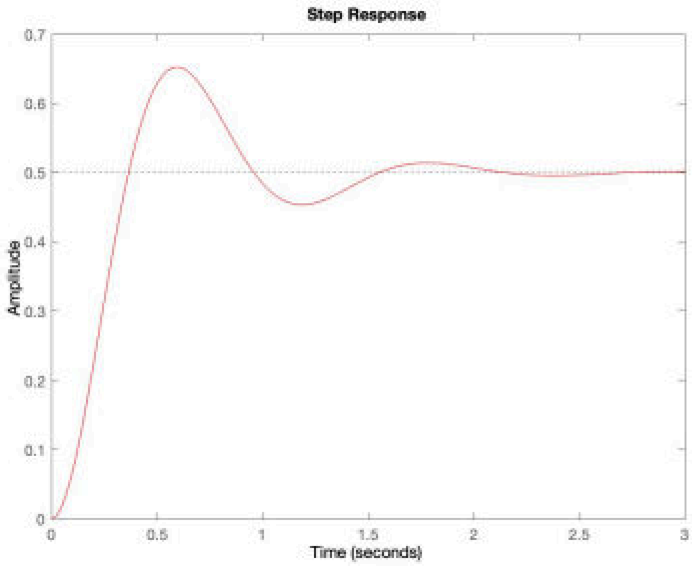
Select the correct unit step response when $\zeta = 0$



1.
CORRECT, a continuous time signal with constant amplitude and frequency

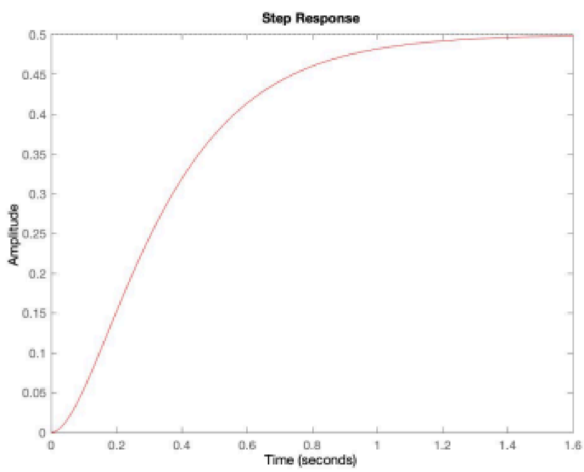


2.
FALSE, $\zeta = 1$, the unit step response of the second order system will try to reach the step input in steady state.



3.

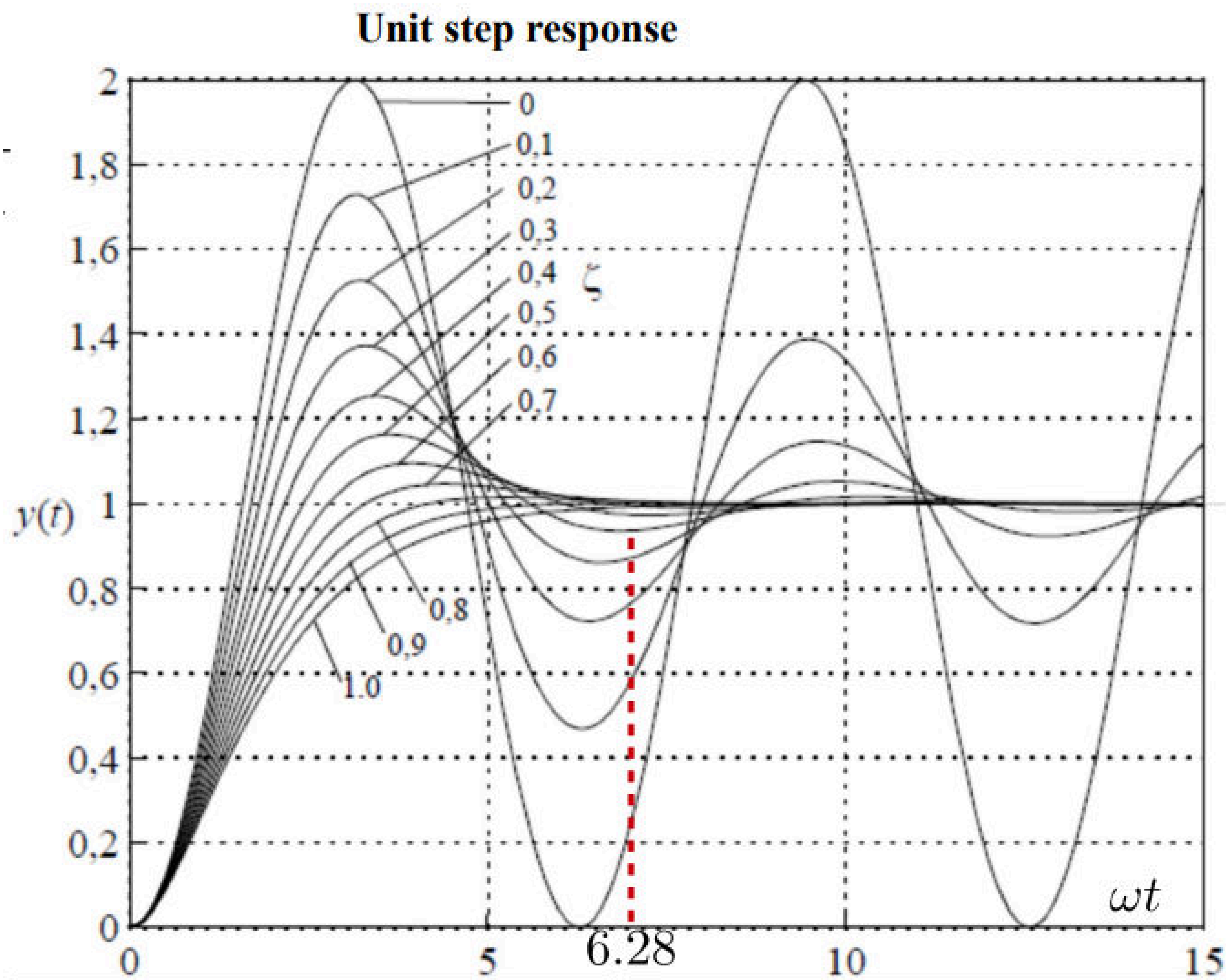
FALSE - the unit step response of the second order system have damped oscillations (decreasing amplitude) when ζ lies between zero and one.



4.

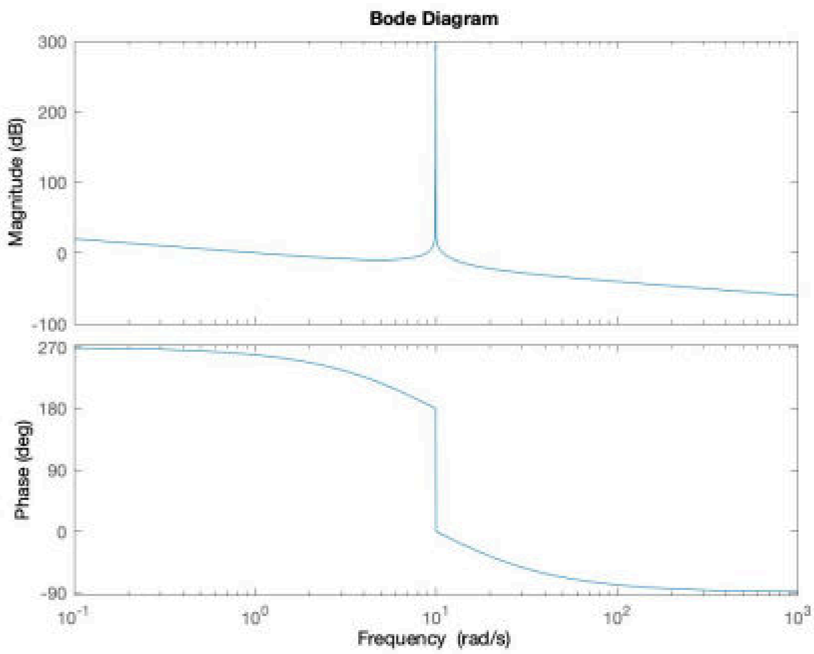
FALSE - Since it is over damped, the unit step response of the second order system when ζ ¿ 1 will never reach step input in the steady state.

Answer:
Q3. [**CORRECT n. 1**]

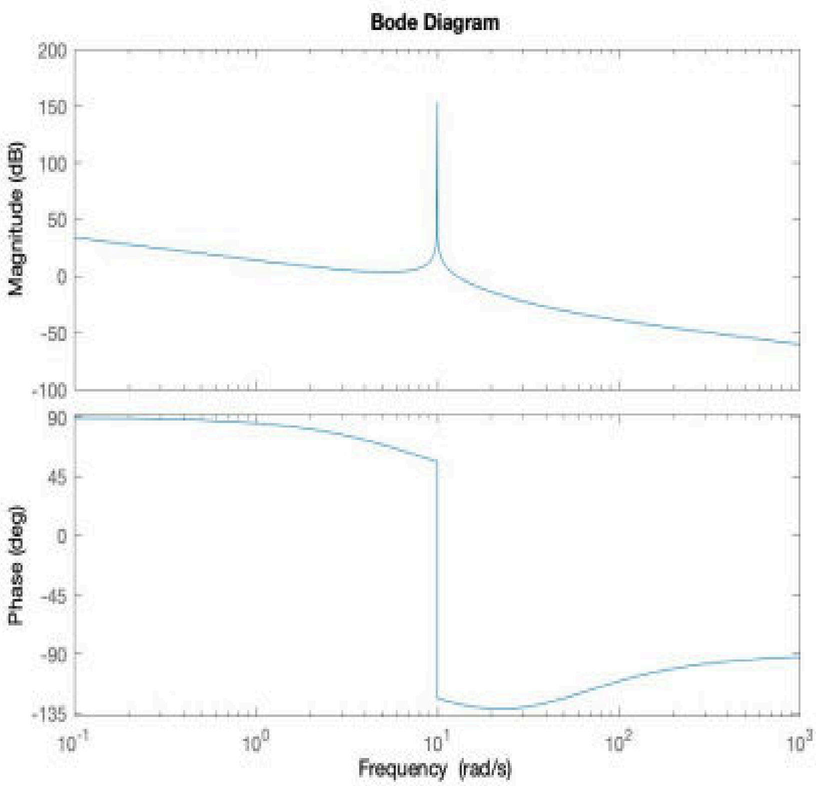


'22 Linear Control Design 1

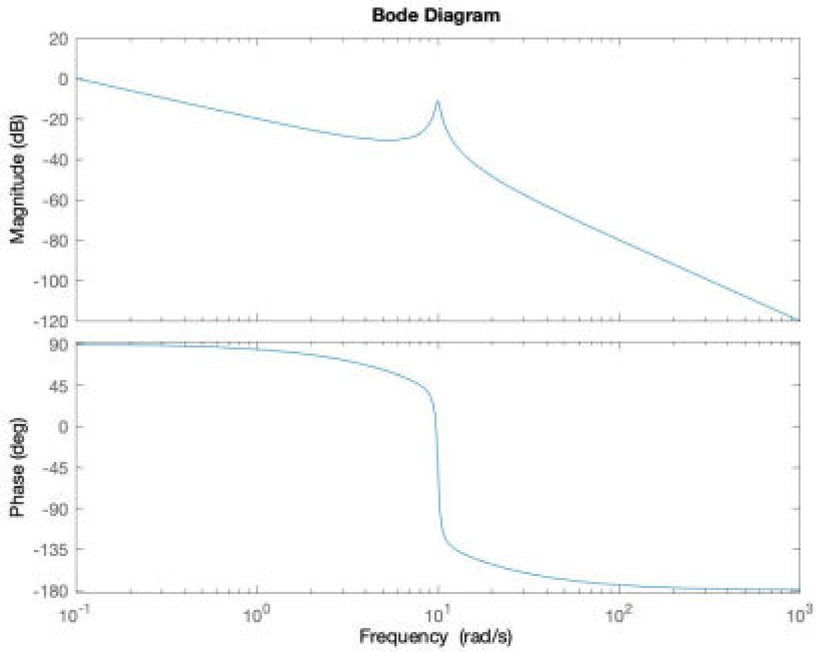
4. Q4. An open-loop transfer function has one real zero in the RHP and two conjugate complex poles (null real part):
Which is the corresponding Bode plot? (1 Point)



1. (double positive zeros z=+10)

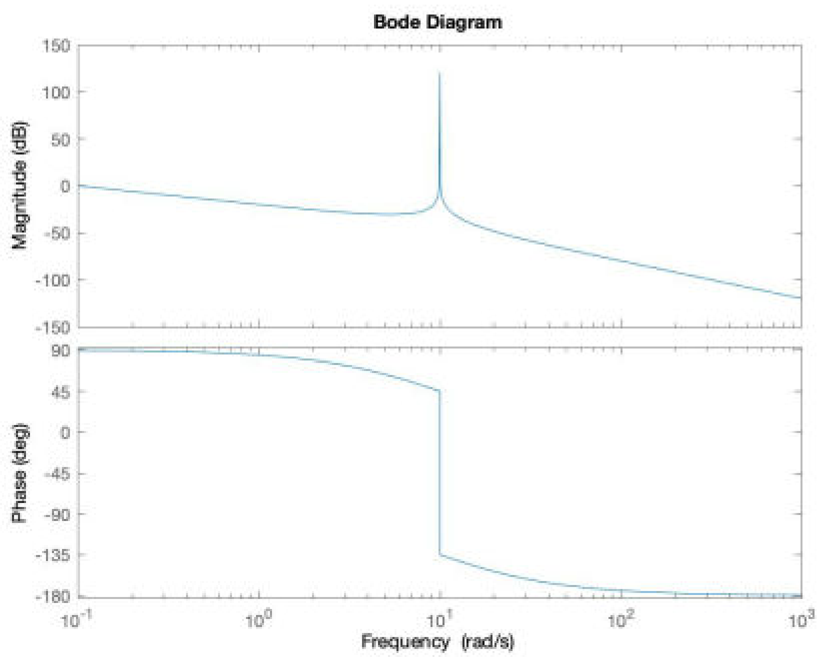


2. (two zeros z=+10, z= -50, two conjugate poles)



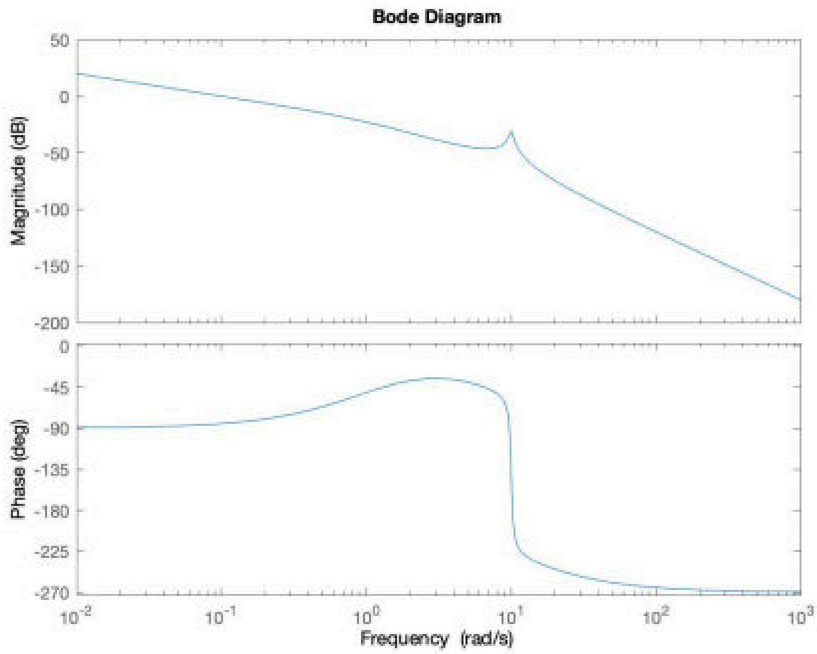
3.

(one real zero, pole in the origin and two complex poles)



4.

(CORRECT)



5. (pole in the origin and two complex poles, and one pole in RHP)

```
% Zero i RHP = +20DB/dc og -90Phase
% Pole på Im-aksen -180phase
```

5. Q5. The Bode plot diagrams of the magnitude and phase of a transfer function G(s) of a linear system is shown in Fig. 2.
Which of the following answers is correct? (1 Point)

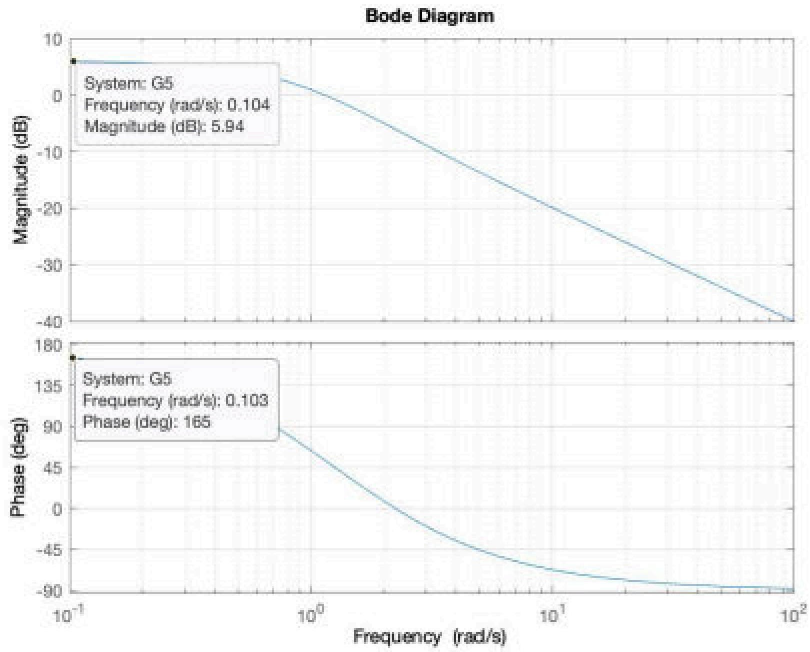


Figure 2: Bode Plot Diagrams - Q5

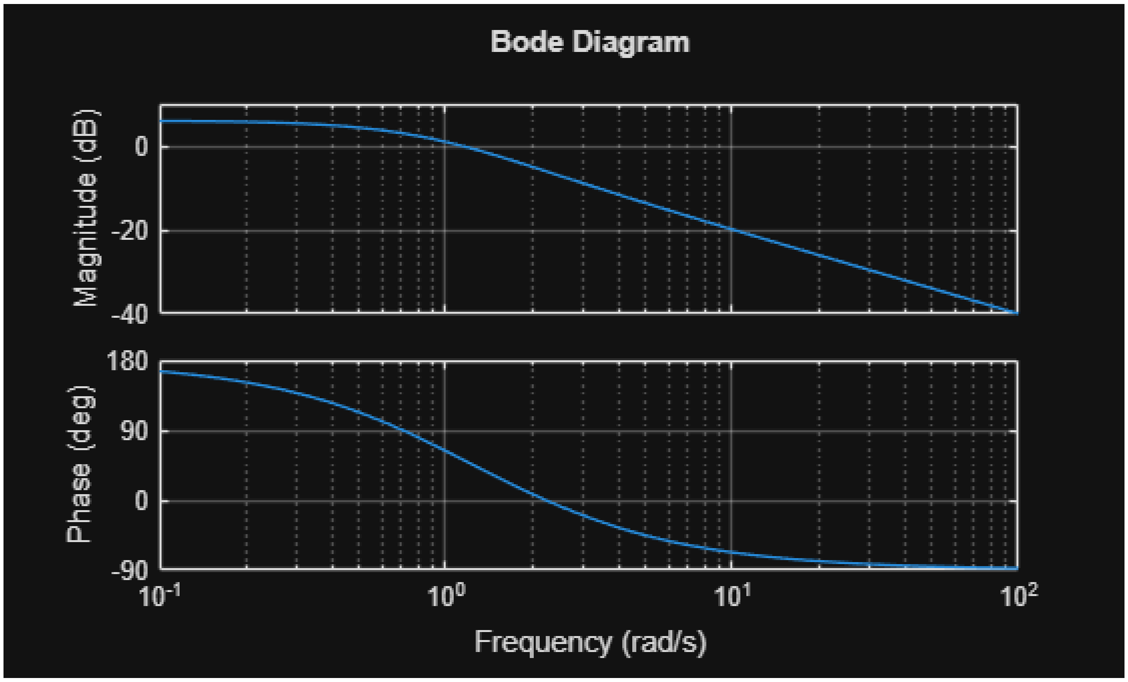
1. G(s) has negative and positive real poles and no zeros
   FALSE - the phase would go up in correspondence of the positive pole
2. G(s) has only 1 negative real pole and 1 negative real zero
   FALSE - Looking at the Bode diagram, there is more than one pole, magnitude decrease -40 db per decade
3. G(s) has only 1 positive real pole and no zeros
   FALSE - The magnitude would start in 0 dB and the phase would start in -180 upwards.
4. G(s) has complex poles and 1 negative real zero
   FALSE - there would be an oscillation and rapid phase shift

5. G(s) has positive real poles and 1 positive real zero
   CORRECT

Answer:
Q5. [**CORRECT n.5**] DC gain 5.9 dB equal to ca. 2 (magnitude horizontal and initial phase is 165 degrees. Magnitude is constant until $\omega = 1$ rad/sec and after it decreases -40 dB/decade, so two poles. Correspondingly, the phase decreases and it goes to -90 which means one zero. There are no peaks, so poles are reals. Dc gain 2, z = 2; p = +1, p=1.

$$G(s) = \frac{s-2}{(1+s)^2}$$

```
clear; clc; close all;

s = tf('s');

G = (s - 2)/(1 + s)^2;

figure;
bode(G);
grid on;
```




```
[z, p, k] = zpkdata(G, 'v');

disp('Zeros:')
```

```
Zeros:
```

```
disp(z)
```

```
     2
```

```
disp('Poles:')
```

```
Poles:
```

```
disp(p)
```

```
    -1
    -1
```

```
disp('Gain:')
```

Gain:

```
disp(k)
```

```
     1
```

```
db2mag(5.94)
```

```
ans =
1.9815
```

6. Q6. A closed-loop system has a loop transfer function

$$G(s) = \frac{K}{s(s+21)}$$

and the following Bode plot in Fig.3

What is the gain K so that the phase margin is PM=40°?
(1 Point)

1. K = 0.1 Wrong, this is -18.5 in mag
2. K = 8.4 Correct, this is 18.5 in mag
3. K = 77.5 Wrong, this is the PM
4. K = 18.5 Wrong, this is in mag. in db
5. K = 40 Wrong, this is the desired PM

Answer:
Q6. [**CORRECT n.2**]

```
clear; clc; close all;

s = tf('s');

% System uden K
G0 = 1/(s*(s+2.1));

% Ønsket phase margin
PM_desired = 40;

% Ved gain crossover skal fasen være:
% PM = 180 + phase
% 40 = 180 + phase
```

```
% phase = -140 deg
phase_gc = -180 + PM_desired;

% Fase for G0:
% phase = -90 - atan(w/2.1)
% -140 = -90 - atan(w/2.1)
% atan(w/2.1) = 50 deg
w_gc = 2.1*tand(50);

% Magnitude uden K ved denne frekvens
mag_G0 = abs(freqresp(G0, w_gc));

% K skal gøre magnituden lig 1 ved gain crossover
K = 1/mag_G0;

fprintf('Gain crossover frequency = %.4f rad/s\n', w_gc);
```

```
Gain crossover frequency = 2.5027 rad/s
```

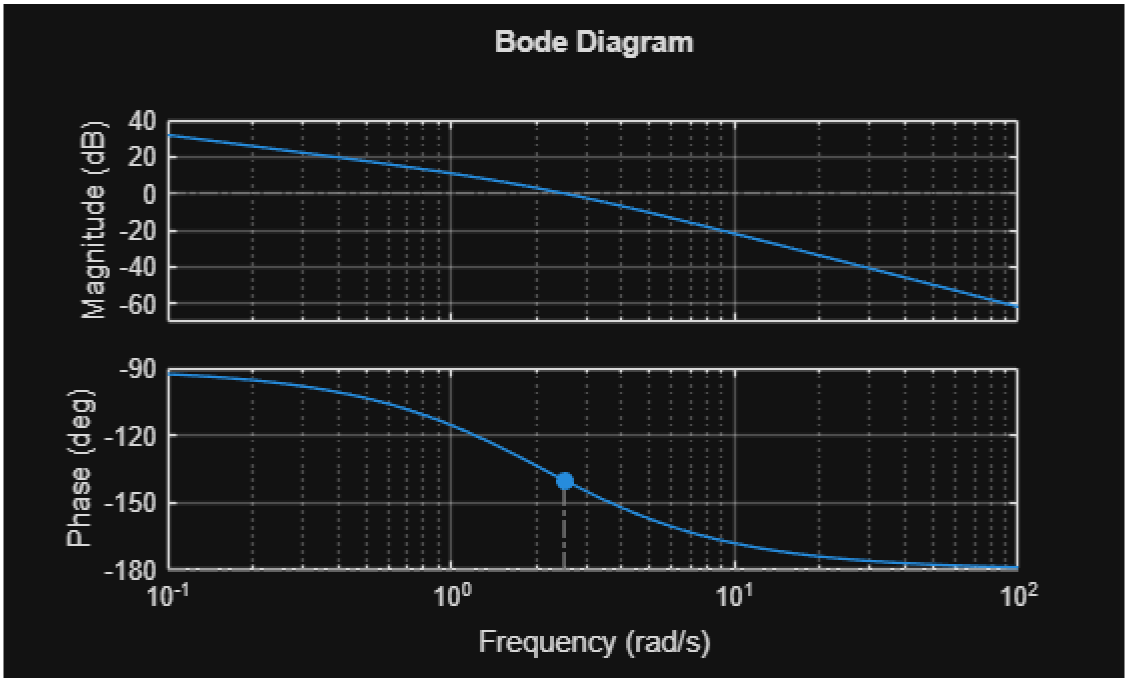
```
fprintf('K = %.4f\n', K);
```

```
K = 8.1763
```

```
fprintf('K in dB = %.2f dB\n', mag2db(K));
```

```
K in dB = 18.25 dB
```

```


% System med korrekt K
G = K*G0;

figure;
margin(G);
grid on;
```




```
[GM, PM, Wcg, Wcp] = margin(G);

fprintf('Phase margin = %.2f deg\n', PM);
```

```
Phase margin = 40.00 deg
```

```
fprintf('Gain crossover frequency from margin = %.4f rad/s\n', Wcp);
```

```
Gain crossover frequency from margin = 2.5027 rad/s
```

7. Q7. What is the DC gain in Db of the following transfer function?

$$G(s) = \frac{12}{(s+2)(s+3)}$$

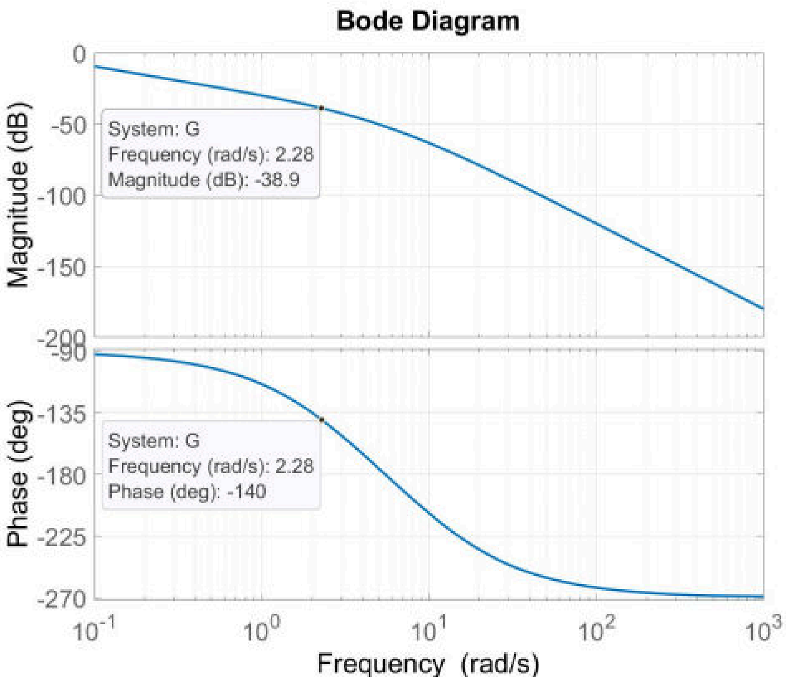


Figure 3: Bode Plot - Q6

1. DC = 2 Wrong, this is in mag
2. DC = 12 this is K
3. DC = 6 Correct
4. DC = 0 Wrong, this is the lim of G(s) multiplied s
5. DC = -15.6 Wrong, this is db of 1/6

Answer:
Q7. [**CORRECT n. 3**] DC Gain of a system is the gain at the steady state which is at t tending to infinity i.e., s tending to zero. Let s tend to zero, it will result in 12/(3*2) = 2, in db is 6

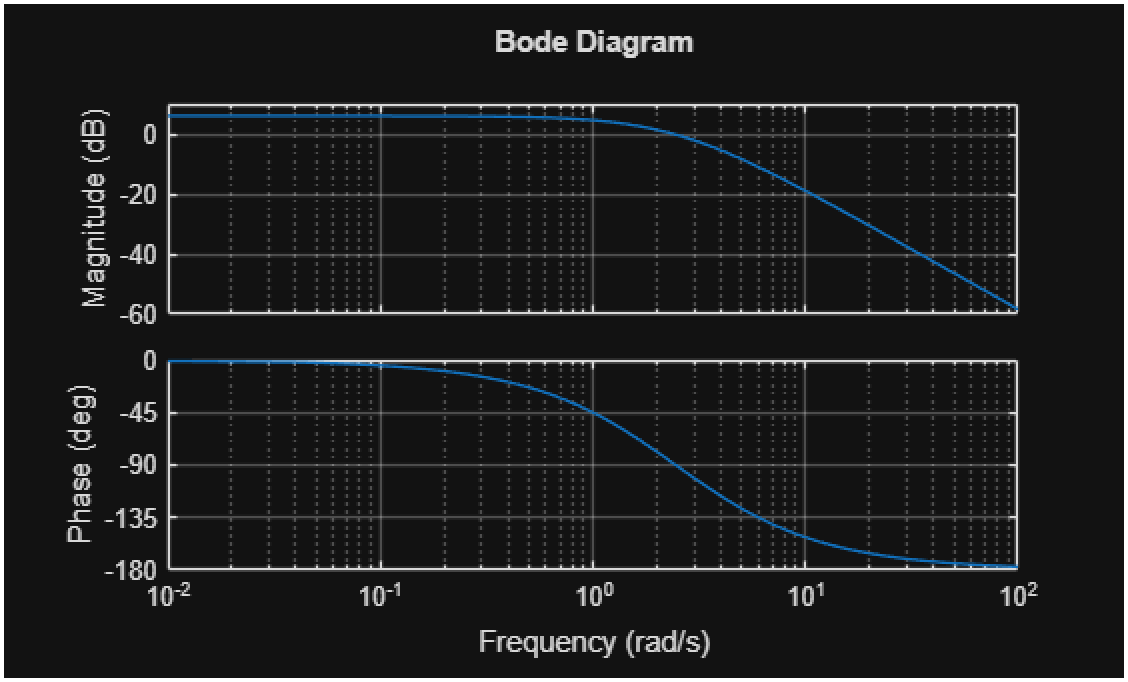
```
clear; clc; close all;

s = tf('s');

G = 12/((s+2)*(s+3));

DC_mag = dcgain(G);
DC_dB = mag2db(DC_mag);

fprintf('DC gain = %.2f\n', DC_mag);
```

```
DC gain = 2.00
```

```
fprintf('DC gain in dB = %.2f dB\n', DC_dB);
```

```
DC gain in dB = 6.02 dB
```

```
figure;
bode(G);
grid on;
```



8. Q8. Consider the differential equation

   $$5\ddot{y}(t) + \dot{y}(t) + 0.5y(t) = 3u(t)$$

   Where $y(0) = \dot{y}(0) = 0$ and u(t) is the input.
   Which are the poles of this system?

1. s1 = -0.1, s2= -0.1
2. s1 = -0.1+0.3j, s2= -0.1-0.3j CORRECT
3. s1 = 0.3j, s2= -0.3j
4. s1 = 0.3, s2= 0.1
5. s1 = 0.1+0.3j, s2= 0.1-0.3j

```
clear; clc;

syms s;

G = 3 / (5*s^2+s+0.5)
```

```
G =
```

$$\frac{3}{5\,s^2+s+\frac{1}{2}}$$

```
poles(G)
```

```
ans =
```

$$\begin{pmatrix} -\frac{1}{10}+\frac{3}{10}\,\mathrm{i} \\ -\frac{1}{10}-\frac{3}{10}\,\mathrm{i} \end{pmatrix}$$

9. Q9. Consider the following Linear Time Varying system:

$$\dot{x}_1 = -x_1 + x_2$$

$$\dot{x}_2 = 2x_1 - wx_2$$

For which values of the parameter w are the poles of the system in the left half-plane? (1 Point)

1. $w \leq 2$
   false
2. $w \geq 1$
   false
3. $w < 1$
   false
4. $w > 2$
   correct
5. $w \geq -2$
   false

Answer:
Q9. [**CORRECT n. 4**] Solve the system, Laplace transform, Transfer function, set the denominator = 0 and find the value of w that makes the root being negative. This happens for w greater than 2.

```
clear; clc;

syms s w

A = [-1  1;
    2 -w];

char_poly = det(s*eye(2) - A);
```

```
disp('Characteristic polynomial:')
```

```
Characteristic polynomial:
```

```
pretty(expand(char_poly))
```

$s + w + s\,w + s^2 - 2$

```
% Find betingelsen manuelt:
% s^2 + (w+1)s + (w-2)
% For LHP:
% w+1 > 0 og w-2 > 0

fprintf('Condition for LHP poles: w > 2\n');
```

```
Condition for LHP poles: w > 2
```

```
clear; clc;

for w = [1 2 3]
    A = [-1  1;
        2 -w];

    poles = eig(A);

    fprintf('w = %.1f, poles = %.4f, %.4f\n', w, poles(1), poles(2));
end
```

```
w = 1.0, poles = 0.4142, -2.4142
w = 2.0, poles = 0.0000, -3.0000
w = 3.0, poles = -0.2679, -3.7321
```

---

10. Q10. A system with the following characteristic equation is $s^2 + 2s + 2 = 0$ of the closed loop system is called (select the right answer): (1 Point)

    1. Over damped
       false
    2. Undamped
       false
    3. Underdamped
       Correct
    4. Critically damped
       false

Answer:
Q10. [**CORRECT n. 3**]
The given equation is: $s^2 + 2s + 2 = 0$ It is a second-order differential equation. The Laplace transform of a standard form of a second-order differential equation is:

$$s^2 + 2w_n\zeta + w_n^2$$

Comparing the values we get:

$$w_n = \sqrt{2}$$

$2w_n\zeta = 2$

$\zeta < 1$

Thus, the system is underdamped.

```
clear; clc; close all;

% Characteristic equation:
% s^2 + 2s + 2 = 0

coeffs = [1 2 2];

poles = roots(coeffs);

disp('Poles:')
```

```
Poles:
```

```
disp(poles)
```

```
  -1.0000 + 1.0000i
  -1.0000 - 1.0000i
```

```

% Standard 2nd order form:
% s^2 + 2*zeta*wn*s + wn^2 = 0

wn = sqrt(coeffs(3));                % wn^2 = 2
zeta = coeffs(2)/(2*wn);             % 2*zeta*wn = 2
```

```
fprintf('wn = %.4f\n', wn);
```

```
wn = 1.4142
```

```
fprintf('zeta = %.4f\n', zeta);
```

```
zeta = 0.7071
```

```
if zeta > 1
    disp('Systemet er overdamped')
elseif zeta == 1
    disp('Systemet er critically damped')
elseif zeta > 0 && zeta < 1
    disp('Systemet er underdamped')
elseif zeta == 0
    disp('Systemet er undamped')
else
    disp('Systemet er unstable')
end
```

```
Systemet er underdamped
```

Question 11

Consider a third order, open-loop stable system described by a transfer function $G(s)$ with Nyquist diagram as shown in the following figure. The Nyquist curve intersects the negative

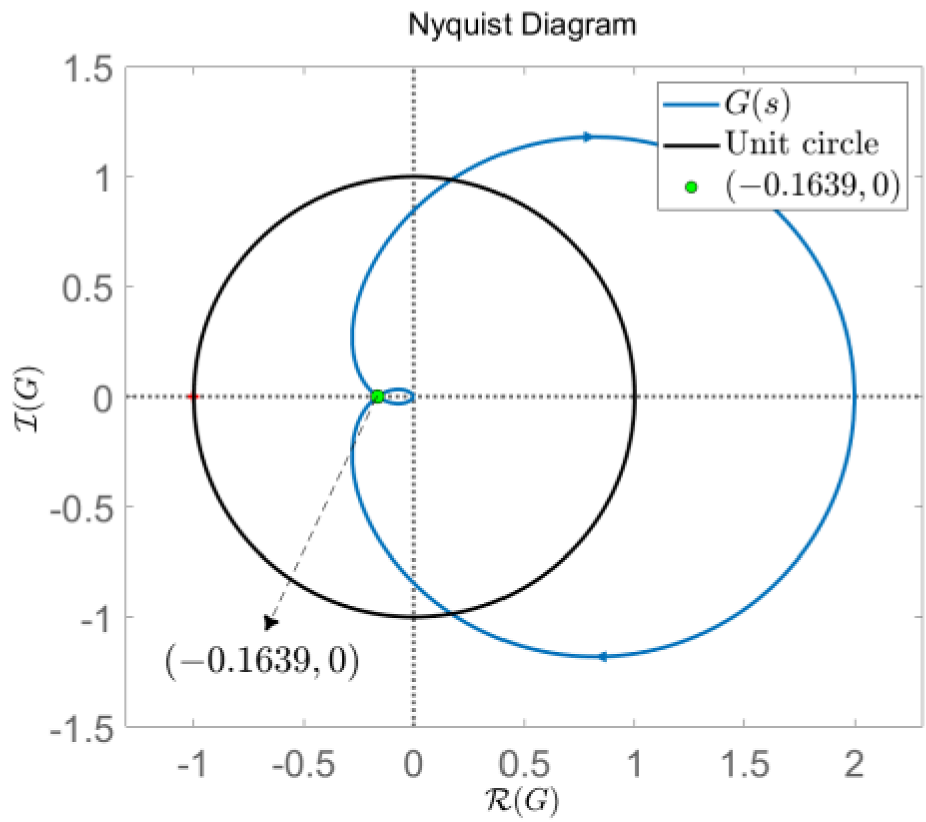


Figure 11: Nyquist diagram

horizontal semi-axis at the point $(-0.1639, 0)$ along the clockwise direction. The gain margin K_M in dB is approximately:

a) $K_M = -15.71$ ($1/K_M$ in dB)

b) $K_M = 15.71$ (Correct)

c) $K_M = -6.1$ ($-K_M$ not converted in dB)

d) $K_M = 6.1$ (Forgot to convert in dB)

e) $K_M = 0.1639$ ($1/K_M$)

Ja — her bruger man kun punktet, hvor Nyquist-kurven skærer den negative reelle akse:

$$(-0.1639, 0)$$

Det betyder, at ved fase -180° er størrelsen:

$$|G(j\omega)| = 0.1639$$

Gain margin er den faktor, man kan gange systemet med, før punktet rammer -1:

$$K_M \cdot 0.1639 = 1$$

Derfor:

$$K_M = \frac{1}{}$$

Solution: The gain margin K_M is the factor with which the magnitude of $G(j\omega)$ at $\omega = -\pi$ needs to be multiplied to reach 1. Hence, $K_M = \frac{1}{0.1639} \approx 6.1$ or 15.7067 dB. Therefore, the only acceptable answer is (b).

```
clear; clc;

% Nyquist-kurven skærer den negative reelle akse ved:
x = -0.1639;

% Magnituden ved phase crossover
mag = abs(x);

% Gain margin i normal faktor
Km = 1/mag;

% Gain margin i dB
Km_dB = 20*log10(Km);

fprintf('Magnitude at phase crossover = %.4f\n', mag);
```

```
Magnitude at phase crossover = 0.1639
```

```
fprintf('Gain margin Km = %.4f\n', Km);
```

```
Gain margin Km = 6.1013
```

```
fprintf('Gain margin Km in dB = %.4f dB\n', Km_dB);
```

```
Gain margin Km in dB = 15.7084 dB
```

Question 12

Consider a system described by a transfer function $G(s)$ with poles $\{4.5,\ -3 \pm j\}$ and open-loop Nyquist plot as shown in the following figure. The Nyquist plot intersects the real axis

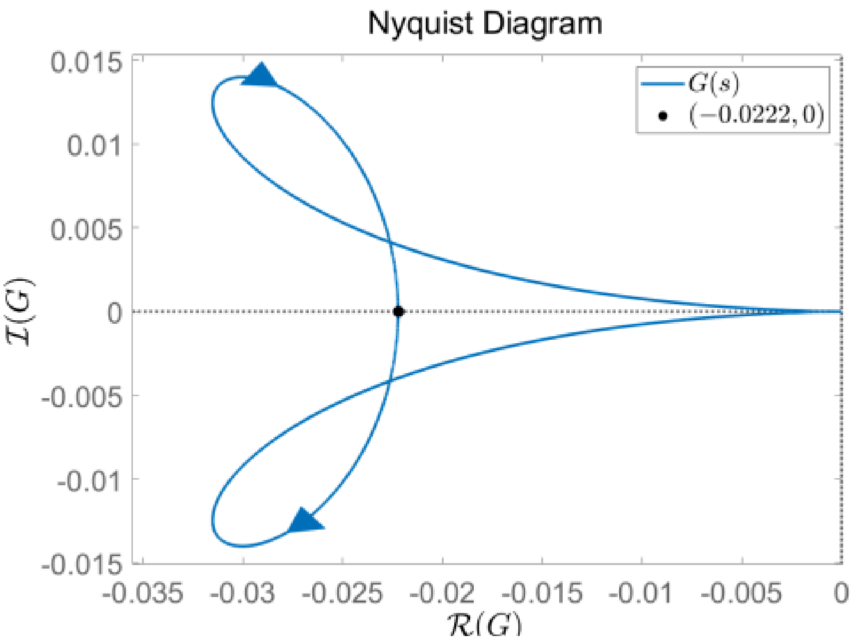


Figure 12: Nyquist plot

along the counter-clockwise direction at the point $(-0.0222, 0)$. A P-controller K_P is applied to the system and the loop is closed with unit feedback. The closed-loop system is stable for:

a) $0.0222 < K_P < 1$ (Too small gain)

b) $K_P = 50$ (Correct)

c) $-45 < K_P < 0$ (Too small negative gain)

d) $-0.0222 < K_P < 0$ (Negative gain)

e) $K_P = 0.0222$ (Too small gain)

Solution: The system is unstable with one pole in the RHP. We need to ensure that the Nyquist curve encircles the point (-1,0) once along the CCW direction. Already the direction is ok, so we only need to adjust the gain K_P. The intersection point of the curve with the negative real axis is at -0.0222, so multiplying G by $K_P = \frac{1}{0.0222} \approx 45.045$ ("gain margin" if we did not have unstable poles), we will make the curve cross exactly the critical point (-1,0). Hence a proportional gain $K_P > 45.045$, will ensure one encirclement of the critical point, which is needed for closed-loop stability. Negative gain will simply flip the curve about the imaginary axis. Therefore, the only acceptable option is (b).

Ja — her er løsningen til Q12 i MATLAB-stil.

Nyquist-kurven skærer den negative reelle akse ved:

$$(-0.0222, 0)$$

Når man ganger med en proportional gain K_P, bliver punktet skaleret til:

$$K_P \cdot (-0.0222)$$

For at ramme -1 skal:

$$K_P \cdot 0.0222 = 1$$

Derfor:

$$K_P = \frac{1}{0.0222}$$

$$K_P \approx 45.05$$

Så for at kurven kommer forbi -1, skal:

$$K_P > 45.05$$

Blandt svarmulighederne er det kun:

$$\boxed{K_P = 50}$$

```
clear; clc;

% Nyquist-kurven skærer realaksen ved:
x = -0.0222;

% Kritisk gain, hvor punktet rammer -1
Kcrit = 1/abs(x);

fprintf('Critical gain Kcrit = %.4f\n', Kcrit);
```

```
Critical gain Kcrit = 45.0450
```

```

% Test svarmuligheder
Kp_values = [0.0222, 1, 50, -45, -0.0222];

for Kp = Kp_values
    scaled_point = Kp*x;
    fprintf('Kp = %.4f gives real-axis crossing at %.4f\n', ...
        Kp, scaled_point);
end
```

```
Kp = 0.0222 gives real-axis crossing at -0.0005
Kp = 1.0000 gives real-axis crossing at -0.0222
Kp = 50.0000 gives real-axis crossing at -1.1100
```

```
Kp = -45.0000 gives real-axis crossing at 0.9990
Kp = -0.0222 gives real-axis crossing at 0.0005
```

Question 13

Consider an open-loop system $G_{ol}(s) = K_P C_{PI}(s) C_D(s) G(s)$, where K_P is the proportional gain, $G(s)$ is the transfer function of the system to be controlled, C_{PI} is the PI controller part and the Lead part $C_D(s)$ is defined as

$$C_D(s) = \frac{0.355s + 1}{\alpha 0.355s + 1} .$$

If the magnitude curve in the Bode plot of $G_{ol}(s)$ is strictly decreasing and $|G_{ol}(10j)| = 1$, what is (approximately) the magnitude contribution M_D in dB of the Lead part to the open-loop transfer function at the crossover frequency?

a) $M_D = 5.5$ dB (Forgot to square when calculating α)

b) $M_D = 3.55$ dB (Forgot to convert to dB)

c) $M_D = 11$ dB (Correct)

d) $M_D = 1.08$ dB (Calculated the phase contribution instead)

e) $M_D = 0.7$ dB (Calculated the phase contribution instead in dB)

Solution: Since the magnitude curve in the Bode plot of $G_{ol}(s)$ is strictly decreasing and $|G_{ol}(10j)| = 1$, $\omega_c = 10$ rad/s is the only crossover frequency of the open-loop system. The time constant of the Lead part is given by its transfer function as $\tau_d = 0.355$ s. Hence, the parameter α can be found from

$$\tau_d = \frac{1}{\omega_c \sqrt{\alpha}} \Leftrightarrow \alpha = \left(\frac{1}{\omega_c \tau_d}\right)^2 = \left(\frac{1}{10 \cdot 0.355}\right)^2 = 0.0793 .$$

The magnitude contribution of the Lead transfer function at the crossover frequency is then

$$M_D = |C_D(10j)| = \frac{1}{\sqrt{\alpha}} = 3.55 \text{ or } 11.0046 \text{ dB}.$$

Therefore, the only acceptable option is (c).

- The transfer function of the D-part of the PID (PI-Lead) controller is given by

$$C_D(s) = \underbrace{1 + \tau_d s}_{\text{“PD”}} \approx \underbrace{\frac{\tau_d s + 1}{\alpha \tau_d s + 1}}_{\text{“P-Lead”}}, \ \alpha < 1$$

- A Lead controller has a zero at $\omega_d = \dfrac{1}{\tau_d}$ and a pole at $\omega_a = \dfrac{1}{\alpha \tau_d}$ (higher than the zero).

```
clear; clc; close all;

s = tf('s');

T = 0.355;
wc = 10;
```

```
alpha = (1/(wc*T))^2;

CD = (T*s + 1)/(alpha*T*s + 1);

Md_mag = abs(freqresp(CD, wc));
Md_dB = mag2db(Md_mag);

fprintf('alpha = %.4f\n', alpha);
```

```
alpha = 0.0793
```

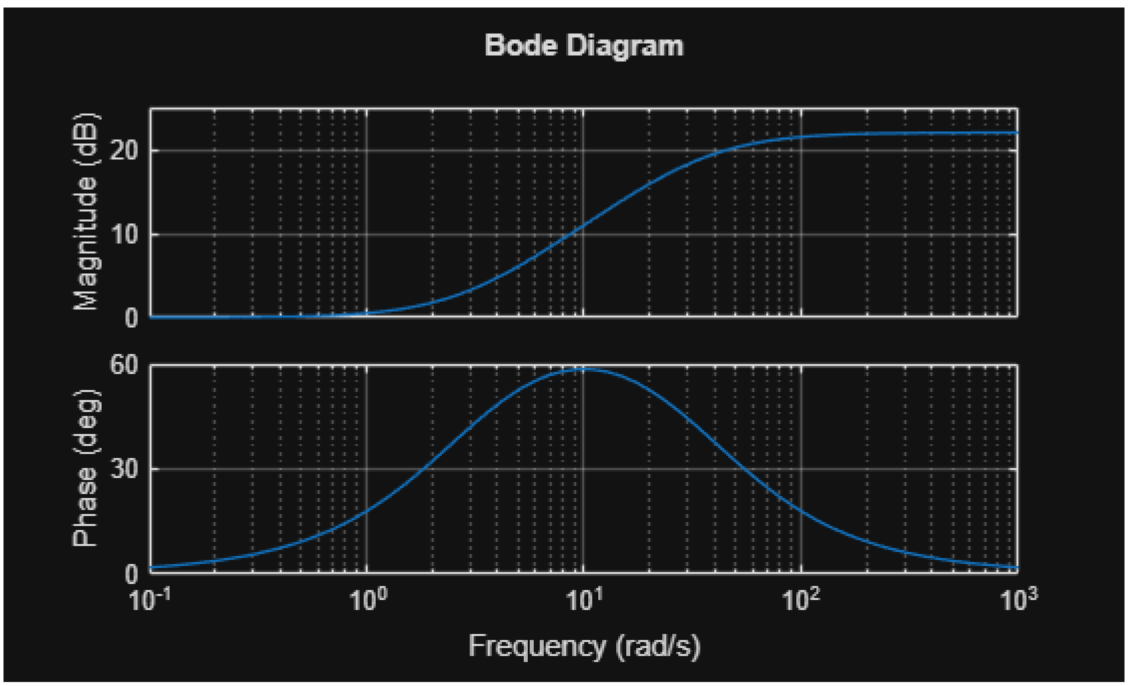
```
fprintf('Magnitude contribution = %.4f\n', Md_mag);
```

```
Magnitude contribution = 3.5500
```

```
fprintf('Magnitude contribution in dB = %.2f dB\n', Md_dB);
```

```
Magnitude contribution in dB = 11.00 dB
```

```
figure;
bode(CD);
grid on;
```



Question 14

The Bode plot of a third-order stable closed-loop system is shown below: Which one of the

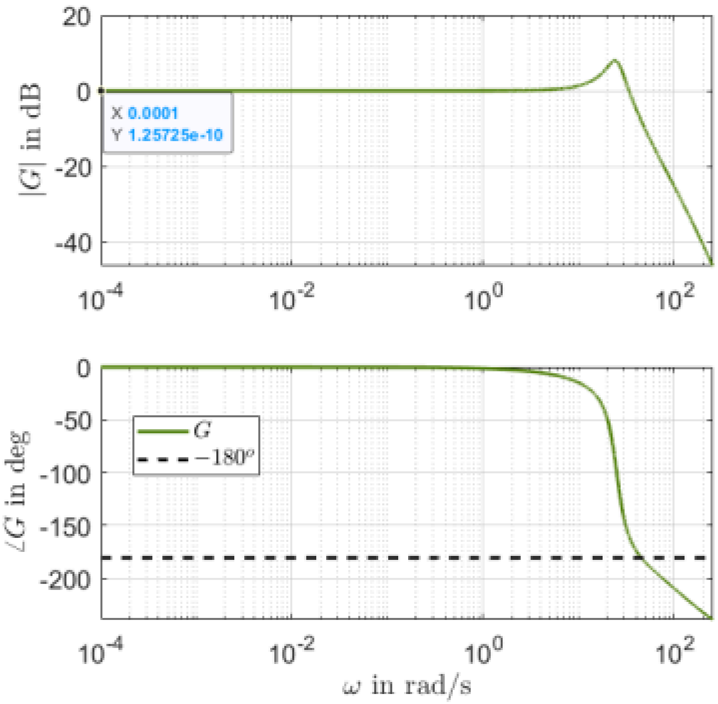


Figure 13: Bode plot

following figures corresponds to the error response to a unit step change in the reference signal?

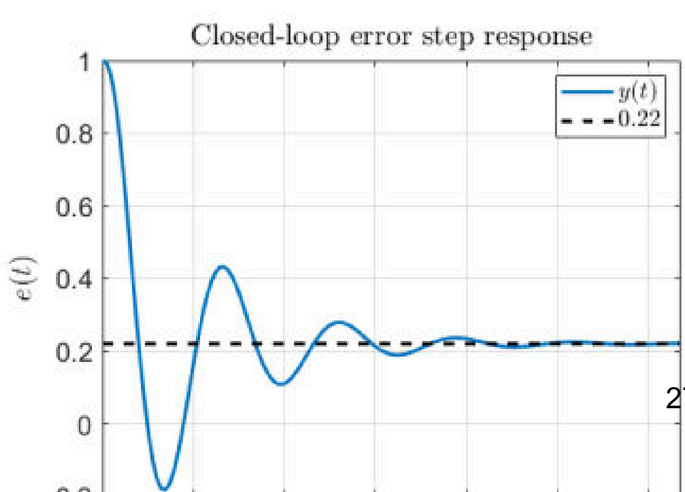
a) (Non zero steady state error)

b) (Non zero steady state error and no oscillations)

c) (Closed-loop output response, not error response)

d) (No oscillations)

e) (Correct. We expect zero steady state error since at low frequencies the closed-loop Bode plot shows magnitude of zero dB (output/reference = 1 so the error is zero). We also see a resonant peak which implies a pair of complex-conjugates poles. This will show up in the output/error step response as oscillations.)



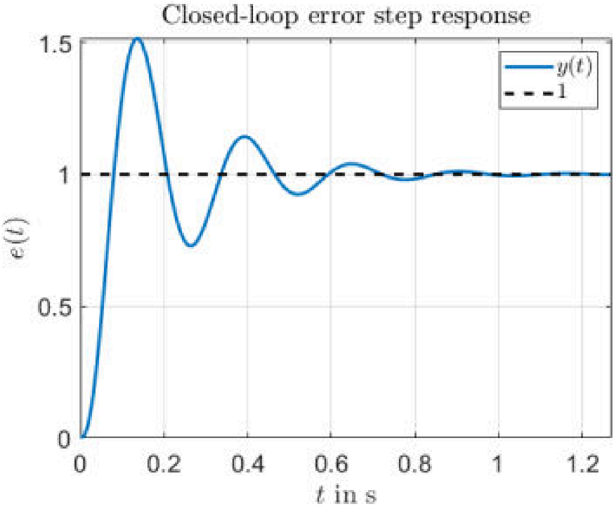


Figure 16: (c) Error step response

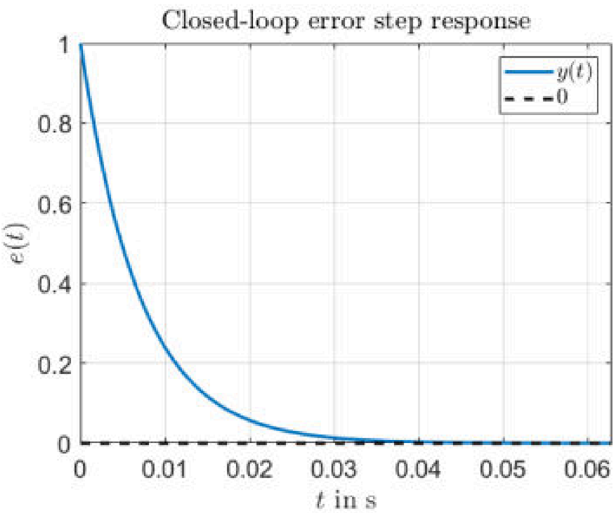
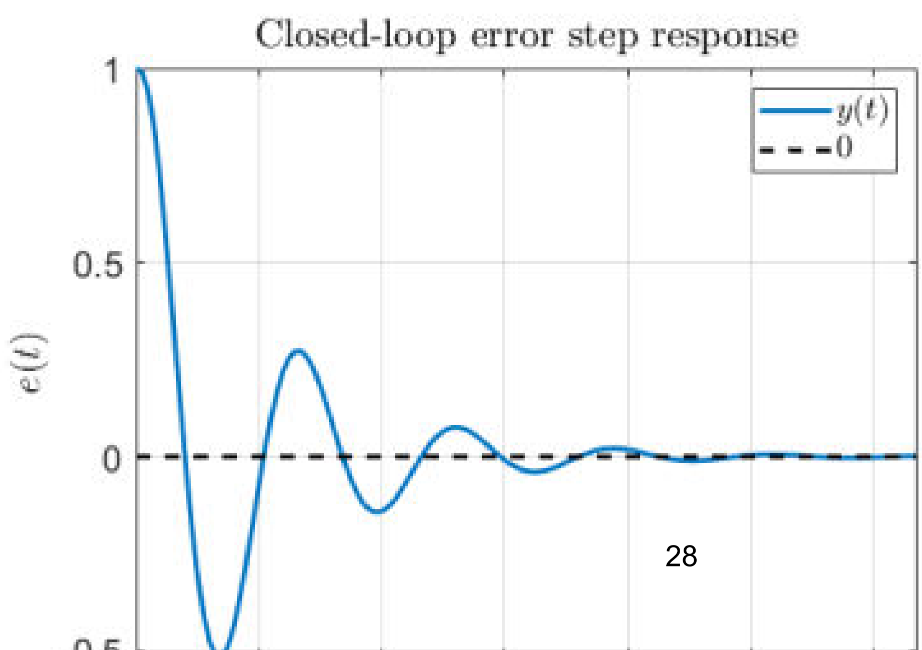


Figure 17: (d) Error step response



Question 15

Consider a fourth-order type-0 system with real poles and a real zero. All poles and the zero are located on the left half complex plane, with the zero located at a lower frequency compared to any of the poles. A P-controller with $K_P > 0$ is applied to the system. Which of the following statements is the correct one?

a) The unit-feedback closed-loop system is stable for any $K_P > 0$. (No, since we have three more poles than zeros, the phase can cross the -180^o, which means there is a finite gain margin for the open-loop system. Any K_P larger than that will lead to an unstable closed-loop system.)

b) Eliminating the steady-state error due to a step change in the reference requires only choosing an appropriate value for K_P. (Type-0 system cannot have zero steady state error if the loop is closed with a finite proportional gain and we cannot infinitely increase K_P. We therefore need a PI term.)

c) There is no positive value of K_P that can give a negative phase margin for the open-loop system. (Since we have three more poles than zeros, the net phase shift is -270^o at infinite frequency. Therefore it is not guaranteed that the open-loop system will have a positive phase margin for any $K_P > 0$.)

d) For some values of K_P it is possible to have two crossover frequencies. (Correct. Since the zero is a lower frequency than the poles, we have an increase in the magnitude

followed by a decrease. This means that there can be a selection of K_P for which the 0 dB line intersects the magnitude curve twice.)

e) Stability of the closed-loop system is ensured if K_P makes the Nyquist plot of the open-loop system encircle the point $(-1, 0)$ once along the counter-clockwise direction. (This would be true if there was an unstable pole in the original system.)

Hvorfor de andre er forkerte:

a) Forkert, fordi høj K_P kan gøre closed-loop ustabilt. Der er flere poler end nulpunkter, så fasen kan komme under $-180°$.

b) Forkert, fordi systemet er type-0. Et type-0 system med endelig proportional gain har normalt ikke nul steady-state error for et step.

c) Forkert, fordi fasen kan gå helt mod cirka:

$$-270°$$

da der er tre flere poler end nulpunkter. Derfor kan phase margin godt blive negativ.

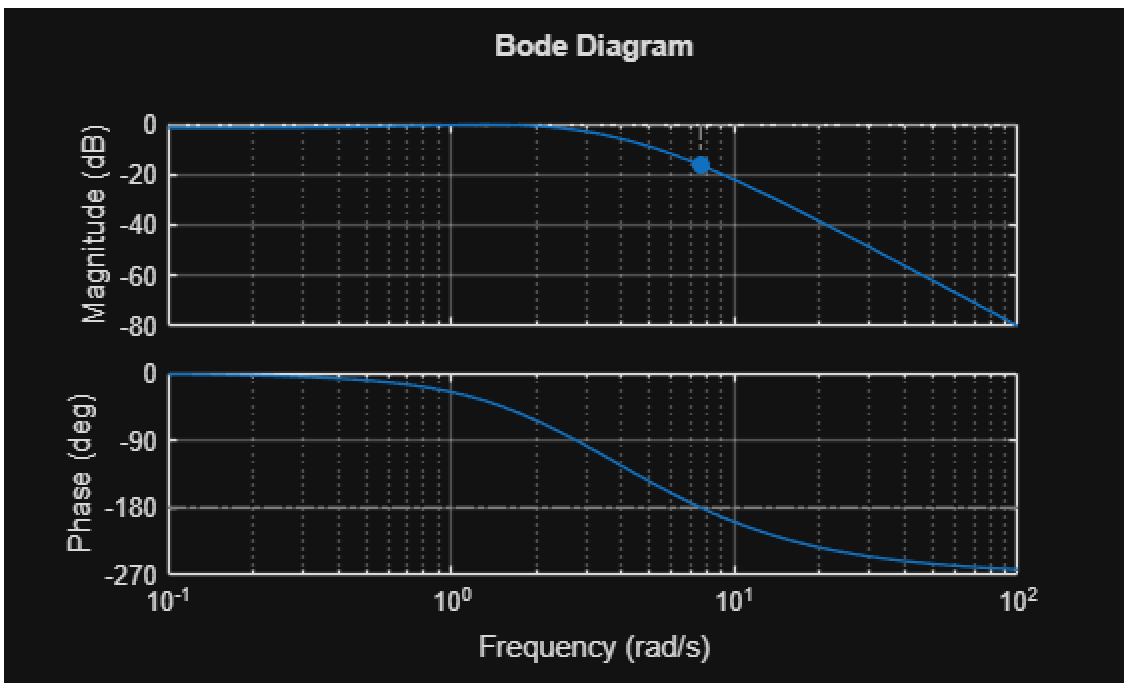
e) Forkert, fordi open-loop systemet er stabilt. For et open-loop stabilt system skal Nyquist-kurven ikke

Et eksempel kunne være:

$$G(s) = \frac{s+1}{(s+2)(s+3)(s+4)(s+5)}$$

hvor nulpunktet $s = -1$ ligger før alle polerne.

```
clear; clc; close all;

s = tf('s');

G = (s+1)/((s+2)*(s+3)*(s+4)*(s+5));

Kp = 100;

L = Kp*G;

figure;
margin(L);
grid on;
```



Question 16

Consider a system described by a transfer function $G(s)$ with a magnitude Bode plot as shown in the figure below: A P-controller with gain K_P is applied to the system and the loop

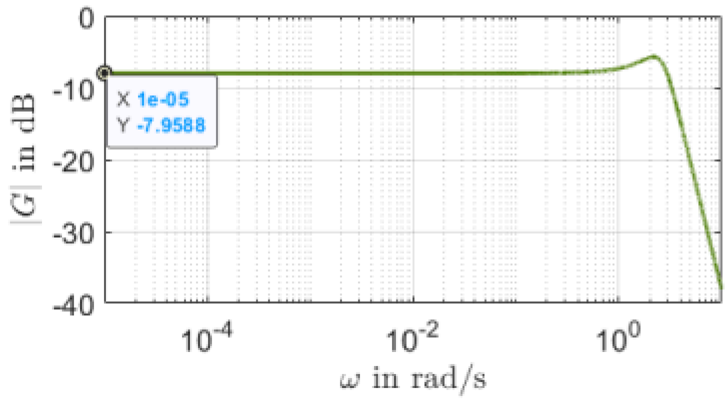


Figure 19: Bode plot.

is closed with unit feedback as in the following figure. If the steady state error $e_{SS} = \lim_{s\to 0} e(s)$

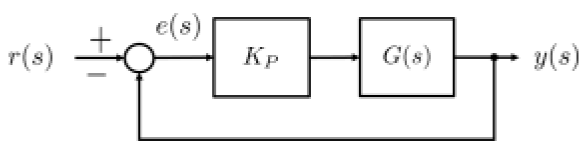


Figure 20: Closed-loop system.

due to a unit step change in the reference signal is $e_{SS} = 0.555$, the proportional gain K_P is approximately:

a) $K_P = 1$ (Unit gain)

b) $K_P = 2$ (Correct)

c) $K_P = 2.5$ (1/DC gain of $G(s)$)

d) $K_P = 0.4$ (DC gain of $G(s)$)

e) $K_P = 7.9588$ (1/DC gain of G in dB)

Solution: The transfer function from the reference signal to the error $e(s)$ is given by

$$G_{er}(s) = \frac{1}{1 + K_P G(s)}\ .$$

The steady-state error is found by applying the final value theorem as

$$e_{ss} = \lim_{s\to 0} e(s) = \lim_{s\to 0} sG_{er}(s)\frac{1}{s} = \frac{1}{1 + K_P G(0)} = 0.555 \Rightarrow K_P = \frac{1}{G(0)}\left(\frac{1}{0.555} - 1\right)\ .$$

From the given Bode plot the DC gain of $G(0)$ can be approximately found to be equal to -7.9588 dB or $G(0) \approx 0.4$. Therefore, $K_P = 2$ and the only acceptable option is (b).

```matlab
clear; clc;

% Aflæst DC gain fra Bode-plottet
G0_dB = -7.9588;

% Konverter fra dB til almindelig gain
G0 = db2mag(G0_dB);

% Opgivet steady-state error
ess = 0.555;

% Formel: ess = 1 / (1 + Kp*G0)
Kp = (1/ess - 1)/G0;

fprintf('G(0) = %.4f\n', G0);
```

```
G(0) = 0.4000
```

```matlab
fprintf('Kp = %.4f\n', Kp);
```

```
Kp = 2.0045
```

Question 17

Consider the system described by a transfer function G with a phase Bode plot as shown in the following figure. The objective is to design a PI-Lead controller $C(s) = K_P \frac{\tau_d s + 1}{\alpha \tau_d s + 1} \frac{\tau_i s + 1}{\tau_i s}$

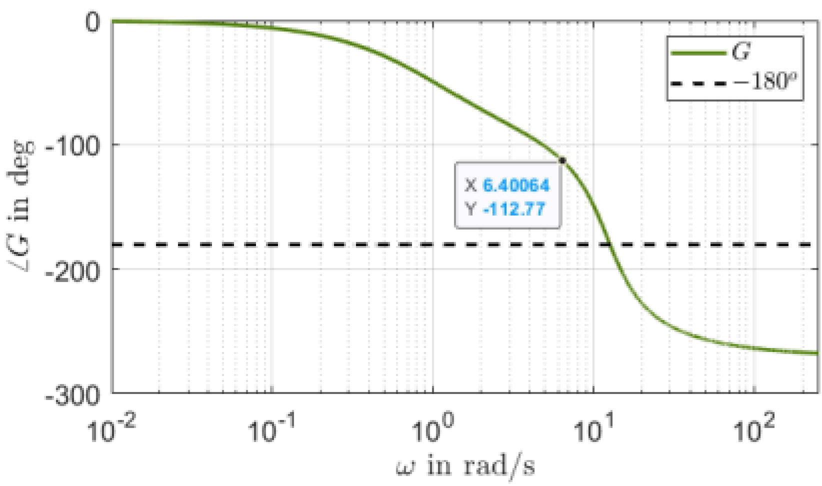


Figure 21: Bode plot

such that the open-loop system has crossover frequency $\omega_c = 6.4$ rad/s and a phase margin $\gamma_M = 75^o$ is achieved. If the zero of the PI part is placed at a frequency 5 times smaller than the crossover frequency, the α parameter of the Lead part is approximately:

a) $\alpha = 0.032$ (Did not convert to degrees when calculating ϕ_m)

b) $\alpha = 2$ ($0 < \alpha < 1$)

c) $\alpha = -1.5$ (α cannot be negative)

d) $\alpha = 1.13$ (Wrong sign when calculating ϕ_i)

e) $\alpha = 0.5$ (Correct)

Solution: Let ϕ_G, ϕ_m, ϕ_i be the phase contributions of G, Lead and PI parts of the open-loop system at the crossover frequency, respectively. The phase of G is given from the Bode plot as $\phi_G = -112.77^o$. Additionally, the zero of the PI is placed at a frequency five times smaller than the crossover frequency, i.e. $N_i = 5$. This means that its phase contribution at the crossover frequency will be $\phi_i = -\arctan\left(\frac{1}{N_i}\right) = -\arctan\left(\frac{1}{5}\right)$ rad or $\phi_i = -11.3099^o$. At $\omega_c = 6.4$ rad/s the following holds:

$$-180^o + \gamma_M = \phi_G + \phi_m + \phi_i \Leftrightarrow \phi_m = -180^o + \gamma_M - \phi_G - \phi_i \Leftrightarrow \phi_m = 19.0797^o \,.$$

converting ϕ_m to rad gives

$$0.333 = \arcsin\left(\frac{1-\alpha}{1+\alpha}\right) \Leftrightarrow \alpha = \frac{1-\sin(0.333)}{1+\sin(0.333)} \approxeq 0.5073 \,.$$

Therefore, the only acceptable option is (e).

```
clear; clc;

wc = 6.4;                    % crossover frequency
phase_G = -112.77;           % phase of G at wc
PM_desired = 75;             % desired phase margin

% Desired total open-loop phase at crossover
phase_desired = -180 + PM_desired;

% PI zero is 5 times smaller than crossover
% Therefore wc*tau_i = 5
phase_PI = atan(5)*180/pi - 90;

% Required lead phase
phi_D = phase_desired - phase_G - phase_PI;

% Alpha from lead phase formula
alpha = (1 - sind(phi_D))/(1 + sind(phi_D));

fprintf('Desired phase at wc = %.2f deg\n', phase_desired);
```

```
Desired phase at wc = -105.00 deg
```

```
fprintf('PI phase contribution = %.2f deg\n', phase_PI);
```

```
PI phase contribution = -11.31 deg
```

```
fprintf('Required lead phase = %.2f deg\n', phi_D);
```

```
Required lead phase = 19.08 deg
```

```
fprintf('alpha = %.4f\n', alpha);
```

```
alpha = 0.5073
```

Question 18

Consider the system described by the following block diagram where $n > 2$ is a positive integer

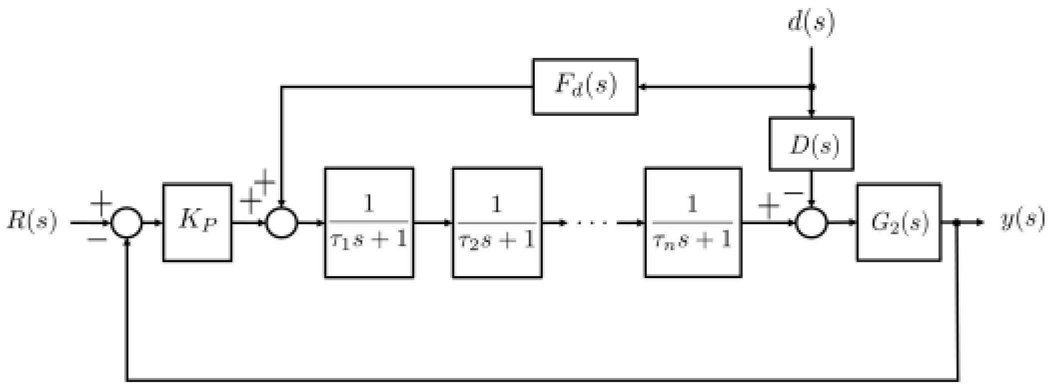


Figure 22: Block diagram.

number and the time constants $\tau_k > 0,\ k = 1, \ldots, n$ are known. The objective is to attenuate the effect of the disturbance $d(s)$ on the system output $y(s)$ by designing a disturbance feed-forward controller $F_d(s)$. If the known transfer function $D(s)$ is defined as

$$D(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \text{ with } \omega_n > 0 \text{ and } 0 < \zeta < 1 \,,$$

which one of the following designs for $F_d(s)$ is the most appropriate?

Hint: *The product operator* $\prod_{k=1}^{n}$ *is defined as* $\prod_{k=1}^{n} A_k = A_1 \cdot A_2 \cdot A_3 \cdot \ldots \cdot A_n$.

a) $F_d(s) = \dfrac{\omega_n^2 \prod_{k=1}^{n} (\tau_k s + 1)}{s^2 + 2\zeta\omega_n s + 1}$ (Did not filter so $F_d(s)$ is improper)

b) $F_d(s) = \dfrac{(s^2 + 2\zeta\omega_n s + 1) \prod_{k=1}^{n} (\tau_k s + 1)}{\omega_n^2 (\tau_f s + 1)^{n+2}}$ with $\tau_f \leq \dfrac{\min(\tau_1, \ldots, \tau_n)}{5}$ (Used the reciprocal of $D(s)$)

c) $F_d(s) = \dfrac{\prod_{k=1}^{n} (\tau_k s + 1)}{(\tau_f s + 1)^n}$ with $\tau_f \leq \dfrac{\min(\tau_1, \ldots, \tau_n)}{5}$ (Did not include $D(s)$)

d) $F_d(s) = \dfrac{\omega_n^2 \prod_{k=1}^{n} (\tau_k s + 1)}{(s^2 + 2\zeta\omega_n s + 1)(\tau_f s + 1)^{n-2}}$ with $\tau_f \leq \dfrac{\min(\tau_1, \ldots, \tau_n)}{5}$ (Correct)

e) $F_d(s) = \dfrac{\omega_n^2 \prod_{k=1}^{n} (\tau_k s + 1)}{(s^2 + 2\zeta\omega_n s + 1)(\tau_f s + 1)^n}$ with $\tau_f \geq \dfrac{\max(\tau_1, \ldots, \tau_n)}{5}$ (Two more unnecessary poles and the filter time constants are slower than the poles of the plant)

Solution: Define $G_1(s) \triangleq \overbrace{\frac{1}{\tau_1 s + 1} \cdot \frac{1}{\tau_2 s + 1} \cdot \ldots \cdot \frac{1}{\tau_n s + 1}}^{n \text{ first-order systems}} = \prod_{k=1}^{n} \frac{1}{\tau_k s + 1}$ to be the transfer function between the two summation points. Then a feed-forward term that would cancel the effect of the disturbance on the system output is given by

$$F_d(s) = \frac{D(s)}{G_1(s)} = \frac{\omega_n^2 \prod_{k=1}^{n} (\tau_k s + 1)}{s^2 + 2\zeta\omega_n s + 1} \,.$$

Of course this transfer function is not proper so we need to filter it with at least a $(n-2)$-order low-pass filter to account for the $n-2$ more zeros we have. This rules out options (a), (b) and (c). Option (e) provides a proper solution but the time constants of the filter are larger (slower) than the largest time constant of $G_1(s)$. This may filter out essential dynamics of the plant and it will give a feed-forward controller with much smaller bandwidth compared to the remaining solution. Therefore, the only acceptable option is (d).

Idéen er, at disturbance-feedforward $F_d(s)$ skal modvirke disturbance-vejen $D(s)$.

Lad plant-delen mellem feedforward-summeren og disturbance-summeren være:

$$P(s) = \prod_{k=1}^{n} \frac{1}{\tau_k s + 1}$$

Altså:

$$P(s) = \frac{1}{\prod_{k=1}^{n}(\tau_k s + 1)}$$

Hvis disturbance-effekten skal fjernes ideelt, skal bidraget fra feedforward-vejen matche disturbance-bidraget:

$$P(s)F_d(s) \approx D(s)$$

Derfor ville den ideelle feedforward-controller være:

$$F_d(s) = \frac{D(s)}{P(s)}$$

Indsæt $P(s)$:

$$F_d(s) = D(s) \prod_{k=1}^{n}(\tau_k s + 1)$$

og da

$$D(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

får vi:

$$F_d(s) = \frac{\omega_n^2 \prod_{k=1}^{n}(\tau_k s + 1)}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Men her er der et problem: tælleren har orden n, og nævneren har kun orden 2. Da $n > 2$, bliver controlleren **improper**, altså ikke fysisk realiserbar.

Derfor tilføjer man et filter i nævneren med orden $n - 2$:

$$(\tau_f s + 1)^{n-2}$$

Så bliver controlleren proper:

$$\boxed{F_d(s) = \frac{\omega_n^2 \prod_{k=1}^{n}(\tau_k s + 1)}{(s^2 + 2\zeta\omega_n s + \omega_n^2)(\tau_f s + 1)^{n-2}}}$$

med

$$\boxed{\tau_f \leq \frac{\min(\tau_1, \ldots, \tau_n)}{5}}$$

Det svarer præcis til svar **d)**.

Hvorfor τ_f skal være lille:

$$\tau_f \leq \frac{\min(\tau_1, \ldots, \tau_n)}{5}$$

betyder, at filterpolerne er hurtigere end plantens poler. Så filteret gør controlleren realiserbar uden at

Question 19

A system is described by the following transfer function

$$G(s) = \frac{900}{(0.25s+1)(s^2+50s+3000)}$$

A PI-Lead controller

$$C(s) = K_P \underbrace{\frac{\tau_i s+1}{\tau_i s}}_{C_{PI}(s)} \underbrace{\frac{\tau_d s+1}{\alpha\tau_d s+1}}_{C_D(s)}$$

is to be designed so that the open-loop system $C(s)G(s)$ has phase margin $\gamma_M = 75^o$. If ω_c is the open-loop crossover frequency, $N_i = \omega_c\tau_i = 3$ and $\alpha = 0.001$, the controller proportional gain K_P is approximately:

a) $K_P = 3.4154$. (Correct)

b) $K_P = 10.67$. (K_P in dB)

c) $K_P = 36$. ($1/|G(\omega_c j)|$)

d) $K_P = 0.293$. (Reciprocal of K_P)

e) $K_P = -10.67$. (Reciprocal of K_P in dB)

Solution: Let ϕ_G, ϕ_d, ϕ_i be the phase contributions of G, Lead and PI parts of the open-loop system at the new crossover frequency, respectively. At ω_c the following holds:

$$\phi_i = \arctan\left(-\frac{1}{N_i}\right) = -0.3218 \text{ rad}$$

$$\phi_m = \arcsin\left(\frac{1-\alpha}{1+\alpha}\right) = 1.3715 \text{ rad}$$

or in degrees $\phi_i = -18.4349^o$ and $\phi_m = 78.5788^o$. Moreover,

$$-180^o + \gamma_M = \phi_G + \phi_d + \phi_i \Leftrightarrow \phi_G = -180^o + \gamma_M - \phi_G - \phi_i \Leftrightarrow \phi_G = -165.1439^o\ .$$

Based on the phase Bode plot of G drown in Matlab, the new crossover frequency ω_c can be easily calculated as $\omega_c = 50.4224$ rad/s. From there we can calculate the time constants of the PI and Lead parts as $\tau_i = \dfrac{N_i}{\omega_c} = 0.0595$ s and $\tau_d = \dfrac{1}{\omega_c\sqrt{\alpha}} = 0.1983$ s, as well as the individual transfer functions (that of the PI and that of the Lead). With the transfer functions of the controller (except for the K_P) and the system known, we can calculate the proportional gain K_P such that the open-loop transfer function has zero dB gain at ω_c:

$$K_P = \frac{1}{|G(j\omega_c)C_{PI}(j\omega_c)C_D(j\omega_c)|} = 3.4154\ .$$

Therefore, the only acceptable option is (a).

PI-Lead design – Proportional-Integral-“Differential”

Design a PI-Lead controller for a given system *G(s)*: **Cookbook recipe**

- **Step 1**: Select γ_M, N_i, α baring in mind the properties of each parameter (see table on next slide)
- **Step 2**: Calculate φ_i and φ_m from (remember to convert to degrees)

$$\varphi_i = -\arctan\left(\frac{1}{N_i}\right),\quad \varphi_m = \arcsin\left(\frac{1-\alpha}{1+\alpha}\right)$$

- **Step 3**: Calculate the phase of *G(s)* at the new crossover frequency from the phase balance equation

```
%Definere kendte variabler
G = 900/((0.25*s+1)*(s^2+50*s+3000))
```

```
G =

                 900
  ----------------------------------
  0.25 s^3 + 13.5 s^2 + 800 s + 3000

Continuous-time transfer function.
Model Properties
```

```
pm = 75
```

```
pm =
75
```

```
a = 0.01
```

```
a =
0.0100
```

```
Ni = 3
```

```
Ni =
3
```

```
%udregner phaser:
phi_i=rad2deg(atan(-1/Ni))
```

```
phi_i =
-18.4349
```

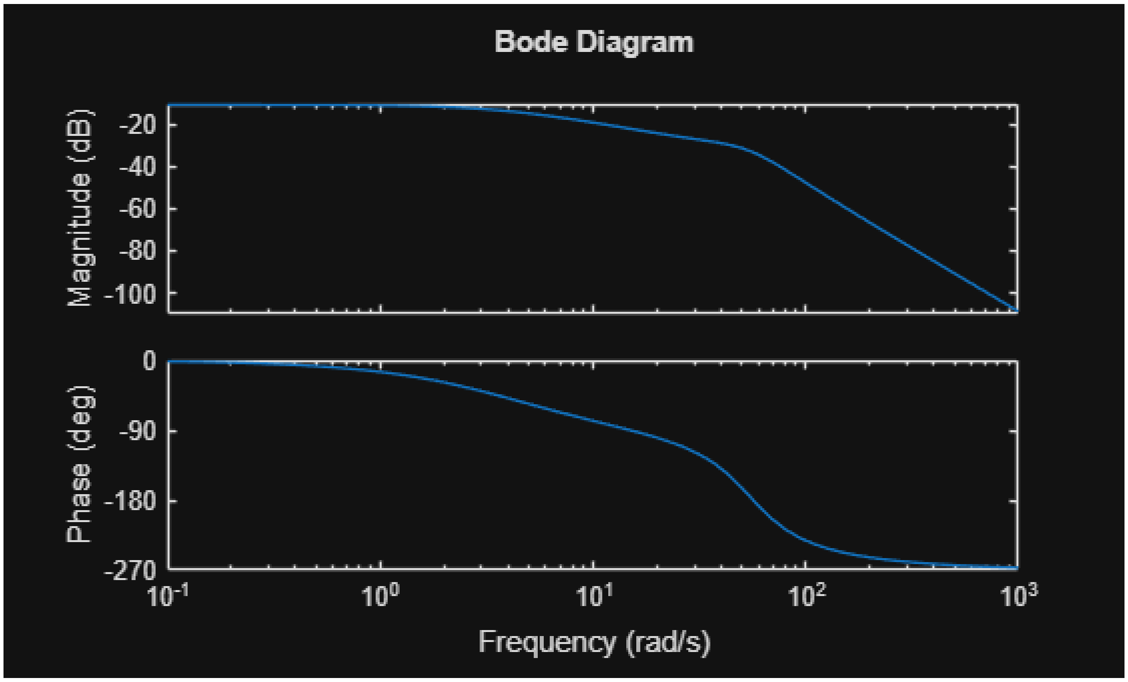
```
phi_m = rad2deg(asin((1-a)/(1+a)))
```

```
phi_m =
78.5788
```

```
phi_g = pm - phi_i -180 - phi_m
```

```
phi_g =
-165.1439
```

```


%Finder omega_c
bodeplot(G)
```




```
w_c = 50.42
```

```
w_c =
50.4200
```

```
%finder tidskonstanter:
tau_i = Ni/w_c
```

```
tau_i =
0.0595
```

```
tau_d = 1/(w_c*sqrt(a))
```

```
tau_d =
0.1983
```

```
C = ((tau_i*s+1)/(tau_i*s))*((tau_d*s+1)/(a*tau_d*s+1))
```

```
C =

  0.0118 s^2 + 0.2578 s + 1
  -------------------------
   0.000118 s^2 + 0.0595 s

Continuous-time transfer function.
Model Properties
```

```
% Udregner Kp
s = w_c*1i
```

```
s =
0.0000 +50.4200i
```

```
Kp = 1/abs(evalfr(C*G,s))
```

```
Kp =
3.4151
```

Question 20

Which of the following statements is correct?

a) Higher phase margin of the open-loop transfer function implies reduced oscillations in the closed-loop step response and more robustness. (Correct)

b) Increasing the parameter α in a PI-Lead controller design improves the phase margin of the open loop system. (No, it actually deteriorates it)

c) Lower bandwidth of a closed-loop system implies faster response of the output to changes in the reference signal. (No it is the opposite)

d) Moving the Lead part of a PI-Lead controller from the feedback branch to the forward branch decreases the effect of an input disturbance (i.e. a disturbance that is added to the input signal) on the step response of the closed-loop system. (No it does not matter at all)

e) Moving the Lead part of a PI-Lead controller from the feedback branch to the forward branch improves oscillation attenuation for the step response of the closed-loop system. (No, it is the opposite)

a) Higher phase margin of the open-loop transfer function implies reduced oscillations in the closed-loop step response and more robustness.

Det er korrekt, fordi **phase margin** fortæller hvor langt systemet er fra ustabilitet. En større phase margin betyder typisk:

$$\text{mindre overshoot}$$

$$\text{færre oscillationer}$$

$$\text{mere robusthed}$$

Så hvis phase margin øges, bliver step-responsen normalt mere dæmpet.

De andre er forkerte:

b) Forkert. I en PI-Lead-controller giver PI-delen typisk **negativ fase**, fordi integratoren giver faseforsinkelse. Derfor kan PI-delen reducere phase margin.

$$C_{PI}(s) = \frac{\tau_i s + 1}{\tau_i s}$$

Integratoren $1/s$ bidrager med cirka:

$$-90^\circ$$

---

c) Forkert. Lavere båndbredde betyder normalt **langsommere respons**, ikke hurtigere.

$$\text{lav bandwidth} \Rightarrow \text{langsomt system}$$

$$\text{høj bandwidth} \Rightarrow \text{hurtigere system}$$

---

d) Forkert. At flytte Lead-delen fra feedback branch til forward branch reducerer ikke nødvendigvis input disturbance-effekten. Det afhænger af hvor forstyrrelsen kommer ind i systemet.

---

e) Forkert. Lead-delen forbedrer normalt fase og hastighed, men kan også give mere højfrekvent forstærkning. At flytte den til forward branch forbedrer ikke automatisk oscillation attenuation.

1. Q1. **What is the transfer function of this linear system in Fig.1**? (1 Point)

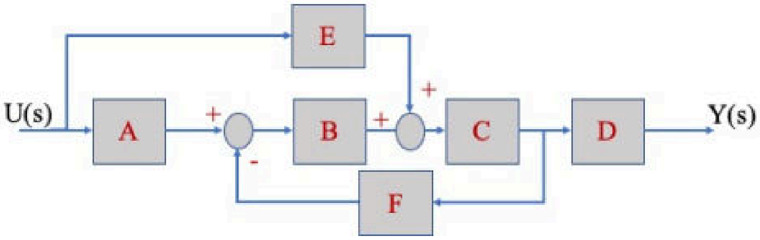


Figure 1: Block Diagram

1. $$\frac{Y}{U} = \frac{ABD + DEF^2}{1 + BCF}$$

2. $$\frac{Y}{U} = \frac{ABCD + ECD}{1 + BCF}$$ *CORRECT*

3. $$\frac{Y}{U} = \frac{ABCD + EF}{BCF}$$

4. $$\frac{Y}{U} = \frac{ABCD + ECD + EC^2FBD}{1 + CDE}$$

5. $$\frac{Y}{U} = \frac{ABCD}{1 + BCF} + ECD$$

```
clear; clc;

syms A B C D E F

FB = (A*B+E)*C*D;

G_cl = FB / (1+B*C*F);
G_cl = expand(simplifyFraction(G_cl))
```

G_cl =

$$\frac{C\,D\,E}{B\,C\,F+1} + \frac{A\,B\,C\,D}{B\,C\,F+1}$$

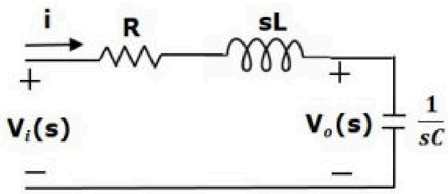
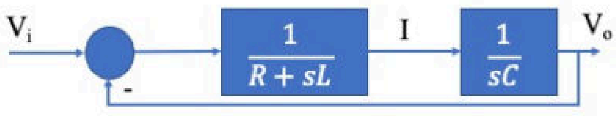
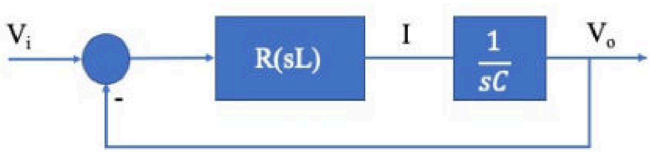
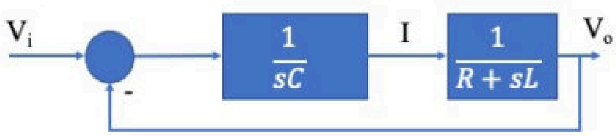


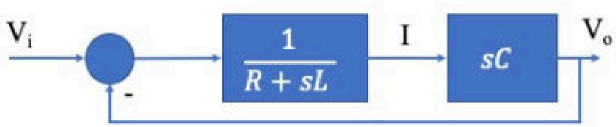
Figure 2: RLC circuit

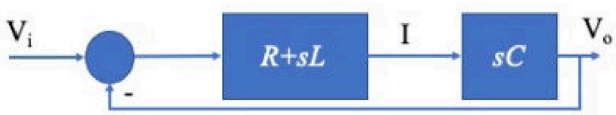
1. 

CORRECT

2. 

3. 

4. 

5. 


```
clc; clear;

syms R L C s Vi Vo I

% KVL:
% Vi = I*(R + s*L) + Vo
% Capacitor:
% Vo = I*(1/(s*C))

eq1 = Vi == I*(R + s*L) + Vo;
eq2 = Vo == I/(s*C);

sol = solve([eq1, eq2], [Vo, I]);

G = simplify(sol.Vo / Vi);
```

```
disp('Transfer function Vo/Vi = ')
```

```
Transfer function Vo/Vi =
```

```
pretty(G)
```

$$\frac{1}{C\,L\,s^2 + C\,R\,s + 1}$$

3. Q3. A system has poles and zeros as shown in Fig. 3 (the circle marks the zero, the cross marks the pole). **Which is the corresponding Bode plot?** (1 Point

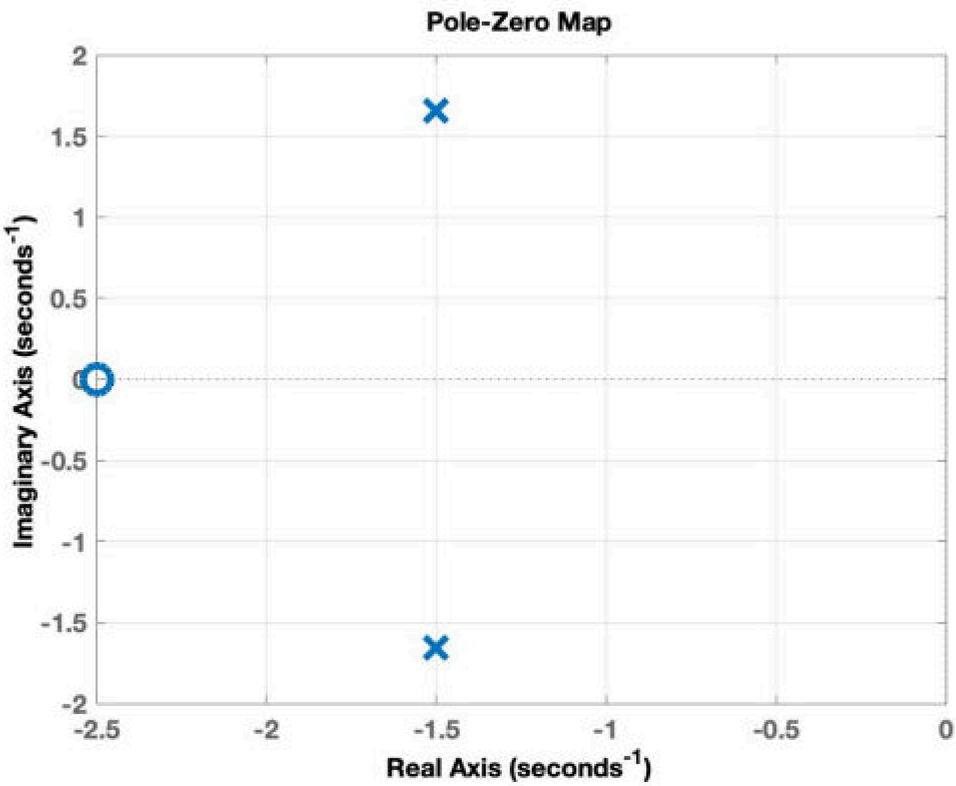


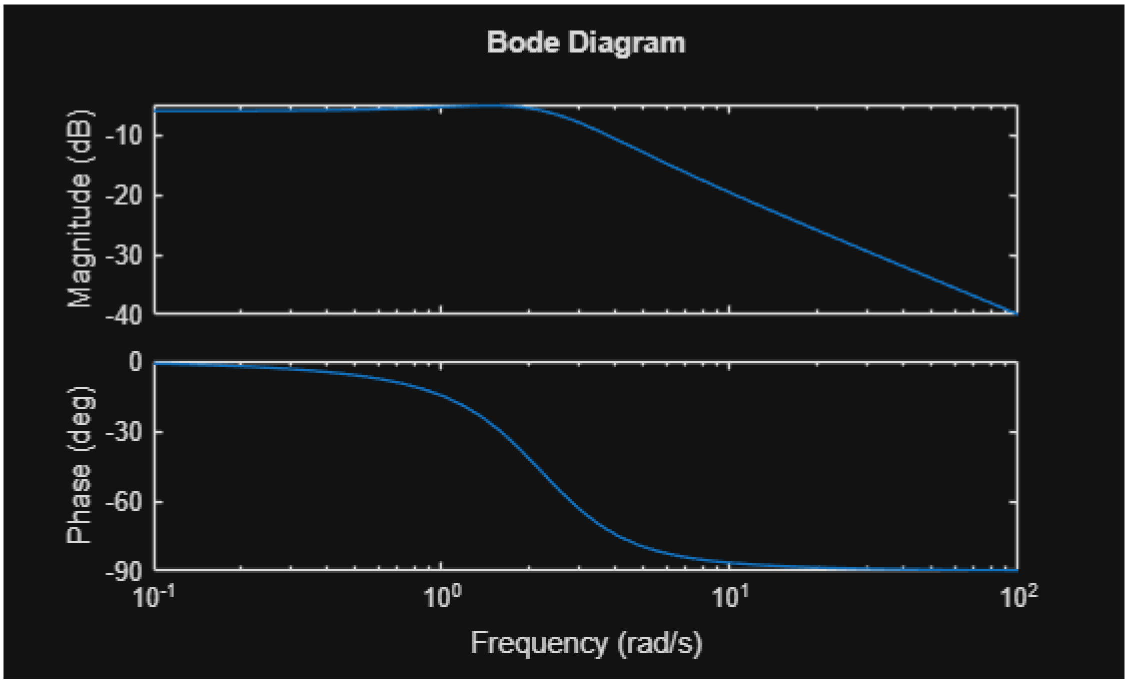
Figure 3: Poles/zeros

```
clear; clc;

s = tf('s');

p1 = -1.5 + (-1.67)*j;
p2 = -1.5 - (-1.67)*j;

z1 = -2.5;
```

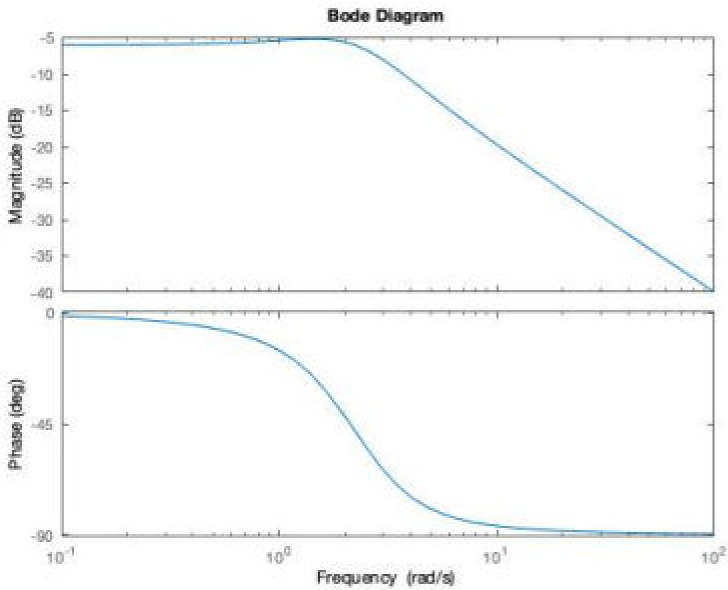
```
G = (s-z1) / ((s-p1)*(s-p2))
```

```
G =

       s + 2.5
  -----------------
  s^2 + 3 s + 5.039

Continuous-time transfer function.
Model Properties
```

```
bode(G)
```



Svaret er 4



4.

(CORRECT - negative zero -2.5, negative complex poles - 1.5)

4. Q4. The following closed-loop system is given in Fig. 4. Where

$$P(s) = \frac{1}{(s+1)^3}$$

is the system transfer function and K is a proportional controller with gain $K \in \mathbb{R}^+$.
For which values of K is the closed loop stable? (1 Point)

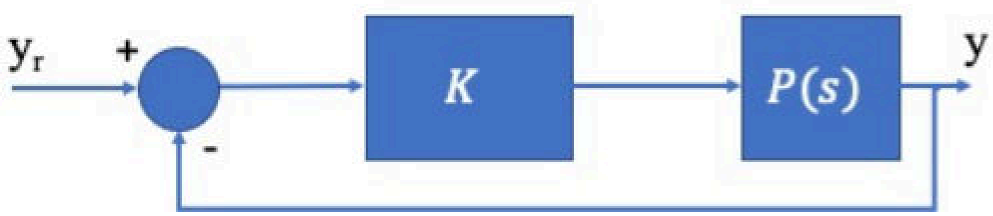


Figure 4: Closed-loop system

1. $K \geq 8$
2. $-1 < K < 6$
3. $0 < K < 8$

```
B = isstable(sys)

B = logical
   1
```

The system sys is stable.

4. $K > -10$

```
clear; clc; close all;

% Closed-loop characteristic equation:
% 1 + K/(s+1)^3 = 0
% => (s+1)^3 + K = 0
% => s^3 + 3s^2 + 3s + (1+K) = 0

Kvals = linspace(-12, 12, 20000);
```

```
stable = false(size(Kvals));

for n = 1:length(Kvals)
    K = Kvals(n);

    den = [1 3 3 1+K];
    poles = roots(den);

    stable(n) = all(real(poles) < 0);
end

% Since the problem says K ∈ R+, only positive K counts
K_positive = Kvals > 0;

true_condition = stable & K_positive;

% Multiple choice conditions
opt1 = K_positive & (Kvals >= 8);
opt2 = K_positive & (Kvals > -1 & Kvals < 6);
opt3 = K_positive & (Kvals > 0 & Kvals < 8);
opt4 = K_positive & (Kvals > -10);

options = {opt1, opt2, opt3, opt4};
names = ["1: K >= 8", ...
         "2: -1 < K < 6", ...
         "3: 0 < K < 8", ...
         "4: K > -10"];

for i = 1:4
    correct = isequal(options{i}, true_condition);
    fprintf('%s  -->  Correct? %d\n', names(i), correct);
end
```

```
1: K >= 8  -->  Correct? 0
2: -1 < K < 6  -->  Correct? 0
3: 0 < K < 8  -->  Correct? 1
4: K > -10  -->  Correct? 0
```

5. Q5. The bode plot diagrams of the magnitude and phase of a transfer function G(s) of a linear system is shown in Fig. 5.
Which of the following answers is correct? (1 Point)

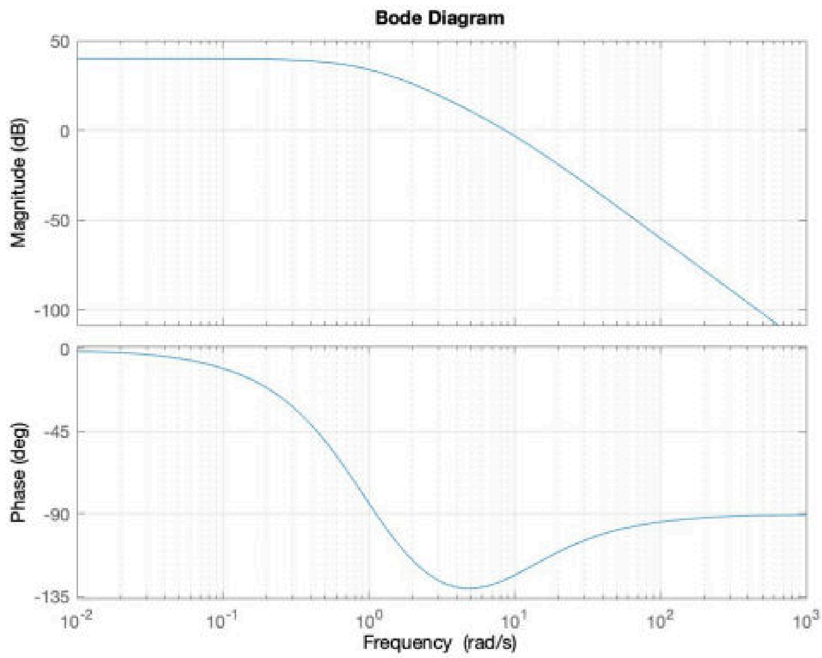


Figure 5: Bode Plot Diagrams - Q5

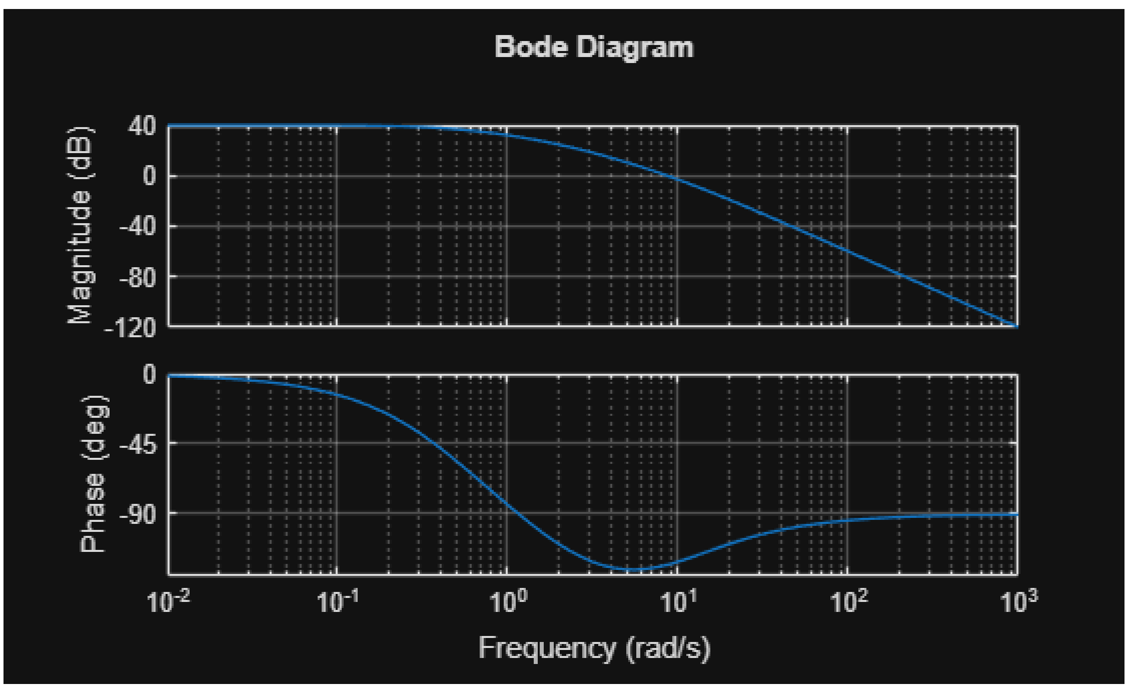
1. G(s) has negative and positive real poles and no zeros
2. G(s) has only 1 negative real pole and 1 negative real zero
3. G(s) has only 1 positive real pole and no zeros
4. G(s) has positive real poles and 1 negative real zero

Answer:
Q5. [**CORRECT n.1**] DC gain 40 dB equal to 100 (magnitude horizontal and initial phase is 0 degrees.. Magnitude decreases from 0 dB/decade to -40 dB/decade at $\omega = 1$. Correspondingly, the phase changes 180 degrees. This means that there are two poles with abs 1. There are no peaks, so poles are reals. moreover, the phase in $\omega = 1$ decreases of 180 degrees so both poles are

negatives. At $\omega = 10$ the slope of the magnitude decreases from -40 dB/decade to -60 dB/decade and the asymptotic phase increases of 90 degrees. This concludes that G(s) has one pole +10. In conclusion, DC gain =10, poles p1=p2=-1 and p3=10, and no zeros.

$$G(s) = \frac{100}{(1+s)^2 * (1-0.1*s)}$$

```
clear; clc; close all;

s = tf('s');

G = 100 / ((s+0.5)*(s+2)*(1 - s/10));

bode(G)
grid on
```



6. Q6. A closed-loop system has a loop transfer function

$$L(s) = \frac{K}{s(s+3)(s+10)}$$

and the following Bode plot in Fig.6

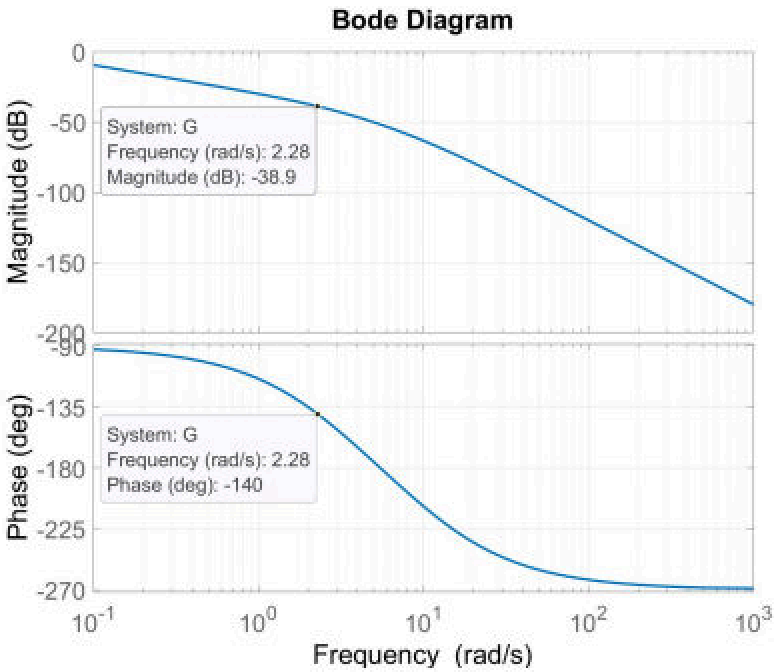


Figure 6: Bode Plot - Q6

What is the gain K so that the phase margin is PM=40°?
(1 Point)

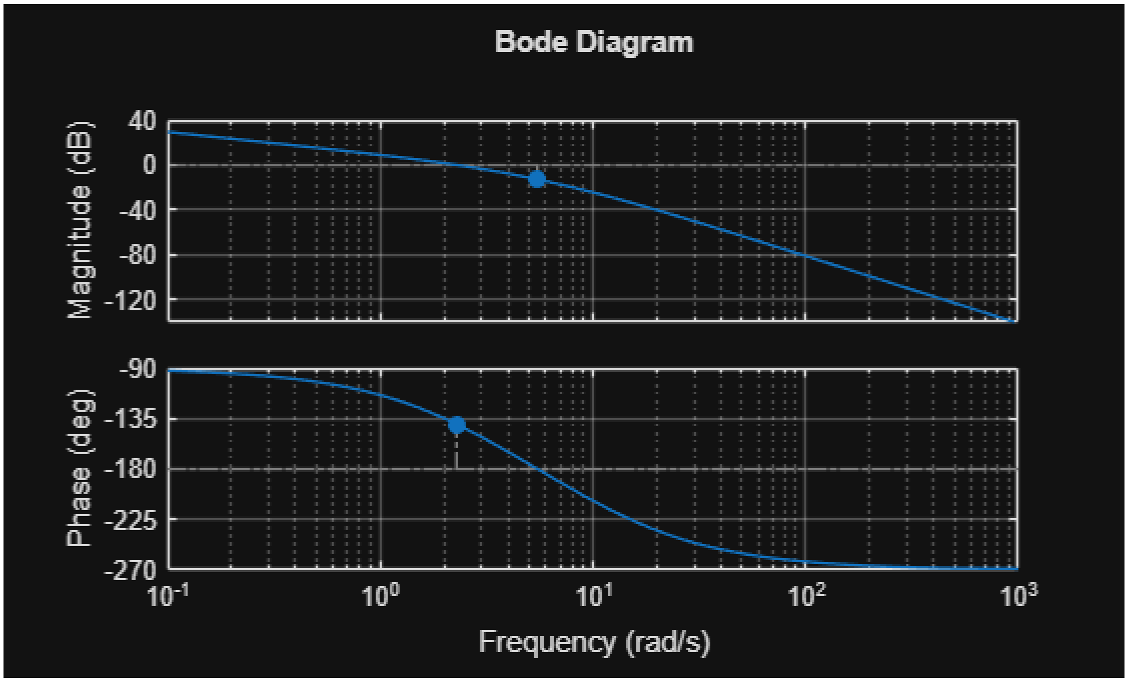
1. K = 19.5
2. K = 90
3. K = 44
4. K = 38.9
5. K = 88 = 88.1

```
clear; clc; close all;

K = 10^(38.9/20)
```

```
K =
88.1049
```

```
s = tf('s');
```

```
L = K/(s*(s+3)*(s+10));

margin(L)
grid on
```



7. Q7. **Which is the transfer function $\frac{X_2(s)}{F(s)}$ for the system shown in Figure 7**? Both masses slide on a frictionless surface and $k = 1$ N/m. (Consider zero initial conditions). [hint = write the equations of motion]

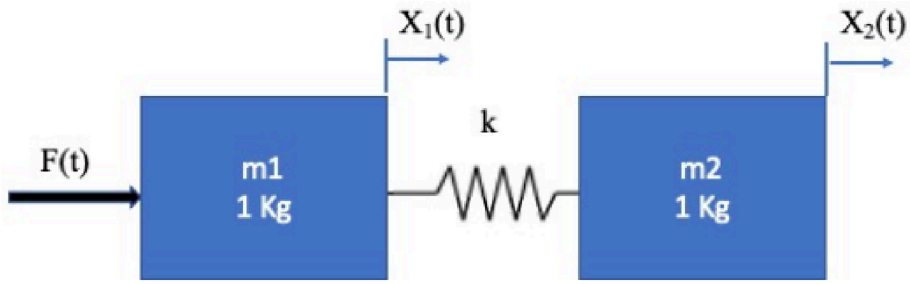


Figure 7: Two connected masses on a frictionless surface

(1 Point)

1. $$\frac{X_2(s)}{F(s)} = \frac{1}{(s^2)*(s^2+1)}$$
2. $$\frac{X_2(s)}{F(s)} = \frac{1}{s^2*(s^2+2)}$$
3. $$\frac{X_2(s)}{F(s)} = \frac{1}{s^2*(s+2)}$$
4. $$\frac{X_2(s)}{F(s)} = \frac{1}{s*(s^2+1)}$$
5. $$\frac{X_2(s)}{F(s)} = \frac{1}{(s^4+1)}$$

```
clear; clc;

syms s X1 X2 F

% m1 = m2 = 1 kg, k = 1 N/m

% Equations of motion in Laplace domain:
% s^2*X1 = F + (X2 - X1)
% s^2*X2 = X1 - X2

eq1 = s^2*X1 == F + X2 - X1;
eq2 = s^2*X2 == X1 - X2;
```

```
sol = solve([eq1, eq2], [X1, X2]);

G = simplify(sol.X2 / F);

disp('Transfer function X2(s)/F(s):')
```

```
Transfer function X2(s)/F(s):
```

```
pretty(G)
```

$$\frac{1}{s^2\ (s^2+2)}$$

8. Q8. Consider the differential equation

   $\ddot{y}(t) + 2\dot{y}(t) + y(t) = u(t)$

   Where y(0) = $\dot{y}(0)$ = 0 and u(t) is the input.
   Which are the poles of this system?

   1. s1 = 2, s2= -1
   2. s1 = -1, s2= -1
   3. s1 = 1j, s2= -1j
   4. s1 = -1, s2= -2
   5. s1 = -j, s2= -j

```
clear; clc;

syms s;

den = 1;
num = s^2+2*s+1;

H = den/num
```

```
H =
```

$$\frac{1}{s^2+2\,s+1}$$

```
pol = solve(num==0,s)
```

```
pol =
```

$$\begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

```
poles(H)
```

ans = -1

Svaret er 2

9. Q9. For the unit feedback system shown in Fig.8, **Which is the gain K of the proportional controller so that the output y(t) has an overshoot of no more than 12% in response to a unit step?**

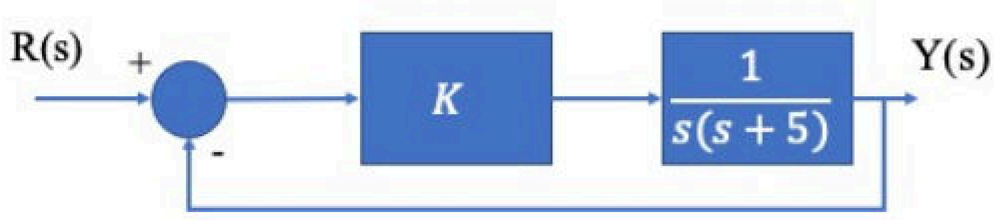


Figure 8: Unity feedback system for Q8.

(1 Point)

1. $0 \leq K \leq 20$
2. $5 \leq K \leq 25$
3. $K \geq 25$
4. $-1.5 \leq K \leq 10$

```
clear; clc;

syms s K;

H = K * 1/(s*(s+5))
```

H =

$$\frac{K}{s\ (s+5)}$$

```
T = H / (1+H);
T = simplifyFraction(T)
```

T =

$$\frac{K}{s^2+5\,s+K}$$

```
Mp = 0.12;    % 12 percent overshoot
```

```
% Required damping ratio
zeta_min = -log(Mp) / sqrt(pi^2 + log(Mp)^2);

% Closed-loop system:
% T(s) = K / (s^2 + 5s + K)
% Compare with s^2 + 2*zeta*wn*s + wn^2
% wn^2 = K
% 2*zeta*wn = 5
% zeta = 5/(2*sqrt(K))

Kmax = (5/(2*zeta_min))^2;

fprintf('Minimum damping ratio = %.4f\n', zeta_min);
```

```
Minimum damping ratio = 0.5594
```

```
fprintf('Maximum K = %.4f\n', Kmax);
```

```
Maximum K = 19.9715
```

```
disp('Therefore:')
```

```
Therefore:
```

```
fprintf('0 <= K <= %.2f\n', Kmax);
```

```
0 <= K <= 19.97
```

10. Q10. A certain system is represented by the Bode diagram (only the magnitude in dB) shown in Fig. 9 (no zeros in the RHP). **Which is the correct answer showing the response of this system to a unit step input (assuming zero initial conditions)?**

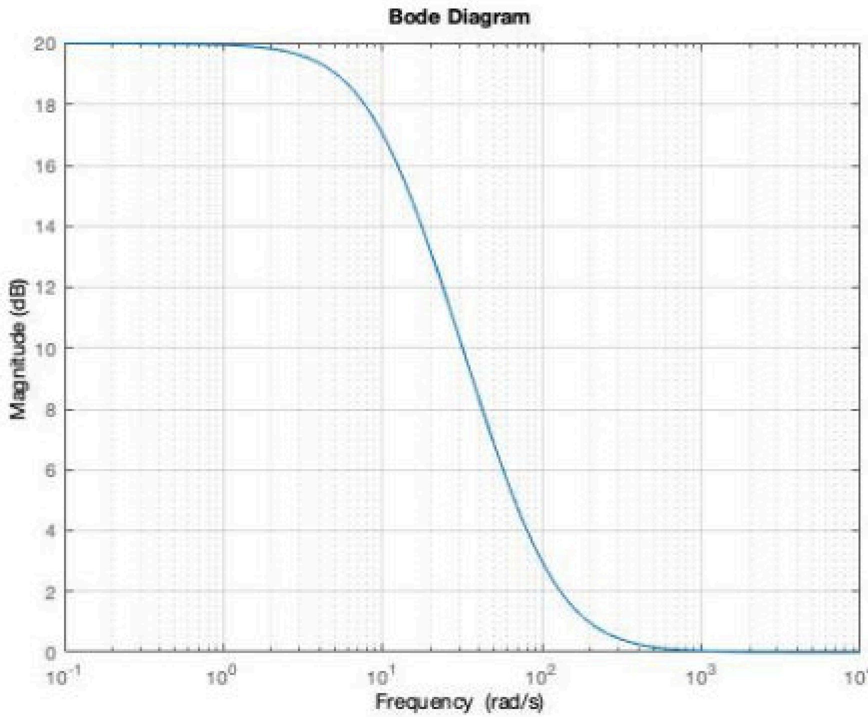
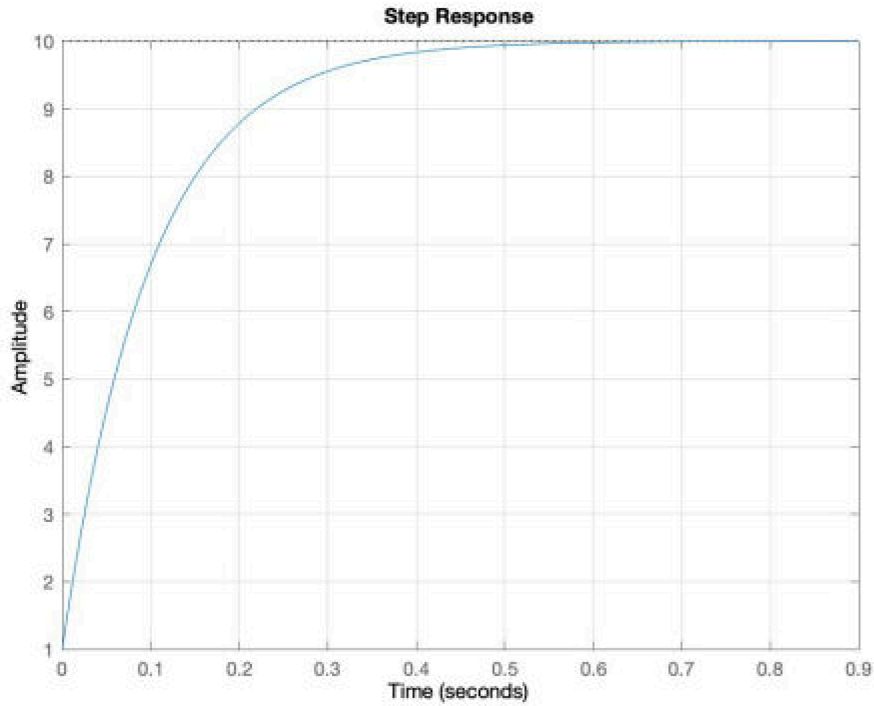
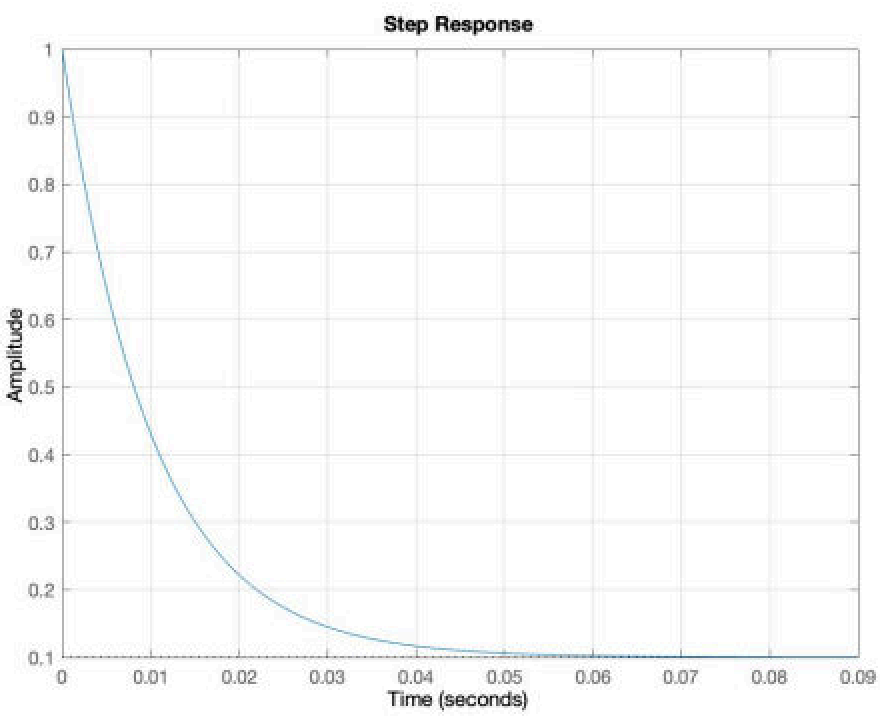


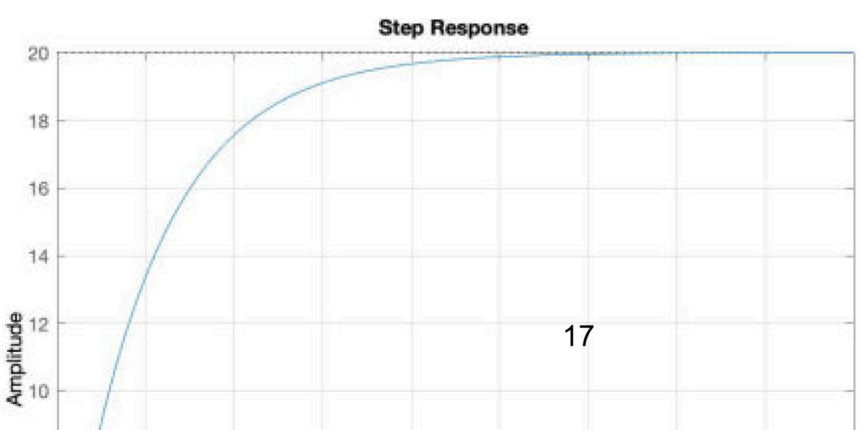
Figure 9: Magnitude portion (dB) of the Bode diagram for Q10.

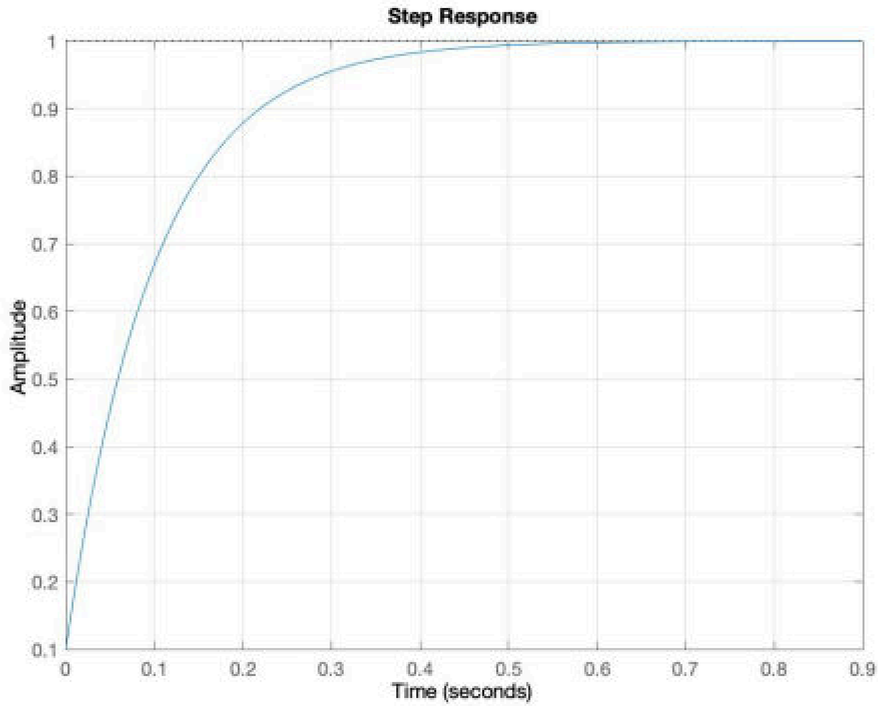

Step Response
Amplitude
Time (seconds)
10
9
8
7
6
5
4
3
2
1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9

1.


Step Response
Amplitude
Time (seconds)
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09

2.


Step Response
Amplitude
20
18
16
14
12
10



5.

Low frequency gain:

$$20\ \mathrm{dB} \Rightarrow |G(0)| = 10^{20/20} = 10$$

High frequency gain:

$$0\ \mathrm{dB} \Rightarrow |G(\infty)| = 10^{0/20} = 1$$

Kurven falder fra 20 dB til 0 dB, altså har systemet én **pol** og én **nulpunkt**. Fordi den falder først, ligger polen ved lavere frekvens end nulpunktet.

Fra plottet kan man cirka aflæse:

$$\omega_p = 10\ \mathrm{rad/s}$$

$$\omega_z = 100\ \mathrm{rad/s}$$

Derfor kan transferfunktionen skrives som:

$$G(s) = 10\frac{1+\frac{s}{100}}{1+\frac{s}{10}}$$

Omskrives:

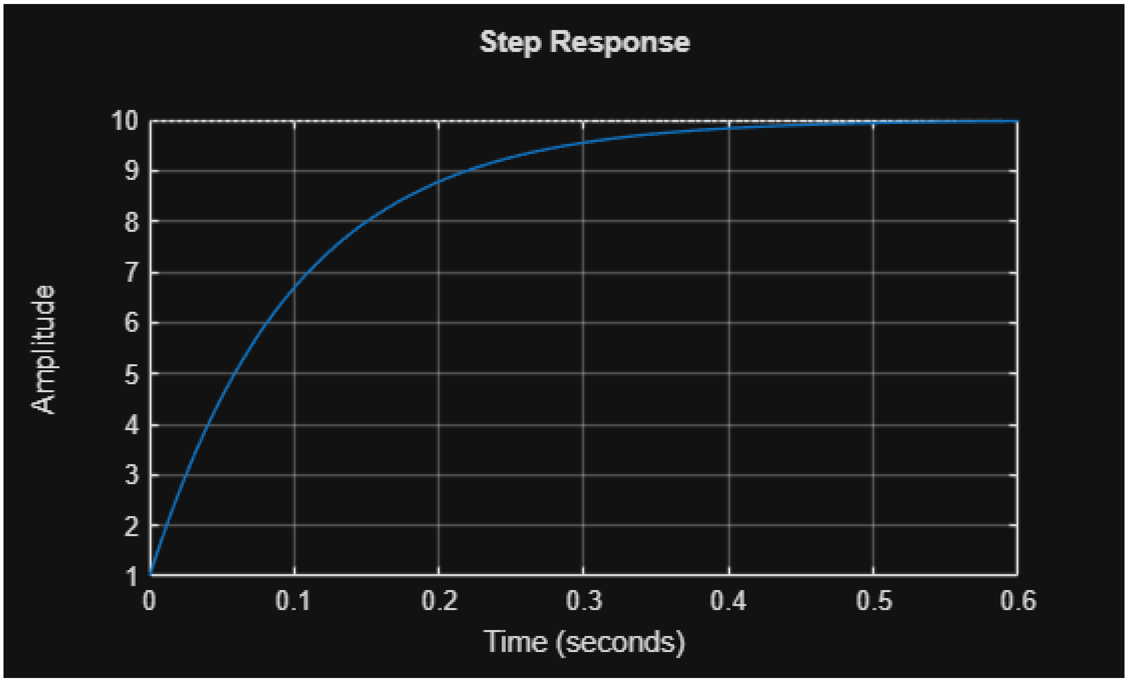
$$G(s) = \frac{s+100}{s+10}$$

Så et godt bud er:

$$\boxed{G(s) = \frac{s+100}{s+10}}$$

```
clear; clc; close all;

s = tf('s');

G = (s + 100)/(s + 10);
```

```
step(G)
grid on
```



Answer:
Q10. [**CORRECT n. 1**]
Through the inspection of the given Bode plot, the transfer function is the following:

$$G(s) = \frac{10 * (s/100 + 1)}{(s/10 + 1)} = \frac{s + 100}{(s + 10)}$$

The response to a unit step input is $Y(s) = G(s)U(s)$

$$Y(s) = \frac{s + 100}{(s + 10)s} = \frac{10}{s} - \frac{9}{(s + 10)} \qquad y(t) = \mathcal{L}^{-1}Y(s) = 10 \times 1(t) - 9e^{-10t}$$

Question 11

Consider a system with transfer function

$$G(s) = \frac{120}{s^3 + 43s^2 + 120s}$$

and Bode plot as shown in the following figure. A P-controller K_P is applied to the system

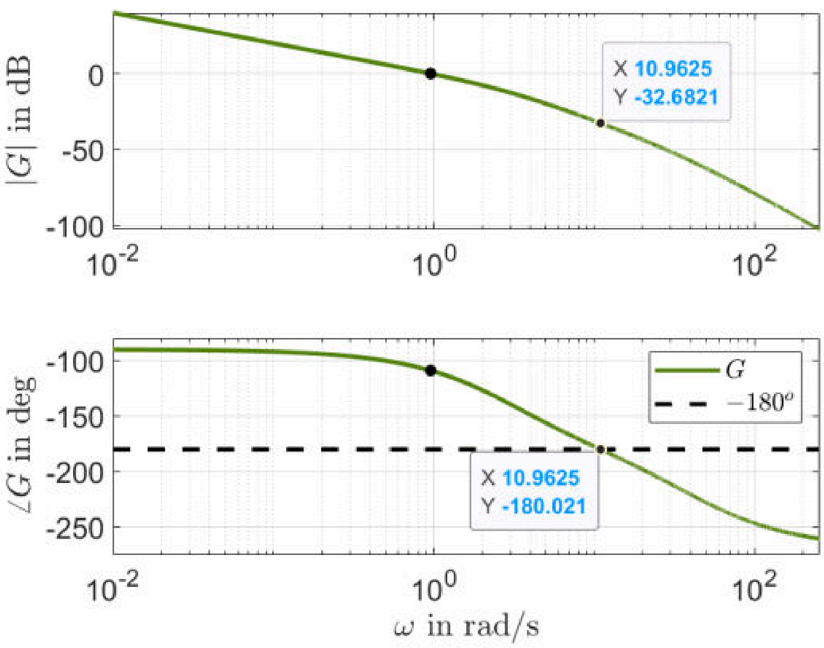


Figure 1: Bode plot

and the loop is closed with unit feedback. The closed-loop system is stable for:

a) $K_P = -1$ (Negative gain)

b) $K_P = 25$ (Correct)

c) $K_P = 71$ (Exceeds gain margin $K_M = 32.6821$ dB $= 43$)

d) $K_P = -32$ (Negative gain)

e) $K_P = 45$ (Exceeds gain margin $K_M = 32.6821$ dB $= 43$)

```
clear; clc;

Kvals = [-1 25 71 -32 45];

for Kp = Kvals
    den = [1 43 120 120*Kp];
    poles = roots(den);

    stable = all(real(poles) < 0);

    fprintf('Kp = %6.1f   Stable = %d   Poles = ', Kp, stable);
    disp(poles.')
```

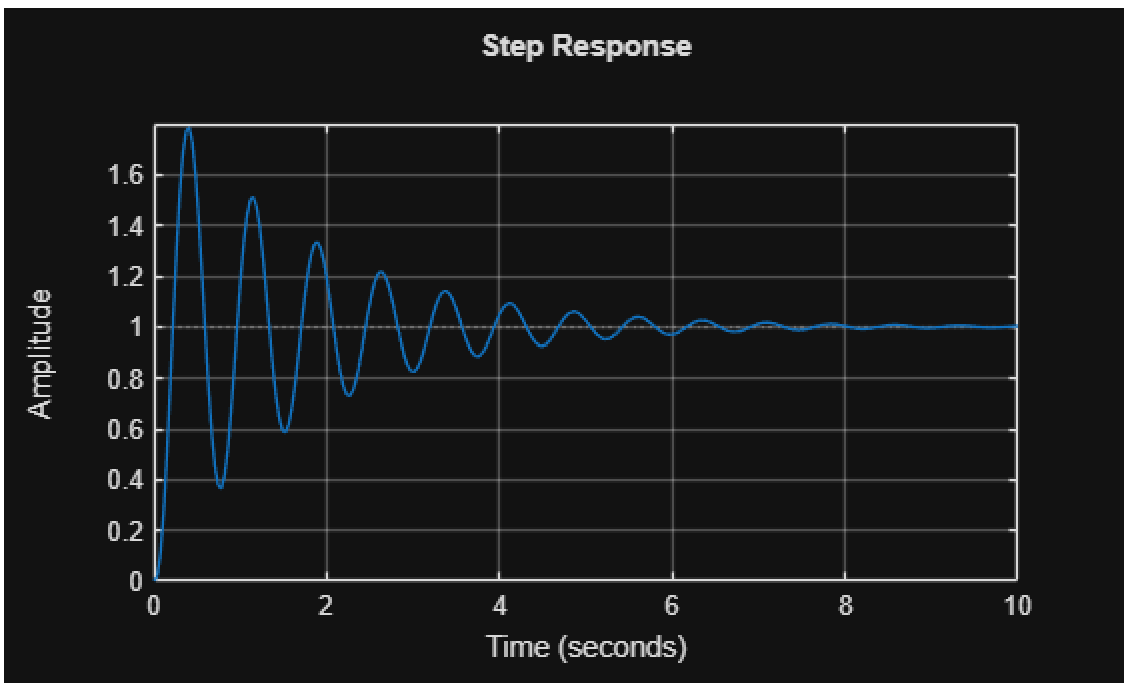
```
end
```

```
Kp =   -1.0   Stable = 0   Poles =
  -39.9186   -3.8602    0.7788
Kp =   25.0   Stable = 1   Poles =
 -41.8456 + 0.0000i  -0.5772 + 8.4474i  -0.5772 - 8.4474i
Kp =   71.0   Stable = 0   Poles =
 -44.5935 + 0.0000i   0.7967 +13.7994i   0.7967 -13.7994i
Kp =  -32.0   Stable = 0   Poles =
  -36.9366  -13.6691    7.6056
Kp =   45.0   Stable = 0   Poles =
 -43.1212 + 0.0000i   0.0606 +11.1904i   0.0606 -11.1904i
```

```
s = tf('s');

G = 120/(s^3 + 43*s^2 + 120*s);

Kp = 25;

T = feedback(Kp*G, 1);

pole(T)
```

```
ans = 3×1 complex
 -41.8456 + 0.0000i
  -0.5772 + 8.4474i
  -0.5772 - 8.4474i
```

```
step(T)
grid on
```



Question 12

Consider the system described by the following differential equation

$$y^{(4)}(t) + 9y^{(3)}(t) + 20\ddot{y}(t) = 71u(t)$$

where $\ddot{y}(t) = \frac{d^2y}{dt^2}$ and $y^{(n)}(t) = \dfrac{d^ny}{dt^n}$ denotes the n^{th} time derivative of $y(t)$. Which of the following statements is the correct one?

a) The system is stable. (Wrong)

b) The system is nonlinear. (It is linear)

c) The system has two zeros. (It has no zeros)

d) The system is unstable. (Correct)

e) The steady state (DC) gain of G is 3.55. (It is infinite)

Solution: The system is written in the Laplace domain as

$$s^4y(s) + 9s^3y(s) + 20s^2y(s) = 71u(s) \Rightarrow G(s) \triangleq \frac{y(s)}{u(s)} = \frac{71}{s^4 + 9s^3 + 20s^2} = \frac{71}{s^2(s^2 + 9s + 20)}$$

The transfer function has a double pole at the origin and the system is therefore unstable.

```
clear; clc;

syms s;
den = 71;
num = s^4+9*s^3+20*s^2;
H = den/num
```

H =

$$\frac{71}{s^4 + 9\,s^3 + 20\,s^2}$$

```
pole = solve(num == 0,s)
```

pole =

$$\begin{pmatrix} -5 \\ -4 \\ 0 \\ 0 \end{pmatrix}$$

Question 13

Consider a system with open-loop transfer function

$$G(s) = \frac{s+1}{s^2 + 43s - 90}$$

and Nyquist plot as shown in the following figure. The Nyquist plot intersects the real axis along the counter-clockwise direction at the point $(-0.0111, 0)$. A P-controller K_P is applied to the system and the loop is closed with unit feedback. The closed-loop system is stable for:

a) $K_P = 100$ (Correct)

b) $0.02 < K_P < 1$ (Too small gain)

c) $0 < K_P < 80$ (Too small gain)

d) $K_P < -0.0111$ (Negative gain)

e) $0 < K_P < 0.02$ (Too small gain)

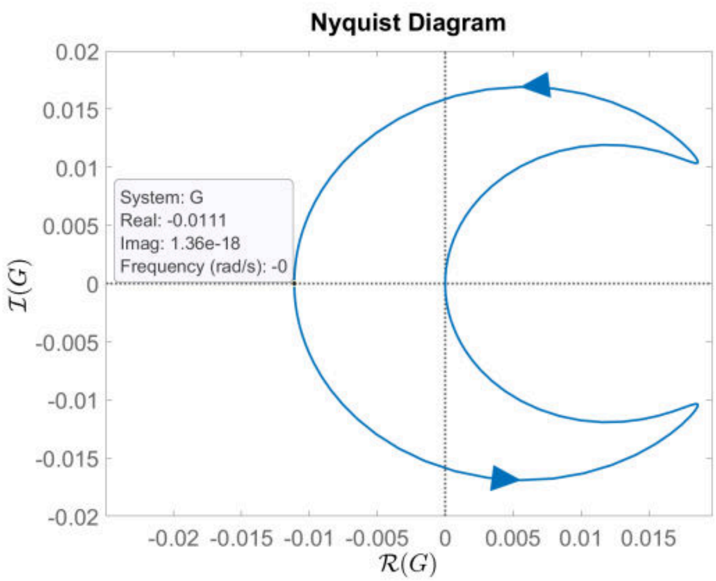


Figure 2: Nyquist plot

Solution: The system is unstable with one pole in the RHP. We need to ensure that the Nyquist curve encircles the point (-1,0) once along the CCW direction. Already the direction is ok, so we only need to adjust the gain K_P. The intersection point of the curve with the negative real axis is at -0.0111, so multiplying G by $K_P = \frac{1}{0.0111} \approx 90$ ("gain margin" if we did not have unstable poles), we will make the curve cross exactly the critical point (-1,0). Hence a proportional gain $K_P > 90$, will ensure one encirclement of the critical point, which is needed for closed-loop stability. Negative gain will simply flip the curve about the imaginary axis. Therefore, the only acceptable option is (a).

```python
s, Kp = symbols("s Kp", real=True)

G = (s + 1)/(s**2 + 43*s - 90)

# Characteristic polynomial for unity feedback:
p = s**2 + 43*s - 90 + Kp*(s + 1)

K_values = [100, 0.5, 40, -0.02, 0.01]

for val in K_values:
    poly = p.subs(Kp, val)
    poles = solve(poly, s)
    stable = all(re(root.evalf()) < 0 for root in poles)
    print(val, poles, stable)

find_stable_gain_ranges(p, Kp, s)
```

Question 14

Consider a system with transfer function

$$G(s) = \frac{1200}{s^3 + 54s^2 + 570s + 400}$$

and Nyquist plot (only positive frequencies) as shown in the following figure. The Nyquist

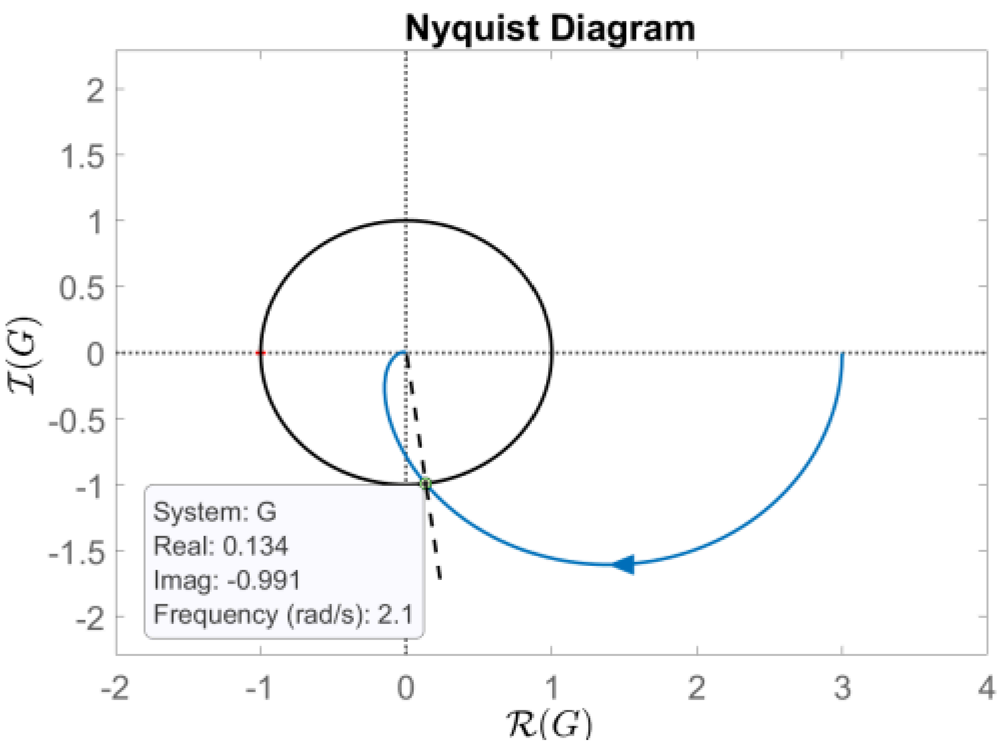


Figure 3: Nyquist plot

plot intersects the unit circle along the clockwise direction at the point $(0.134, -0.99)$. The phase margin of the system is approximately

a) $\gamma_M = 0.1354^o$ (Wrong)

b) $\gamma_M = 1.705^o$ (In rad)

c) $\gamma_M = -1.44^o$ (ϕ_c in rad)

d) $\gamma_M = 82.29^o$ ($-\phi_c$)

e) $\gamma_M = 97.7^o$ (Correct)

Solution: The phase margin is defined as $\gamma_M = 180^o + \phi_c$, where ϕ_c is the phase of the transfer function at the crossover frequency. This angle is confined between the positive real axis (0^o) and the dashed line in the Nyquist diagram, which starts from the origin and crosses the point of intersection of the Nyquist curve with the unit circle (0 dB magnitude). It is calculated as $\phi_c = \arctan\left(\frac{-0.99}{0.134}\right) \approxeq -82.29^o$. The phase margin is then found by $\gamma_M \approxeq 180^o - 89.29^o = 97.7^o$. Therefore, the only acceptable option is (e).

```
clear; clc; close all;

s = tf('s');

G = 1200/(s^3 + 54*s^2 + 570*s + 400);

z = 0.134 - 0.991i;

phi_c = rad2deg(angle(z));
PM = 180 + phi_c;

fprintf('Phase at crossover = %.2f degrees\n', phi_c);
```

```
Phase at crossover = -82.30 degrees
```

```
fprintf('Phase margin = %.2f degrees\n', PM);
```

```
Phase margin = 97.70 degrees
```

Question 15

Consider a second-order type-0 system with two stable real poles and no zeros. A P-controller with $K_P > 1$ is applied to the system. Which of the following statements is the correct one?

a) Increasing K_P will shift the phase curve in the open-loop Bode plot upwards. (K_P does not affect the phase curve in the Bode plot)

b) Increasing K_P will increase the crossover-frequency of the open-loop system. (Correct)

c) Increasing K_P too much can theoretically make the closed-loop system unstable. (An open-loop stable second-order type-0 system with no zeros has infinite gain margin)

d) After a finite value of K_P (i.e. $K_P < \infty$), the steady state error in the output of the closed-loop system due to a step change in the reference signal will be zero. (Type-0 system cannot have zero steady state error if the loop is closed with a finite proportional gain)

e) The open-loop system can have multiple crossover-frequencies. (The system has real poles and therefore no resonance peaks that could cause the phenomenon of multiple crossings of the 0 dB line)

Question 16

Consider a system with transfer function

$$G(s) = \frac{1224}{s^3 + 30s^2 + 257s + 612}$$

[redacted]

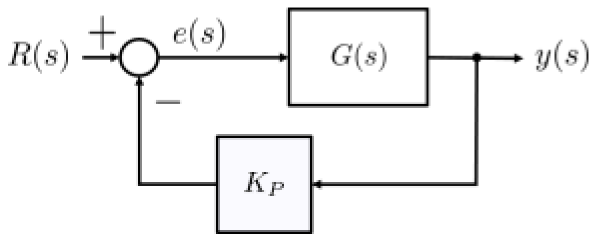


Figure 4: Closed-loop system.

A P-controller with $K_P = 2$ is applied to the system but it is placed in the feedback branch as in the following figure. The steady state error $e(s)$ in the closed-loop system due to a unit step change in the reference signal is:

a) $e_{SS} = 0$ (Type-0 system)

b) $e_{SS} = 2$ (DC gain of $G(s)$)

c) $e_{SS} = 0.2$ (Correct)

d) $e_{SS} = 0.4$ (DC gain of closed-loop transfer function)

e) $e_{SS} = 0.8$ (DC gain of $\frac{K_P G(s)}{1+K_P G(s)}$)

Solution: The transfer function from the reference signal to the error $e(s)$ is given by

$$G_{er}(s) = \frac{1}{1 + K_P G(s)} = \frac{s^3 + 30s^2 + 257s + 612}{s^3 + 30s^2 + 257s + 3060} \, .$$

The steady-state error is found by

$$e_{ss} = \lim_{s \to 0} e(s) = \lim_{s \to 0} sG_{er}(s)\frac{1}{s} = 0.2 \, .$$

Therefore, the only acceptable option is (c).

```
clear; clc;

syms s;
```

```
G = 1224 / (s^3+30*s^2+257*s+612)
```

G =

$$\frac{1224}{s^3 + 30\,s^2 + 257\,s + 612}$$

```
Kp = 2;

Ger = 1 / (1+Kp*G);
Ger = simplifyFraction(Ger)
```

Ger =

$$\frac{s^3 + 30\,s^2 + 257\,s + 612}{s^3 + 30\,s^2 + 257\,s + 3060}$$

```
R = 1/s;               % unit step
E = Ger*R;

ess = limit(s*E, s, 0)
```

ess =

$$\frac{1}{5}$$

Question 17

Consider the system described by the following transfer function and Bode plot

$$G(s) = \frac{5(s+7)(s+12)}{(s+4)(s+9)(s+17)(s+5)(s+70)}$$

A P-Lead-**Lag** controller

$$C(s) = K_P \frac{\tau_d s + 1}{\alpha \tau_d s + 1} \frac{\tau_i s + 1}{\tau_i s + \frac{1}{\beta}}$$

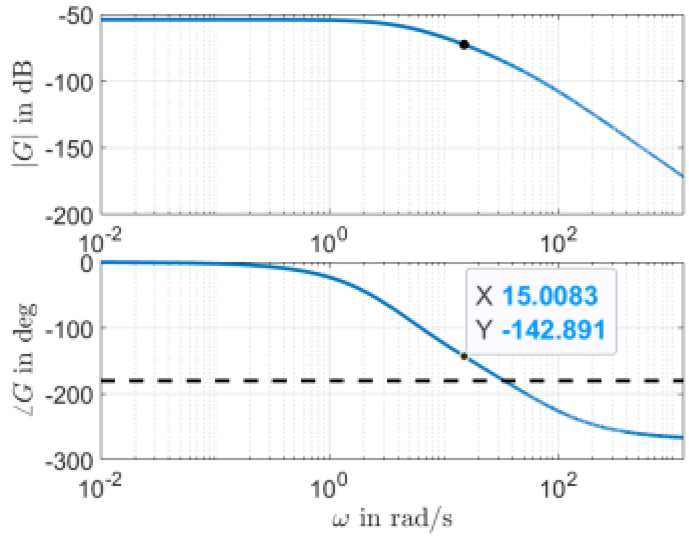


Figure 5: Bode plot

with $\alpha = 0.2$ needs to be designed for the system. The new crossover frequency (after applying the controller) is selected as $\omega_c = 15$ rad/s and the phase contribution of the Lag part at that frequency is given in radians by

$$\phi_L = \arctan\left(N_i \frac{1-\beta}{1+\beta N_i^2}\right),$$

where $N_i = \omega_c \tau_i = 3$. If a phase margin $\gamma_M = 70^o$ is desired for the open-loop system $C(s)G(s)$, what is the value of the parameter β of the Lag-part?

a) $\beta = 0$ (β must be positive)

b) $\beta = 2$ (Correct)

c) $\beta = -1.5$ (β cannot be negative)

d) $\beta = 0.3$ (Used degrees instead of radians for Lag phase)

e) $\beta = 0.62$ (Included I-part too)

Solution: Let ϕ_G, ϕ_d, ϕ_L be the phase contributions of G, Lead and Lag parts of the open-loop system at the crossover frequency, respectively. The phase of G is given from the Bode plot (or can easily be calculated in Matlab) as $\phi_G = -142.891^o$. At $\omega_c = 15$ rad/s the following

holds:

$$-180^o + \gamma_M = \phi_G + \phi_d + \phi_L \Leftrightarrow \phi_L = -180^o + \gamma_M - \phi_G - \phi_d \Leftrightarrow$$

$$\arctan\left(N_i \frac{1-\beta}{1+\beta N_i^2}\right) = -180^o + 70^o + 142.891^o - \arcsin\left(\frac{1-\alpha}{1+\alpha}\right) \Leftrightarrow$$

$$N_i \frac{1-\beta}{1+\beta N_i^2} = \tan(-8.9193^o) \Leftrightarrow \beta = \frac{N_i - \tan(-8.9193^o)}{N_i(1+N_i\tan(-8.9193^o))} \Leftrightarrow \beta = 1.9886 \approx 2$$

```
clear; clc;

alpha = 0.2;
PM = 70;
phiG = -142.891;   % degrees
Ni = 3;

% Desired total phase at crossover
phi_desired = -180 + PM;

% Required controller phase contribution
phiC = phi_desired - phiG;

% Lead phase contribution
phiD = asind((1-alpha)/(1+alpha));

% Required lag phase contribution
phiL_deg = phiC - phiD;
phiL_rad = deg2rad(phiL_deg);

% Solve for beta
syms beta positive

eq = atan(Ni*(1-beta)/(1 + beta*Ni^2)) == phiL_rad;

beta_sol = double(solve(eq, beta));

fprintf('Required controller phase = %.3f deg\n', phiC);
```

```
Required controller phase = 32.891 deg
```

```
fprintf('Lead phase = %.3f deg\n', phiD);
```

```
Lead phase = 41.810 deg
```

```
fprintf('Lag phase = %.3f deg\n', phiL_deg);
```

```
Lag phase = -8.919 deg
```

```
fprintf('beta = %.4f\n', beta_sol);
```

```
beta = 1.9886
```

Question 18

A system with transfer function

$$G(s) = \frac{0.7(s+0.5)}{(5s+1)(s^2+0.2s+0.6)(0.01s+1)}$$

has the following Bode plot

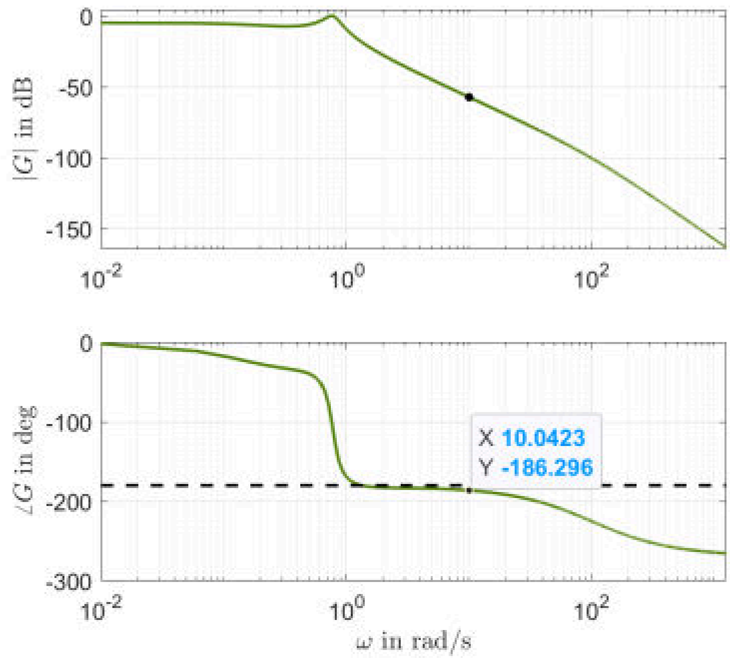


Figure 6: Bode plot

A PI-Lead controller

$$C(s) = K_P \frac{\tau_i s + 1}{\tau_i s} \frac{\tau_d s + 1}{\alpha \tau_d s + 1}$$

is to be designed so that the open-loop system $C(s)G(s)$ has phase margin $\gamma_M = 45^o$ and crossover frequency $\omega_c = 10$ rad/s. If $N_i = \omega_c \tau_i = 8$, a good selection for α and K_P is:

a) $\alpha = 0.5$ and $K_P = 500$. (Phase margin too low, 6^o)

b) $\alpha = 2$ and $K_P = 1000$. (Unstable due to negative phase margin)

c) $\alpha = 0.7$ and $K_P = 591$. (Unstable due to negative phase margin)

d) $\alpha = 0.08$ and $K_P = 200$. (Correct)

e) $\alpha = 0.07$ and $K_P = -154$. (Unstable due to negative gain)

Solution: Let ϕ_G, ϕ_d, ϕ_i be the phase contributions of G, Lead and I parts of the open-loop system at the crossover frequency, respectively. The phase of G can easily be calculated in Matlab as $\phi_G = -184.9765^o$. At ω_c the following holds:

$$-180^o + \gamma_M = \phi_G + \phi_d + \phi_i \Leftrightarrow \phi_d = -180^o + \gamma_M - \phi_G + \arctan\left(\frac{1}{N_i}\right) \Leftrightarrow$$

```
clear; clc; close all;

s = tf('s');

G = 0.7*(s + 0.5)/((5*s + 1)*(s^2 + 0.2*s + 0.6)*(0.01*s + 1));

wc = 10;
PM_desired = 45;
Ni = 8;

[magG, phaseG] = bode(G, wc);
magG = squeeze(magG);
phaseG = squeeze(phaseG);

% MATLAB may return phase wrapped, so force it near the plot value
if phaseG > 0
    phaseG = phaseG - 360;
end

phi_PI = atand(Ni) - 90;

phi_total_desired = -180 + PM_desired;

phi_lead_needed = phi_total_desired - (phaseG + phi_PI);

alpha = (1 - sind(phi_lead_needed))/(1 + sind(phi_lead_needed));

Ti = Ni/wc;
Td = 1/(wc*sqrt(alpha));

PI = (Ti*s + 1)/(Ti*s);
Lead = (Td*s + 1)/(alpha*Td*s + 1);

[magPI, ~] = bode(PI, wc);
[magLead, ~] = bode(Lead, wc);

magPI = squeeze(magPI);
magLead = squeeze(magLead);

Kp = 1/(magG*magPI*magLead);

fprintf('phaseG = %.3f deg\n', phaseG);
```

```
phaseG = -186.275 deg
```

```
fprintf('PI phase = %.3f deg\n', phi_PI);
```

```
PI phase = -7.125 deg
```

```
fprintf('Needed lead phase = %.3f deg\n', phi_lead_needed);
```

```
Needed lead phase = 58.400 deg
```

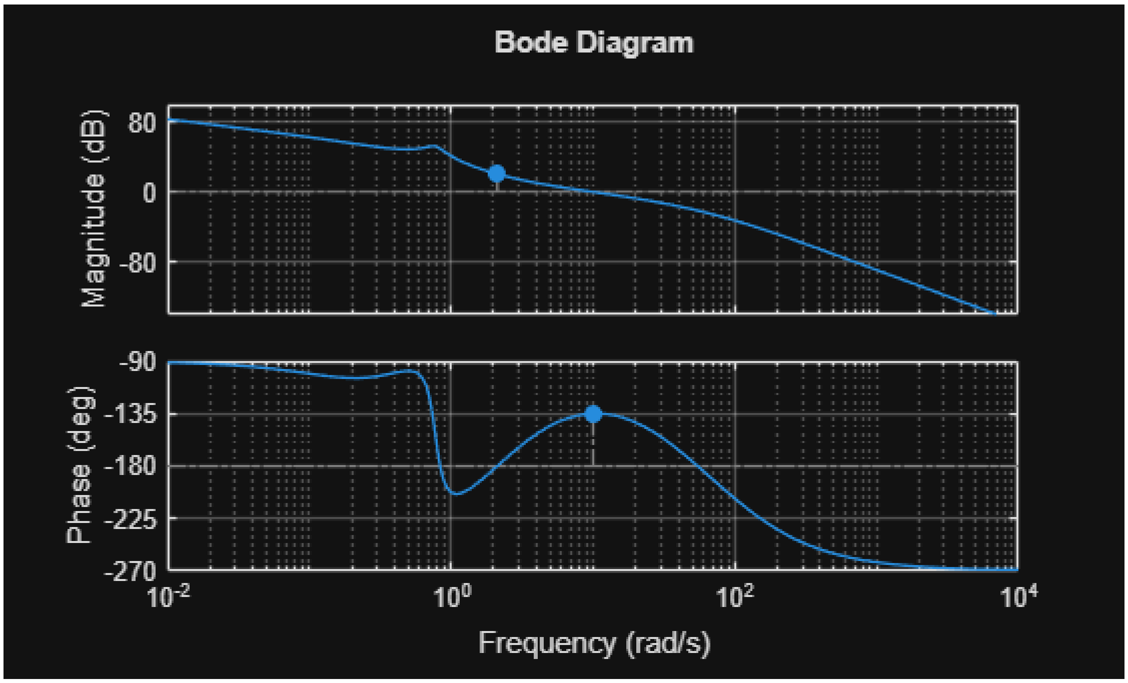
```
fprintf('alpha = %.4f\n', alpha);
```

```
alpha = 0.0801
```

```
fprintf('Kp = %.2f\n', Kp);
```

```
Kp = 200.19
```

```
C = Kp * PI * Lead;
L = C*G;

margin(L)
grid on
```



Question 19

Consider the system described by the following block diagram where $G_1(s)$ and $G_2(s)$ are type-0 systems.

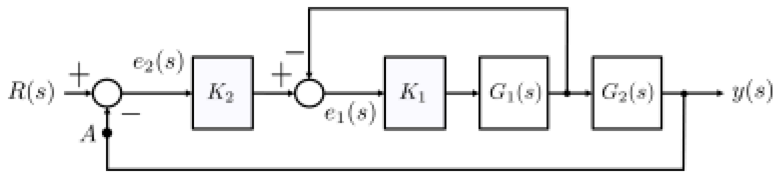


Figure 7: Block diagram.

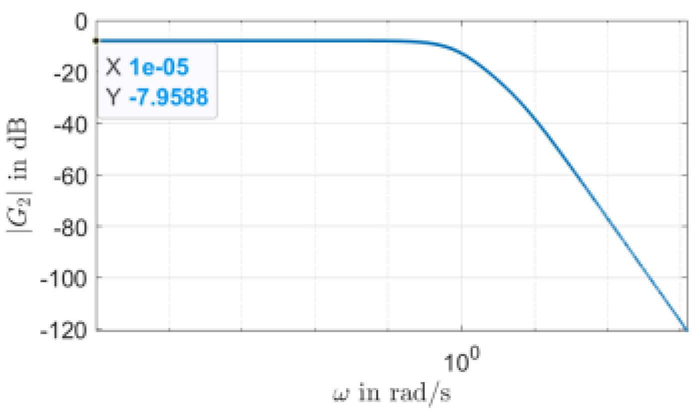


Figure 8: Magnitude Bode plot.

The magnitude Bode plot of $G_2(s)$ is shown below:

When the outer loop is opened at point A, a step change in $R(s)$ equal to $\frac{1}{K_2}$ causes a steady state error $e_1(0) = 0.4$. What should be the value of K_2, such that a unit step change in the reference $R(s)$, when the loop is closed, causes a steady state error $e_2(0) = 0.05$?

a) $K_2 = 118.75$. (Used closed-loop system DC gain instead of that from e_2 to e_1)

b) $K_2 = -3.98$. (Used dB for $G_2(0)$)

c) $K_2 = -5.97$. (Used closed-loop system DC gain instead of $G_{e_2,e_1}(0)$ and dB for $G_2(0)$)

d) $K_2 = 91.25$. (Gain margin of entire open-loop)

e) $K_2 = 79.17$. (Correct)

Solution: Let $\varepsilon_1 \triangleq e_1(0)$ and $\varepsilon_2 \triangleq e_2(0)$. The transfer function from e_2 to e_1 is

$$G_{e_2e_1}(s) = \frac{K_2}{1 + K_1G_1(s)}$$

Then

$$\varepsilon_1 = \lim_{s\to 0} sG_{e_2e_1}(s)\frac{1}{sK_2} \Leftrightarrow \varepsilon_1 = \frac{1}{1 + K_1G_1(0)} \Leftrightarrow K_1G_1(0) = \frac{1-\varepsilon_1}{\varepsilon_1} \quad (1)$$

Moreover, the transfer function $G_{d,1}(s)$ from the output of the gain K_2 to the output of $G_1(s)$ is given by

$$G_{d,1}(s) = \frac{K_1G_1(s)}{1 + K_1G_1(s)} \Rightarrow G_{d,1}(0) = \frac{K_1G_1(0)}{1 + K_1G_1(0)} \overset{(1)}{=} \frac{\frac{1-\varepsilon_1}{\varepsilon_1}}{\frac{1-\varepsilon_1}{\varepsilon_1}} = 1 - \varepsilon_1 \quad (2)$$

```
clear; clc;

% Fra Bode-plottet for G2:
G2_dB = -7.9588;                  % |G2(0)| in dB
G2_0 = 10^(G2_dB/20);             % Convert dB to linear gain

% Givet:
e1_ss = 0.4;                      % steady-state error in inner loop
e2_ss_desired = 0.05;             % desired outer loop steady-state error

% Når outer loop er åben, input til inner loop er 1,
% fordi R step = 1/K2 og derefter gennem K2 bliver det 1.
%
% Derfor:
% inner output DC gain = 1 - e1_ss

T_inner_0 = 1 - e1_ss;

% For outer closed loop:
% e2_ss = 1 / (1 + K2*T_inner_0*G2_0)
%
% Solve for K2:

K2 = (1/e2_ss_desired - 1)/(T_inner_0 * G2_0);

fprintf('G2(0) = %.4f\n', G2_0);
```

```
G2(0) = 0.4000
```

```
fprintf('Inner closed-loop DC gain = %.4f\n', T_inner_0);
```

```
Inner closed-loop DC gain = 0.6000
```

```
fprintf('K2 = %.4f\n', K2);
```

```
K2 = 79.1667
```

Question 20

Consider the system described by the following block diagram,

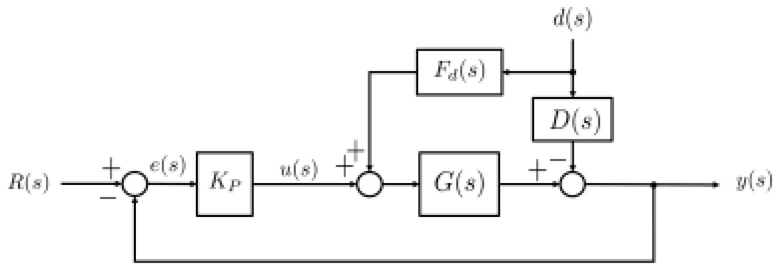


Figure 9: Block diagram.

where $K_P = 1$, $d(s)$ is an external disturbance and the transfer functions $G(s)$ and $D(s)$ are defined as

$$G(s) = \frac{21(0.5s+1)}{s^2+4s+21}$$
$$D(s) = \frac{1}{0.01s+1}$$

If $F_d(s)$ is an unfiltered dynamic feed-forward controller designed to compensate for the disturbance, then the sensitivity function related to the disturbance $d(s)$ is:

a) $G_{yd}(s) = 0$. (Correct)

b) $G_{yd}(s) = \frac{100s^2+400s+2100}{s^3+114.5s^2+1492s+4200}$. (Used $\frac{D}{1+K_PG}$)

c) $G_{yd}(s) = -\frac{100s^2+400s+2100}{s^3+114.5s^2+1492s+4200}$. (Used $\frac{-D}{1+K_PG}$)

d) $G_{yd}(s) = \frac{100}{s+1}$. (Used F_dG)

e) $G_{yd}(s) = \frac{10.5s^2+1071s+2100}{s^3+104s^2+1471s+4200}$. (Used $\frac{D}{1+DG}$)

Solution: The dynamic feed-forward controller is given by

$$F_d(s) = \frac{D(s)}{G(s)} = \frac{s^2+4s+21}{21(0.5s+1)(0.01s+1)}$$

Since the ratio is a stable and proper transfer function, we do not need any filtering. This means that $F_d(s)G(s) = D(s)$ at all frequencies. The sensitivity of the output to the disturbance, i.e. transfer function from $d(s)$ to $y(s)$ is given by

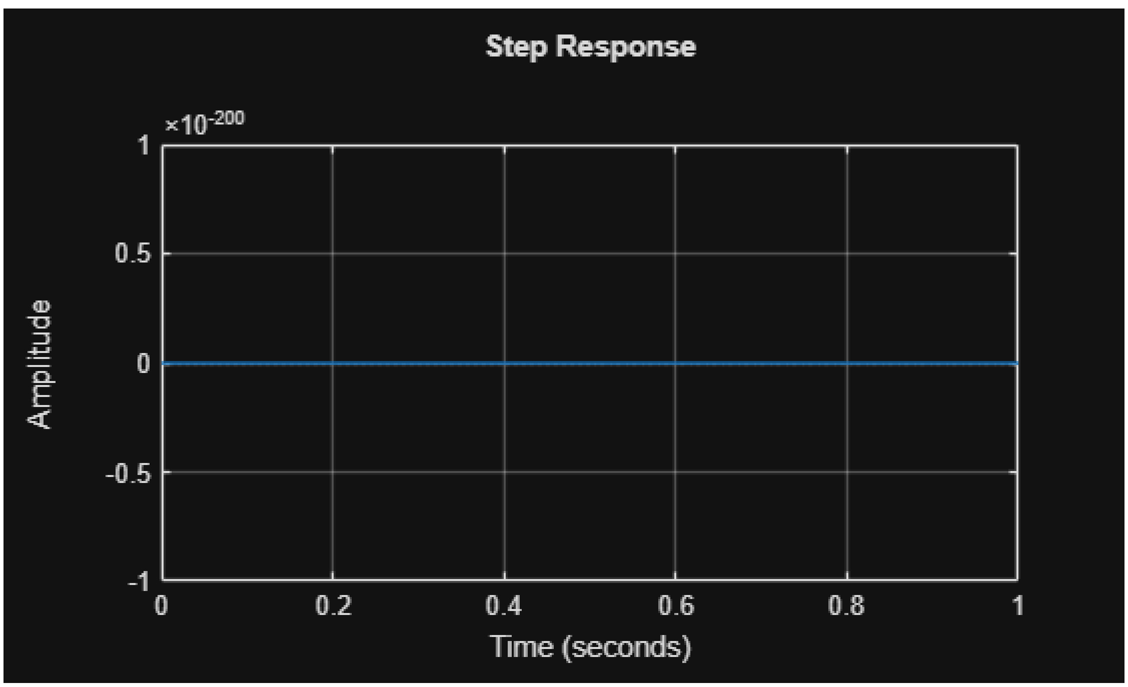
$$G_{yd}(s) = \frac{-D(s)+F_d(s)G(s)}{1+K_PG(s)} = 0\,.$$

Therefore, the only acceptable option is (a).

```
clear; clc;

s = tf('s');

Kp = 1;

G = 21*(0.5*s + 1)/(s^2 + 4*s + 21);
D = 1/(0.01*s + 1);

Fd = -D/G;

Gyd = minreal((G*Fd + D)/(1 + Kp*G))
```

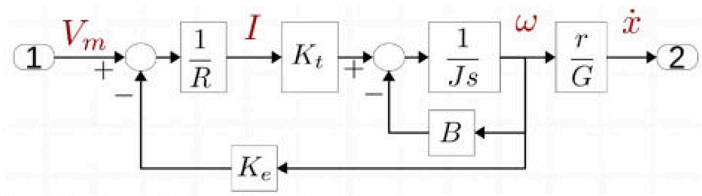
```
Gyd =

  0

Static gain.
Model Properties
```

```
step(Gyd)
grid on
```



S20 Question 1

A brushless motor drives a skateboard with a person of 58 Kg. The motor is controlled with a voltage V_m between 0 and 22 volts, and the system is modelled with the following block diagram:



The following values have been found:
K_t = 0.035 N m A^{-1} (constant torque motor),
K_e = 0.035 Vs (motor back-emf constant),
R = 0.05 Ω (motor internal resistance),
J = 0.015 Kg m^2 (inertia from, among other things, person weight),
r = 0.05 m (wheel radius),
G = 3.2 (gearing),
I is the motor current [A], Vm = is the motor voltage [V], ω is the motor angular velocity [rad s-1] and $\dot{x}$ is speed [m s^{-1}].
Thus, only the friction B is missing, which is determined on the basis of a measurement at the operating point. Operating point is $\dot{x}$ = 3 m s^{-1}. At this speed the motor draws I = 8.5 A at a motor voltage of V_m = 7.2 V.

What value does B have at this operating point?

At steady state, the input to integrator:

$$I_0 K_t = B\omega_0$$

and

$$\omega_0 = \dot{x}_0 \frac{G}{r} \Rightarrow$$

$$I_0 K_t = B\dot{x}_0 \frac{G}{r} \Rightarrow$$

$$B = \frac{I_0 K_t}{\dot{x}_0 \frac{G}{r}} \Rightarrow B = \frac{8.5 \times 0.035}{3\frac{3.2}{0.05}} \Rightarrow \qquad B = 0.0016$$

Select an answer option:

- ❑ B = 0.1
- ❑ B = 57.7
- ★ B= 0.0016
- ❑ B=0.014

```
% Given values
Kt = 0.035;   % Nm/A
I  = 8.5;     % A
r  = 0.05;    % m
G  = 3.2;     % gearing
xdot = 3;     % m/s

% Motor angular velocity at operating point
omega = xdot * G / r;

% Friction coefficient
B = (I * Kt) / omega;

fprintf('omega = %.2f rad/s\n', omega);
```

```
omega = 192.00 rad/s
```

```
fprintf('B = %.4f\n', B);
```

```
B = 0.0015
```

A steam engine is fed with steam through a valve.
Where:
ω is the engine angular velocity in rad/sec,
a is the valve opening - between 0 and 1,
J = 0.23 Kg m^2 (inertia of steam engine and load),
B = 0.12 N m s (load and friction),
K = 0.056 N m /$\sqrt{Pa}$ (system constant),
P = 3 x 105 Pa (steam overpressure),
H = 1600 Pa s (pressure drop due to flow)
The following (simplified) relationship between valve position (a) and the steam engine angular velocity (ω) is stated:

$$J\dot{\omega} + B\omega = a\,K\,\sqrt{P - H\omega}$$

What will be the transfer function at an operating point where the motor is running at 600 rpm (revolutions per minute)?

- ○ $G(s) = \frac{\omega}{a} = \frac{0.26}{s+0.52}$
- ○ $G(s) = \frac{\omega}{a} = \frac{0.11}{s+0.52}$
- ○ $G(s) = \frac{\omega}{a} = \frac{0.08}{s+195}$
- ● $G(s) = \frac{\omega}{a} = \frac{109}{s+0.65}$
- ○ $G(s) = \frac{\omega}{a} = \frac{25}{s+0.15}$

Operating point 600 RPM

$$\omega_0 = \frac{600 \times 2\pi}{60} = 62.8\,\mathrm{rad/s}$$

And corresponding valve position (steady state)

$$B\omega_0 = K a_0 \sqrt{P - H\omega_0}$$

$$a_0 = \frac{B\omega_0}{K\sqrt{P - H\omega_0}} \Rightarrow a_0 = \frac{0.12 \times 62.8}{0.056\sqrt{300000 - 1600 \times 62.8}} \Rightarrow a_0 = 0.3$$

Linearizing for variables $\dot{\omega}\ \omega\ a$

$$J\Delta\dot{\omega} + B\Delta\omega = K\sqrt{P - H\omega_0}\Delta a - 0.5\frac{K a_0 H}{\sqrt{P - H\omega_0}}\Delta\omega$$

Insert values and Laplace transform

$$0.23s\omega + 0.12\omega = 25a - 0.03\omega$$

Isolate and normalize s in denominator

$$G(s) = \frac{\omega}{a} = \frac{25}{0.23s + 0.12 + 0.03}$$

```matlab
% Given values
J = 0.23;         % kg m^2
B = 0.12;         % N m s
K = 0.056;        % N m / sqrt(Pa)
P = 3e5;          % Pa
H = 1600;         % Pa s

% Operating point: 600 rpm
omega0 = 600 * 2*pi / 60;   % rad/s

% Pressure term
Q = sqrt(P - H*omega0);

% Steady-state valve opening a0
a0 = (B*omega0) / (K*Q);

% Linearized transfer function:
% J*d(omega)/dt + (B + a0*K*H/(2*Q))*omega = K*Q*a

num = K*Q / J;
den_const = (B + a0*K*H/(2*Q)) / J;

G = tf(num, [1 den_const])
```

```
G =

    108.7
  ----------
  s + 0.6532

Continuous-time transfer function.
Model Properties
```

```matlab
fprintf('omega0 = %.2f rad/s\n', omega0);
```

```
omega0 = 62.83 rad/s
```

```matlab
fprintf('a0 = %.3f\n', a0);
```

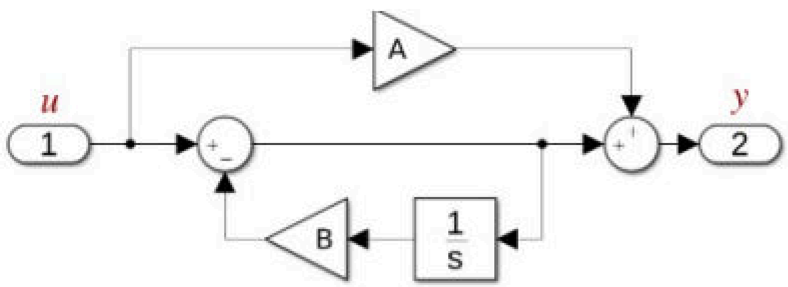
```
a0 = 0.301
```

```matlab
fprintf('G(s) = %.2f / (s + %.2f)\n', num, den_const);
```

```
G(s) = 108.74 / (s + 0.65)
```

S20 Question 3

What is the transfer function for this block diagram?



There is a sum of two branches.

The upper branch

$$G_U(s) = A$$

The lower branch is a loop, that can be

$$G_L(s) = \frac{1}{1 + \frac{B}{s}}$$

$$G_L(s) = \frac{s}{s + B}$$

In total

$$G(s) = G_U(S) + G_L(s)$$

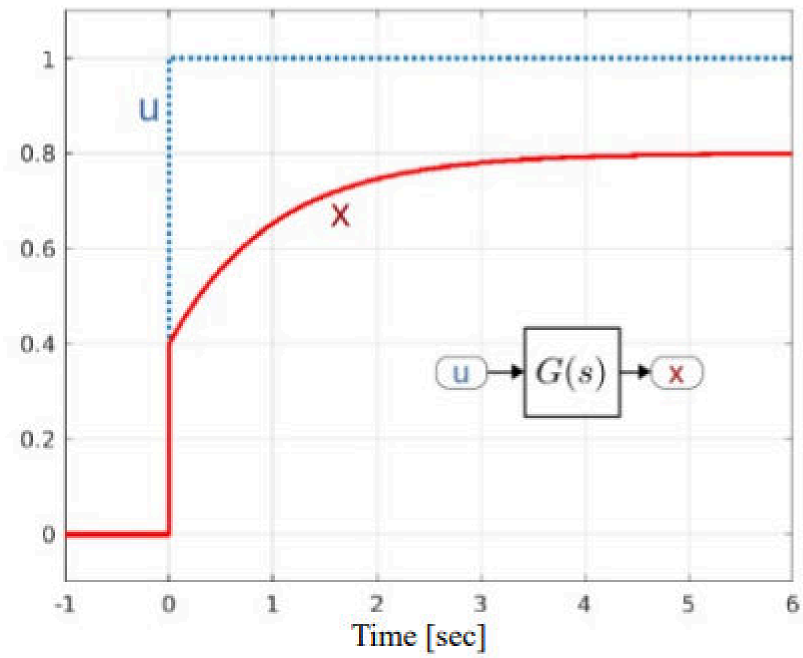
$$G(s) = A + \frac{s}{s + B}$$

$$G(s) = \frac{A(s + B) + s}{s + B}$$

$$G(s) = \frac{(1 + A)s + AB}{s + B}$$

S20 Question 4

A system has the following step response



What type of transfer function can provide this response?

Select an answer option:

- $G(s) = \dfrac{b_0}{s + a_0}$
- $G(s) = \dfrac{b_0}{s^2 + a_1 s + b_0}$
- $G(s) = \dfrac{b_1 s + b_0}{s + a_0}$
- $G(s) = \dfrac{s + b_0}{s^2 + a_1 s + a_0}$

The initial value is 0.4 and the initial value can be found as

$x(t = 0) = \lim_{s \to \infty} (s\ u(s) G(s))$

And for a unit step $\quad u = \dfrac{1}{s}$

$x(t = 0) = \lim G(s) = 0.4$

$$\boxed{G(s) = \frac{b_1 s + b_0}{s + a_0}}$$

So the **starred answer is correct.**

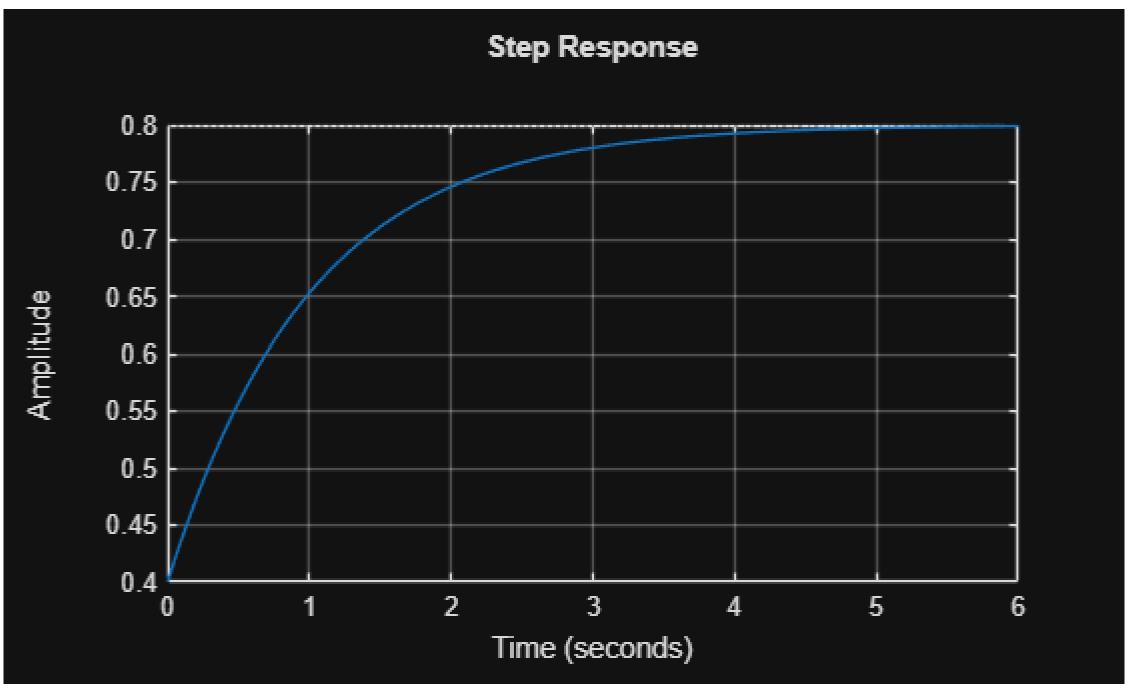
Reason: the output x has an **instant jump** at $t = 0$, then continues with a **first-order exponential rise.** An instant jump means the transfer function must have a numerator with an s-term, giving direct feedthrough.

A first-order transfer function without an s-term,

$$G(s) = \frac{b_0}{s + a_0}$$

would start from zero with no jump, so it cannot match the plot.

```
s = tf('s');

a0 = 1;
b1 = 0.4;
b0 = 0.8;

G = (b1*s + b0)/(s + a0);

step(G)
grid on
```



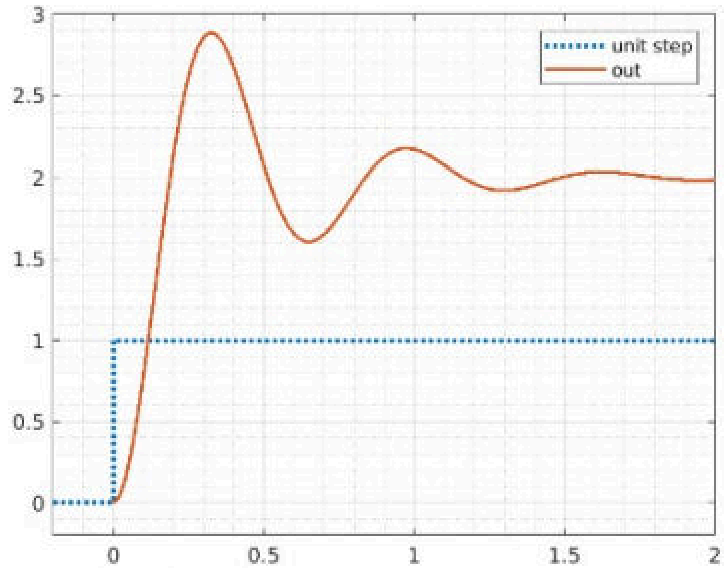
S20 Question 5

A system with the transfer function

$$G(s) = \frac{200}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

and $\omega_n = 10\ \mathrm{rad\,s^{-1}}$

has the following unit step response:



Where the peak value is 2.9.

What value has ζ in the transfer function?

Select an answer option:

- $\zeta = 0.1$
- $\zeta = 0.2$
- $\zeta = 0.4$
- $\zeta = 0.7$

Damping factor ζ determines the overshoot

The overshoot is:

$$M_P = \frac{0.9}{2} = 0.45 \quad (45\%)$$

This corresponds to ζ 0.2 (see slides).

Or it can be calculated from

$$M_P = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}}$$

By using MATLAB:

$\zeta = 0.2$

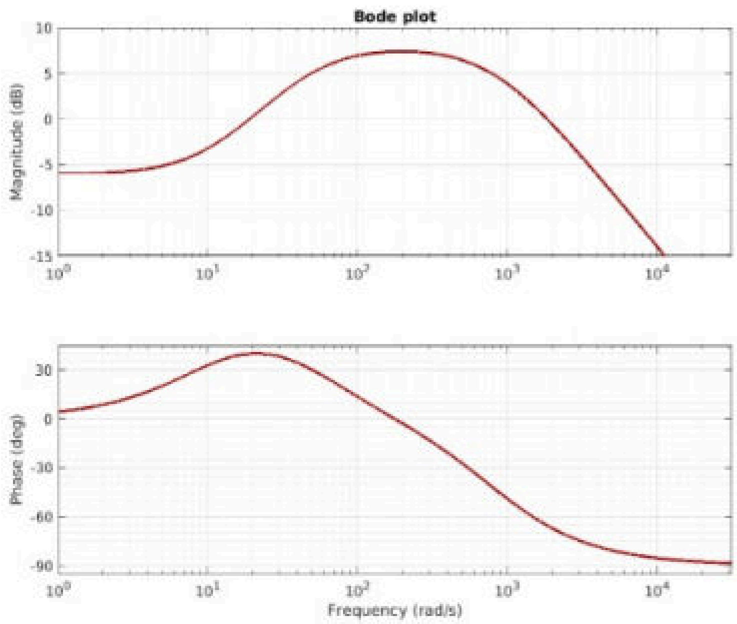
Select an answe

```
clear; clc;

Mp = (2.9 - 2)/2;

zeta = -log(Mp) / sqrt(pi^2 + log(Mp)^2);

fprintf('zeta = %.3f\n', zeta)
```

```
zeta = 0.246
```

S20 Question 6

A transfer function has the following bode plot

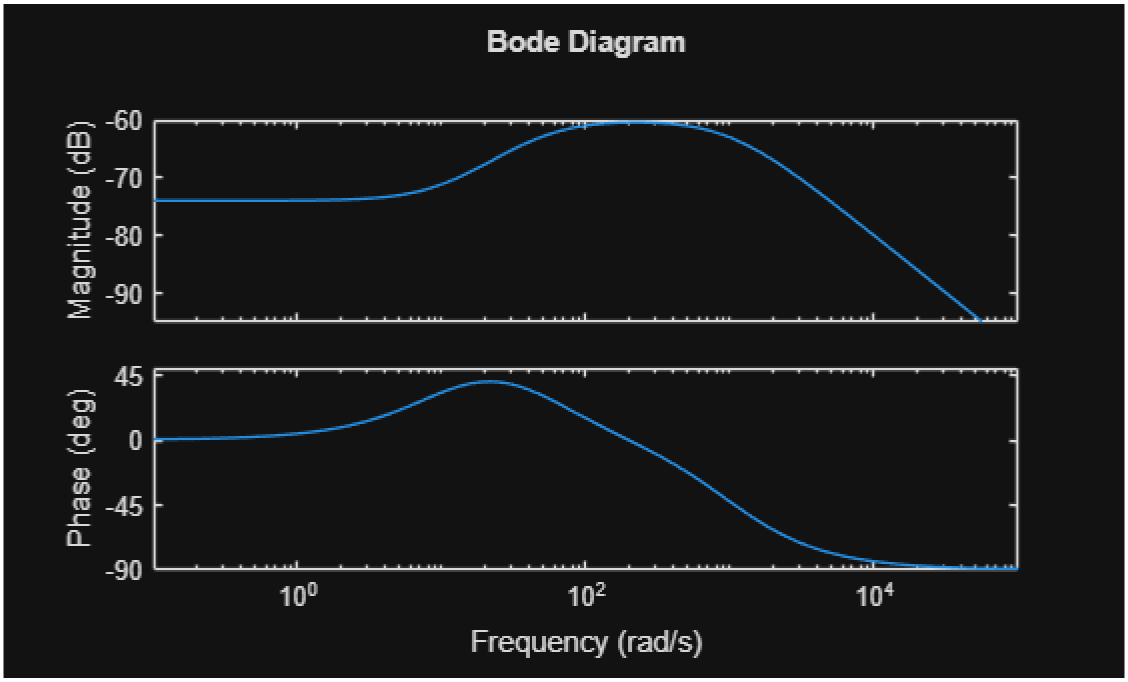


Where does the transfer function have poles and zeros?

Select an answer option:

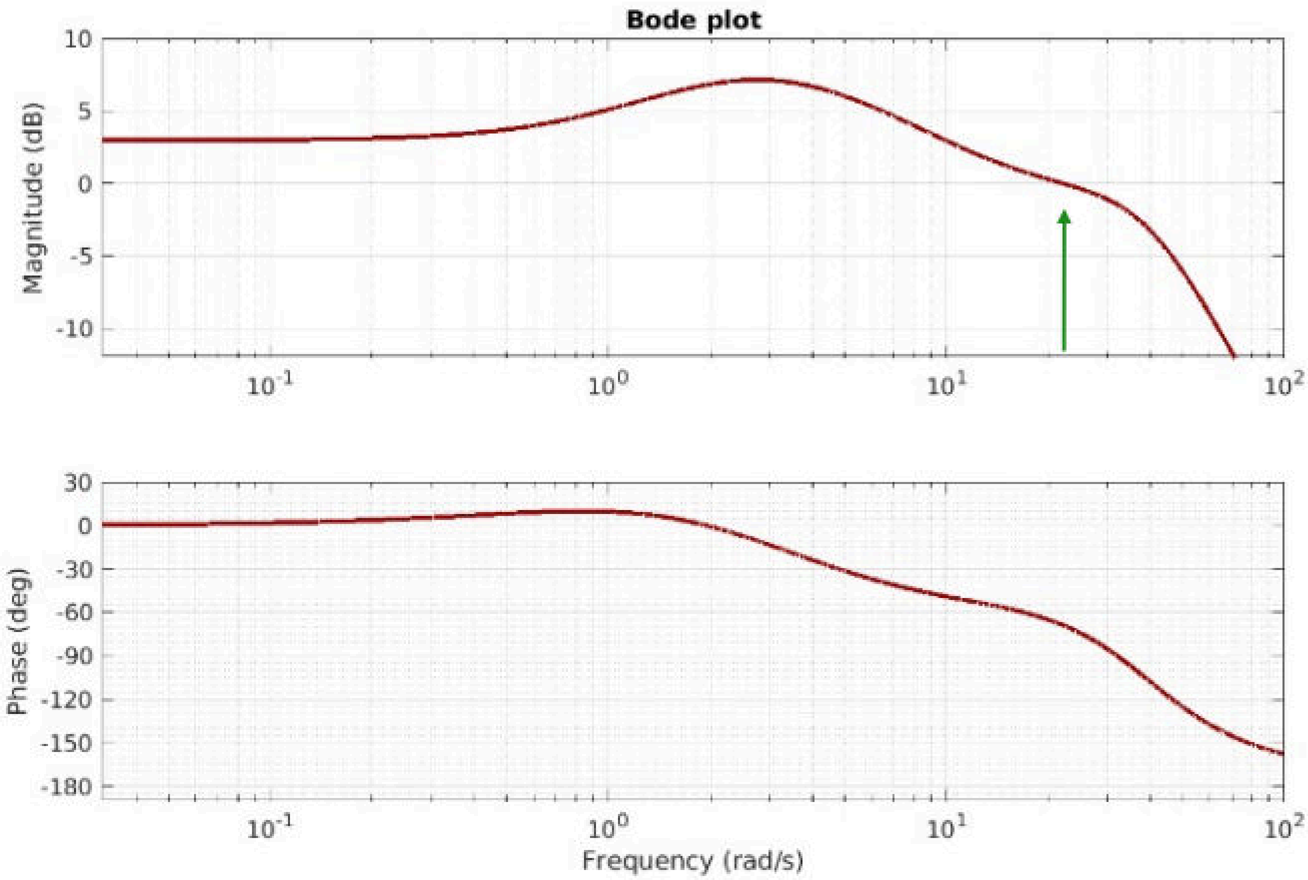
- ❑ Poles s=-50 and s=-1000
- ❑ Poles s=-10 and s=-1000
- ❑ Poles s=-50 and s=-200
- ❑ Zeros s=-10
- ❑ Zeros s = -10 s=-200
- ❑ Zeros s=-10, s=-1000 s=-5000
- ❑ No zero points

```
clear; clc;

s = tf('s');

G = (s+10) / ((s+50)*(s+1000));

bode(G)
```



S20 Question 7

What is the bandwidth (BW) for this (closed loop) transfer function?



Select an answer option:

- ○ $\mathrm{BW} = 1.4\ \mathrm{rad\,s^{-1}}$
- ★ $\mathrm{BW} = 22\ \mathrm{rad\,s^{-1}}$
- ○ $\mathrm{BW} = 10\ \mathrm{rad\,s^{-1}}$
- ○ $\mathrm{BW} = 39\ \mathrm{rad\,s^{-1}}$
- ○ $\mathrm{BW} = 2.7\ \mathrm{rad\,s^{-1}}$

The bandwidth is where the amplitude is fallen to -3 dB compared to the low frequency asymptote.

This starts at +3 dB, so crossing of 0 dB first time is the bandwidth.

$$\boxed{BW = 22\ \mathrm{rad/s}}$$

For a **closed-loop transfer function**, the bandwidth is found at the frequency where the magnitude has dropped **3 dB below the low-frequency/DC gain.**

From the plot:

$$|G(0)| \approx 3\ \mathrm{dB}$$

So the bandwidth occurs when:

$$|G(j\omega)| \approx 3 - 3 = 0\ \mathrm{dB}$$

The plot crosses about **0 dB** at roughly:

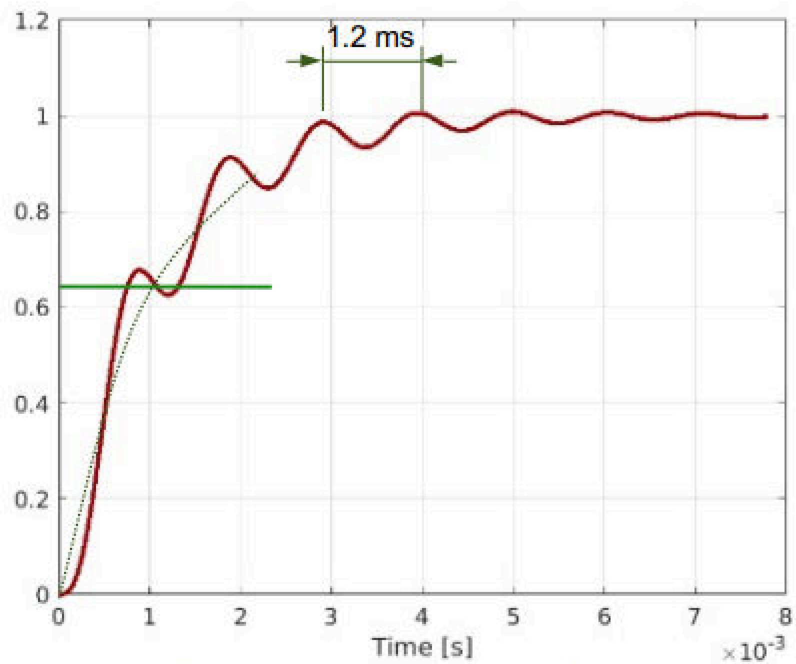
$$\omega \approx 22\ \mathrm{rad/s}$$

So the starred answer is correct:

$$\boxed{BW = 22\ \mathrm{rad/s}}$$

S20 Question 8

A closed loop system has the following unit step response



Which of the following closed loop amplitude Bode plot fits with this response?

Step analysis:
Ignoring the fast damped frequency, it looks like a 1st order pole system. The amplitude rises fast at time 0 and levels out exponentially towards 1, and it has reached 63% after about 1ms.

A 1st order break frequency of 1000 rad/s.

The damped fast frequency must correspond to an amplitude peak at this frequency.

$$T \approx 1.2 \times 10^{-3} \text{ sec}$$

$$\omega_n = \frac{2\pi}{T} \Rightarrow \omega_n \approx \frac{2\pi}{1.2 \times 10^{-3}} = 5236 \text{ rad/sec}$$

```
T = 1.2*10^-3
```

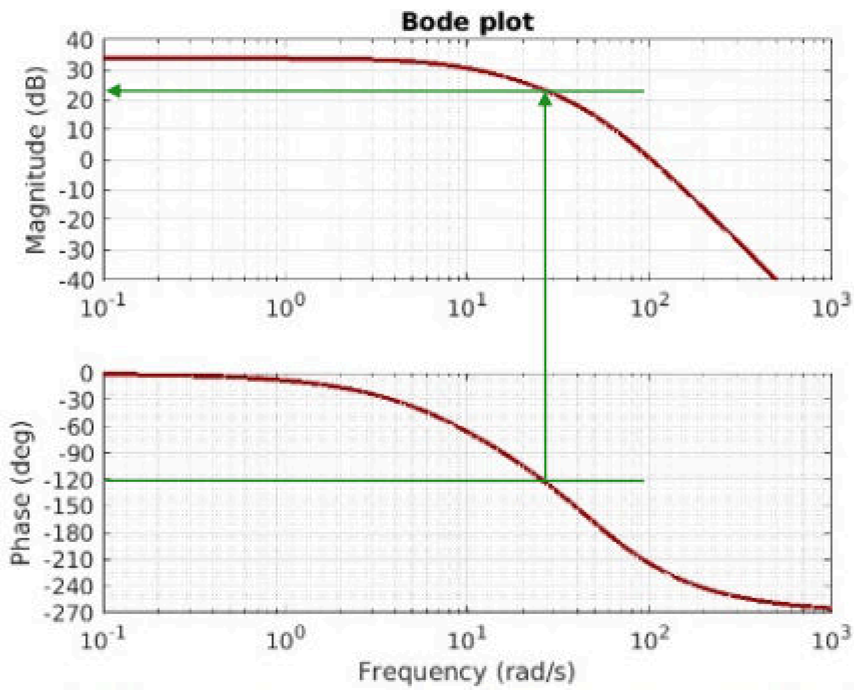
```
T =
0.0012
```

```
omega_n = (2*pi) / T
```

```
omega_n =
5.2360e+03
```

S20 Question 9 - design 1

A system to be regulated has the following Bode plot



You are asked to design a P-controller to achieve a fast and stable system. What value should Kp have?

Select an answer option:

- $K_P = 0.06$
- $K_P = 26$
- $K_P = 16$
- The system does not become stable with a P-controller
- $K_P = 100$

P controller design.

With phase margin of 60 degrees, the crossover frequency should be

$$\gamma_M = 60^o$$

$$\omega_c \approx 25 \text{ rad/sec}$$

The magnitude at this frequency is

$$|G(\omega = 25)| \approx 23dB$$

K_p must then be -23 dB

$$K_P = -23dB$$

$$K_P = 10^{\frac{-23}{20}} = 0.0708$$

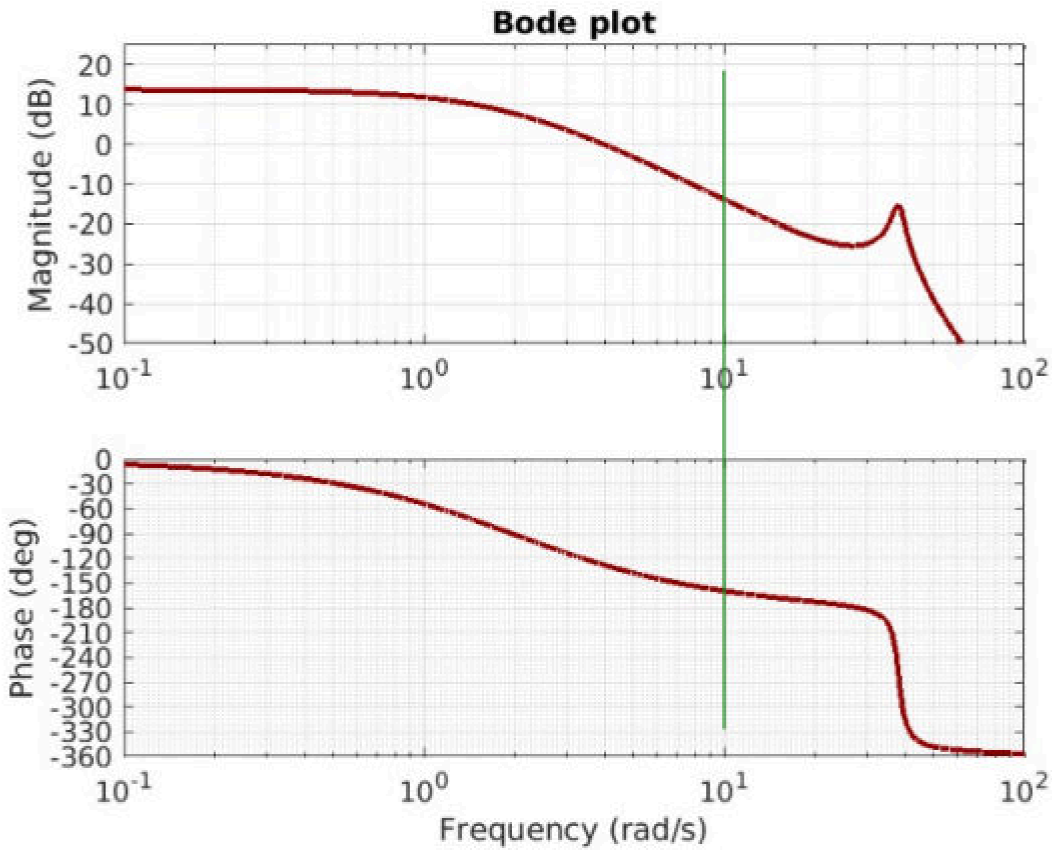
```
mag_dB = 25;

Kp = 10^(-mag_dB/20);

fprintf('Kp = %.3f\n', Kp)
```

```
Kp = 0.056
```

S20 Question 10 - design 2

A system to be regulated has the following transfer function and Bode Plot

$$G(s) = \frac{28880}{s^4 + 7.04s^3 + 1460s^2 + 5788s + 5776}$$



A crossover frequency is required $\omega_c = 10$ rad / sec

To improve the phase margin, a Lead term with $\alpha = 0.1$ is designed.

What will be the zero point frequency of the Lead term?

Select an answer option:

- $\omega = 0.32$ rad/ sec
- $\omega = 3.2$ rad/ sec
- $\omega = 1$ rad/ sec
- $\omega = 6.5$ rad/ sec

Lead design

$$\tau_d = \frac{1}{\omega_c\sqrt{\alpha}} \Rightarrow \tau_d = \frac{1}{10\sqrt{0.1}} = 0.32 \text{ sec}$$

As is the time constant for the Lead zero

$$\omega_z = \frac{1}{\tau_d} \Rightarrow \omega_z = \frac{1}{0.32} = 3.1 \text{ rad/sec}$$

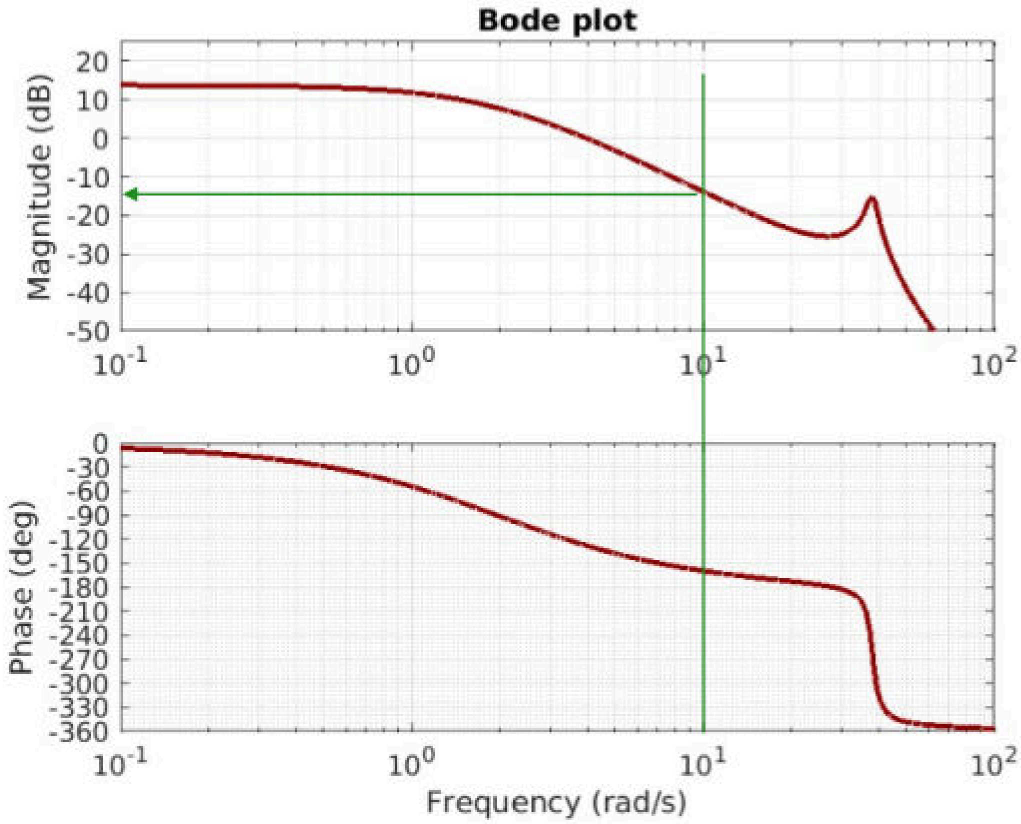
```
clear; clc;

omega_c = 10; %rad/s
alpha = 0.1;

tau_d = 1/(omega_c * sqrt(alpha));
omega_z = 1/tau_d
```

```
omega_z =
3.1623
```

S20 Question 11 - Design 2

A system to be regulated has the following transfer function and Bode plot

$$G(s) = \frac{28880}{s^4 + 7.04s^3 + 1460s^2 + 5788s + 5776}$$



A crossover frequency of $\omega_c = 10$ rad/sec is desired.

To improve the phase margin, a Lead term with $\alpha = 0.1$ is planned.

When this Lead term is part of a P-Lead regulator, what value should K_p have?

Select an answer option:

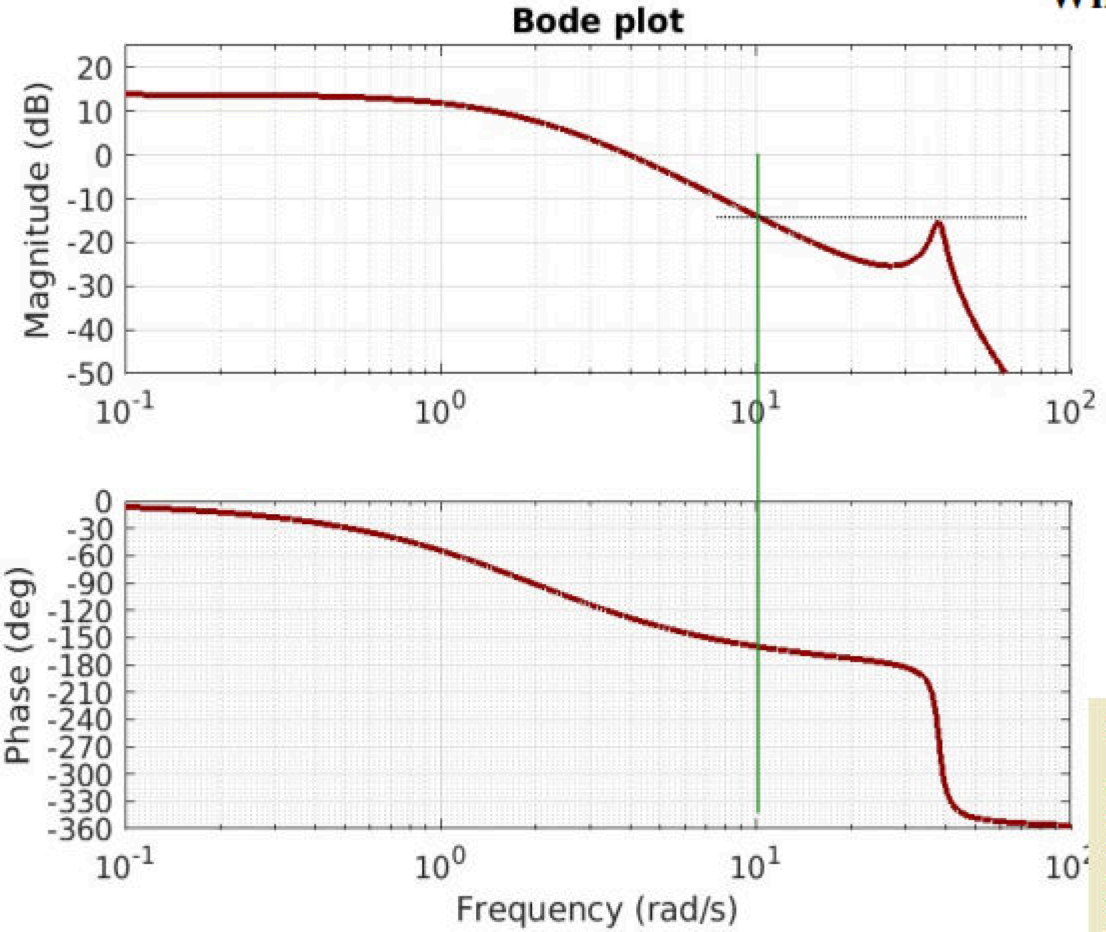
- $K_P = 1.5$
- $K_P = 5$
- $K_P = 14$
- $K_P = 0.76$

```
clear; clc;
s = tf('s');
G = 28880 / (s^4+7.04*s^3+1460*s^2+5788*s+5776);
wc = 10;
al = 0.1;
td = 1/(wc*sqrt(al));
Gd = (td*s+1) / (td*al*s+1);
[M,P] = bode(G*Gd, wc);
kp0 = 1/M
```

```
kp0 =
1.5307
```

S20 Question 12 - Design 2

A crossover frequency of
To improve the phase marg
Will the system be stable



The Lead term inc
frequency around
then also the magn
This peak will the
the phase is below
- You can check th

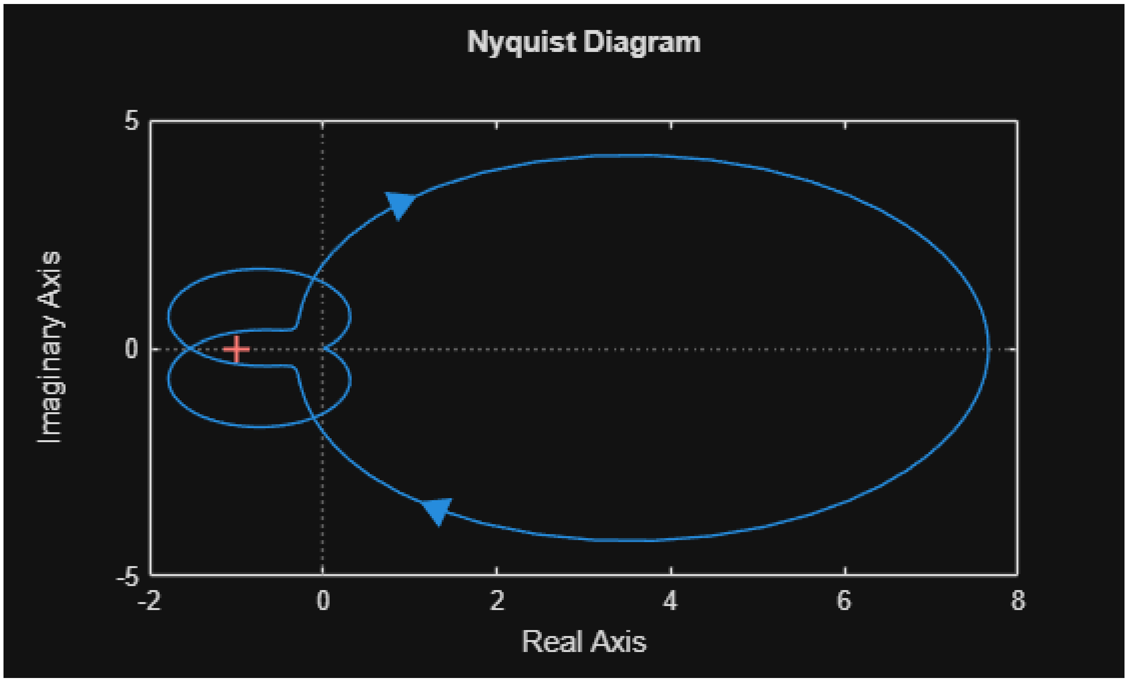
Could also be solved by MATLAB

```
G = tf([28880],[1 7.09 1460 5788 577
wc = 10;
al = 0.1;
td = 1/(wc*sqrt(al));
Gd = tf([td 1],[td*al 1]);
[M,P] = bode(G*Gd, wc);
kp0 = 1/M
% Nyquist plot
figure(922)
nyquist(G*Gd*kp0)
```

```
clear; clc;
s = tf('s');
G = 28880 / (s^4+7.04*s^3+1460*s^2+5788*s+5776);
wc = 10;
al = 0.1;
td = 1/(wc*sqrt(al));
Gd = (td*s+1) / (td*al*s+1);
[M,P] = bode(G*Gd, wc);
kp0 = 1/M
```

```
kp0 =
1.5307
```

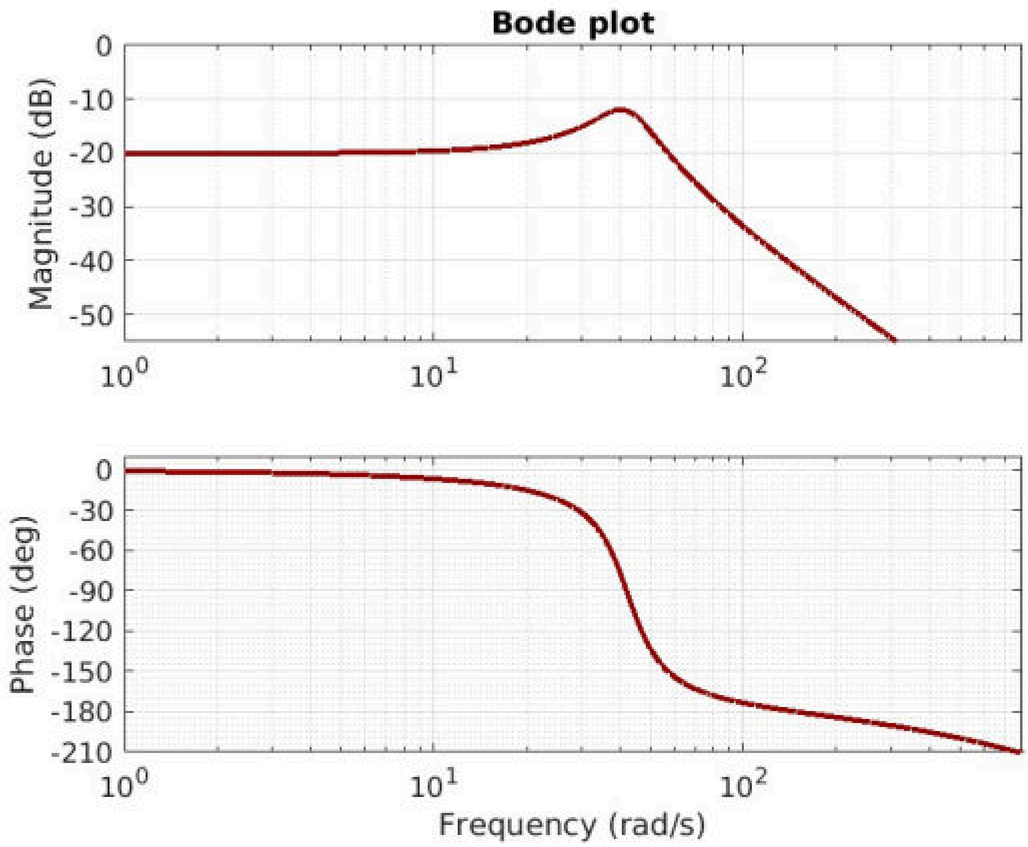
```
% Nyquist plot
figure(922)
nyquist(G*Gd*kp0)
```



S20 Question 13 - Design 3

A system to be regulated has the following transfer function and Bode plot

$$\frac{229320}{s^3 + 1317s^2 + 23604s + 2.29 \times 10^6}$$



The system h
them comple

With poles onl
magnitude pea

Select an answer o

- ❑ Yes
- ❑ No

```
clear; clc;

syms s;
den = 229320;
num = s^3+1317*s^2+23604*s+2.29*10^6;
G = den/num;

poles(G)
```

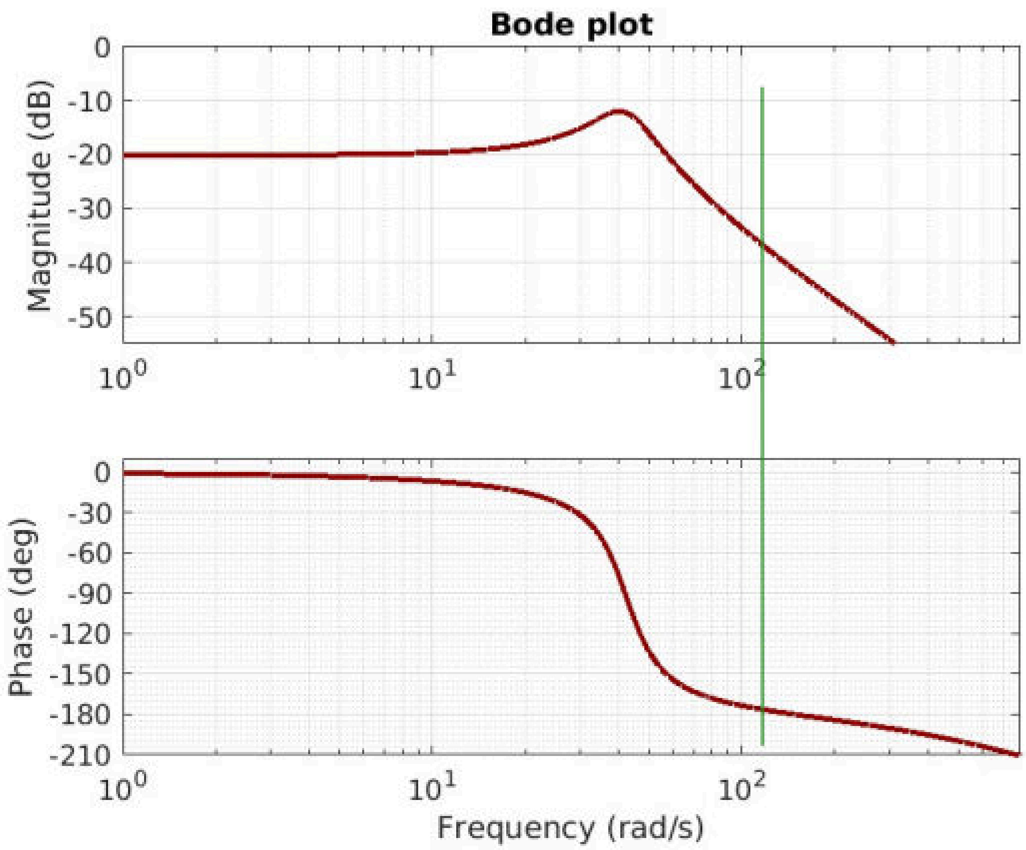
ans =

$$\begin{pmatrix} -8.3997557910812011259996240265779 + 41.118255536215574705237052422106\,i \\ -8.3997557910812011259996240265779 - 41.118255536215574705237052422106\,i \\ -1300.2004884178375977480007519468 \end{pmatrix}$$

S20 Question 14 - Design 3

A system to be regulated has the following transfer function and Bode plot

$$\frac{229320}{s^3 + 1317s^2 + 23604s + 2.29 \times 10^6}$$



A controller must be des
An I-link should elimina
term should be able to c
frequency than the ampl
is acceptable. The follow
$N_i = 3$, $\alpha = 0.1$, $\gamma_M = 40°$
What value should K_p b

Normal design job

$\varphi_{PI} = -90 + \text{arc}$

$\varphi_m = \arcsin \frac{1 -}{1 +}$

So, the crossing fre

$\angle G(\omega_c) = -180$

$\angle G(\omega_c) = -180$

$\omega_c = 122 \text{ rad/se}$

$\tau_d = \frac{1}{122\sqrt{0.1}} =$

I would use MATL

```
G = tf([229320],[1 1317 236
td = 0.026;
ti = 0.025;
al = 0.1;
Gd = tf([td 1],[al*td 1]);
Gi = tf([ti 1],[ti 0]);
[M,P] = bode(G*Gi*Gd,122)
Kp = 1/M
```

```
clear; clc;

s = tf('s');
den = 229320;
num = s^3+1317*s^2+23604*s+2.29*10^6;
G = den/num;

Ni = 3;
al = 0.1;
gamma_m = 40;
phi_PI = -90 + atand(Ni);
```

```
phi_m = asind((1-al) / (1+al));
degG = -180 + gamma_m - phi_m - phi_PI;

% Find omega hvor phase(G) = deg
w = logspace(-2, 5, 10000);

[mag, phase] = bode(G, w);
mag = squeeze(mag);
phase = squeeze(phase);

% Find punktet hvor fasen er tættest på deg
[~, idx] = min(abs(phase - degG));

omega_c = w(idx);

tau_d = 1/(omega_c * sqrt(al));
tau_i = Ni/omega_c;

Gd = (tau_d*s+1) / (tau_d*al*s+1)
```

```
Gd =

  0.02597 s + 1
  --------------
  0.002597 s + 1

Continuous-time transfer function.
Model Properties
```

```
Gi = (tau_i*s+1) / (tau_i*s)
```

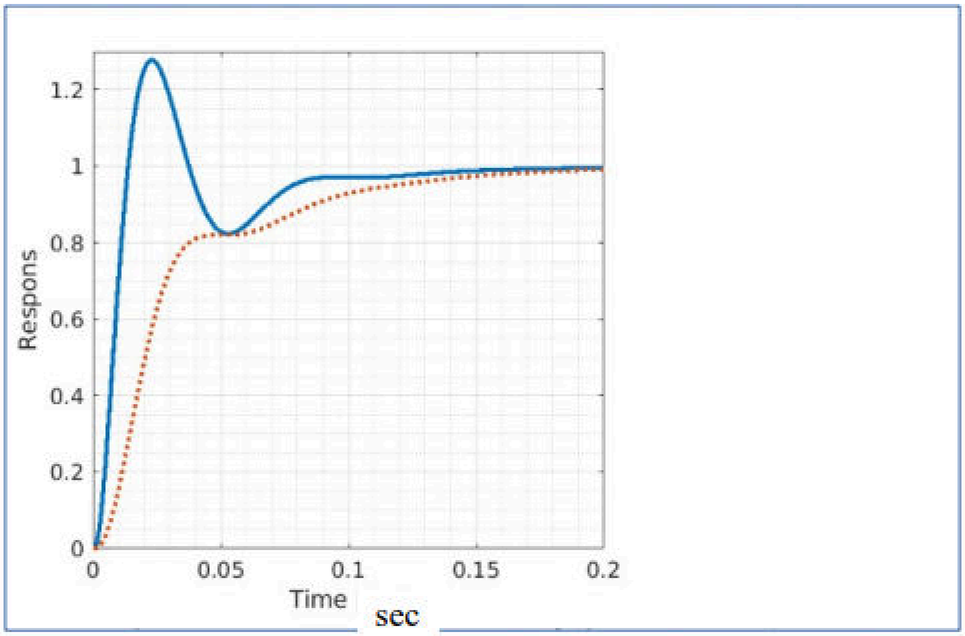
```
Gi =

  0.02463 s + 1
  -------------
    0.02463 s

Continuous-time transfer function.
Model Properties
```

```


[M,P] = bode(G*Gi*Gd,omega_c);
Kp = 1/M
```

```
Kp =
22.6028
```

S20 Question 15 - Design 3

The following closed loop unit-step response is obtained with either the lead term in the forward or back branches.



Which of the couplings fits with the blue (solid) curve?

A Lead enhances cl
branch, it will enha
This gives a steepe
the same lead in the

The blue curve mu
forward branch.

Select an answer o

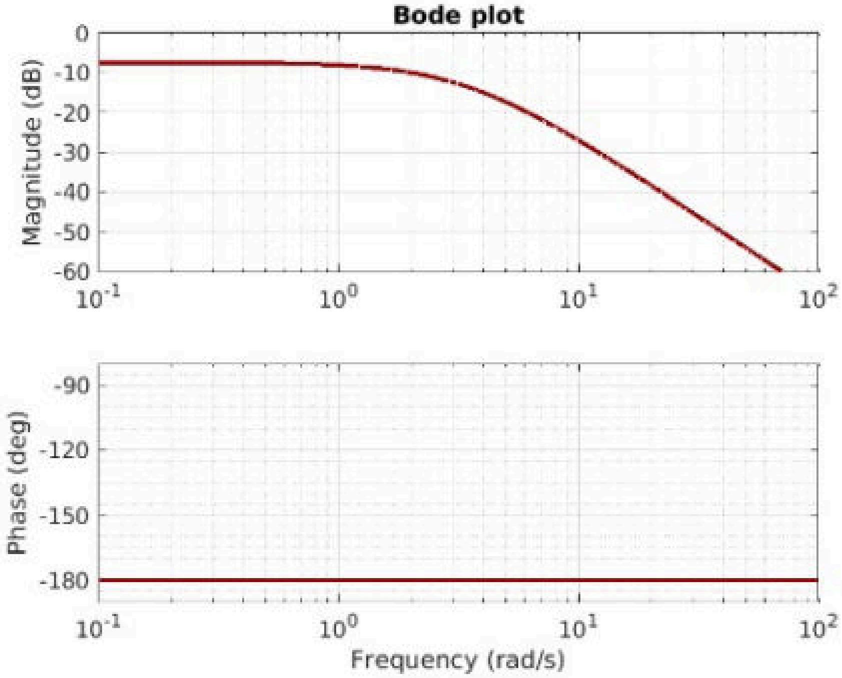
- [x] The blue (fully
the forward bra
- [ ] The blue (fully
the back branch

S20 Question 16 - Stability

A system is modelled with the following transfer function

$$G(s) = \frac{5}{s^2 - 11.5}$$

with the following Bode plot



Does the system have poles in the right hemisphere?

Sign change or m
indication of pole

(at low frequenci
degrees)
At high frequenci

The denominator

$G(s) = \frac{}{(s + \surd}$

One pole in the le

Or MATLAB

G = tf([5],[1 0 -11.5])
pole(G)
ans =
3.3912
-3.3912

```
clear; clc;

syms s;

G = 5 / (s^2-11.5);

eval(poles(G))
```

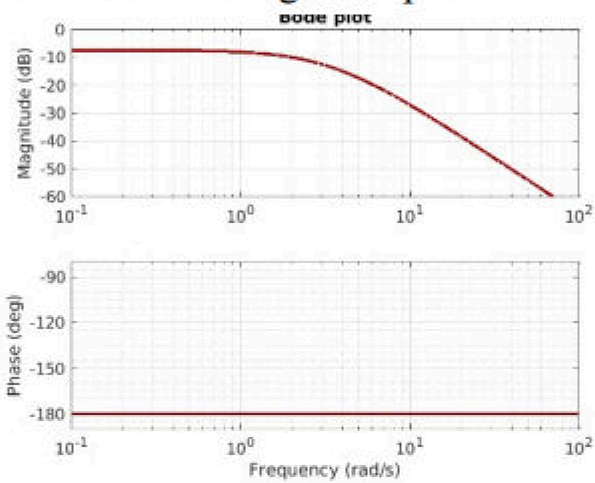
```
Warning: The function sym/eval is deprecated and will be removed in a future release. Depending on the
usage, use subs or double instead.
ans = 2×1
   -3.3912
    3.3912
```

S20 Question 17 - Stability

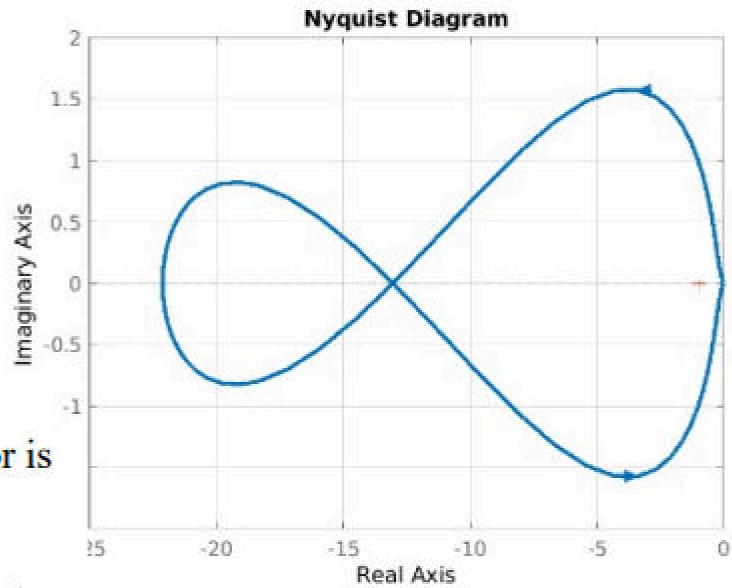
A system is modelled with the following transfer function

$$G(s) = \frac{5}{s^2 - 11.5}$$

with the following Bode plot



Overall, the open loop provides the following Nyquist plot



Based on the Bode plot, a P-lag-lead regulator is designed with the following parameters:
$\omega_c = 20$ rad/sec, $\alpha = 0.1$, $N_i = 4$, $\beta = 2$
It provides the following controller components

$$Cd = \frac{0.16\,s + 1}{0.016\,s + 1} \quad C_i = \frac{0.2s + 1}{0.4s + 1} \qquad K_p = 51$$

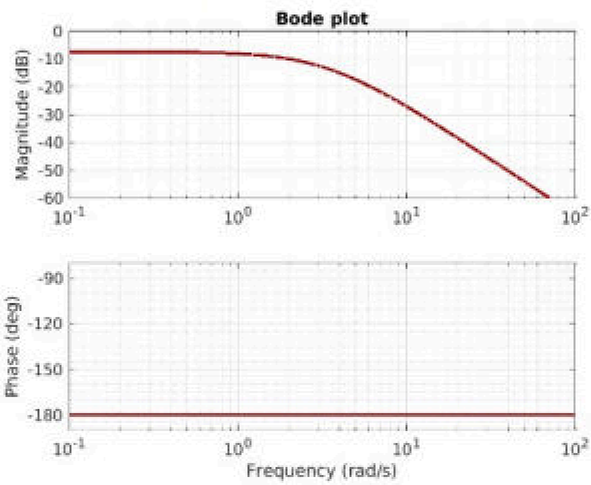
Will closed loop with this regulator be stable?

S20 Question 18 - Stability

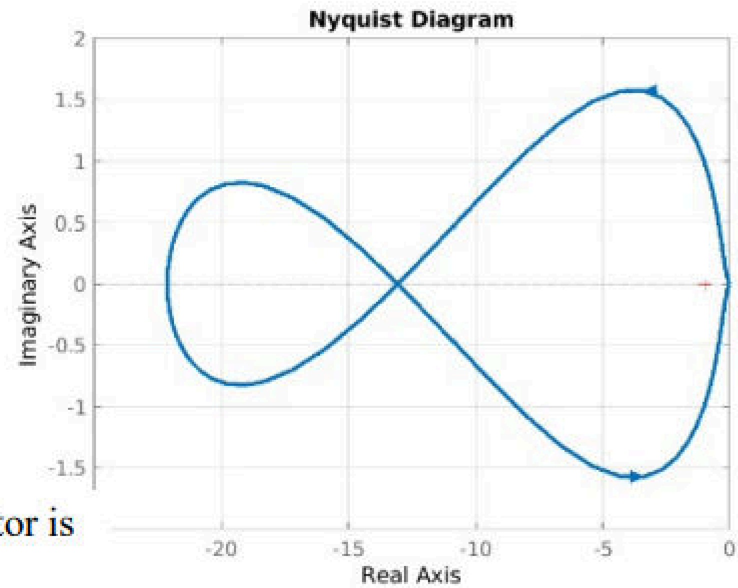
A system is modelled with the following transfer function

$$G(s) = \frac{5}{s^2 - 11.5}$$

with the following Bode plot



Overall, the open loop provides the following Nyquist plot



Based on the Bode plot, a P-lag-lead regulator is designed with the following parameters:
$\omega_c = 20$ rad/sec, $\alpha = 0.1$, $N_i = 4$, $\beta = 2$
It provides the following controller components

$$Cd = \frac{0.16\,s + 1}{0.016\,s + 1} \quad C_i = \frac{0.2s + 1}{0.4s + 1} \quad K_p = 51$$

Will the system be stable if P-gain is reduced to $K_p = 1$ (and other parts - C_i and C_d - remain unchanged)?

```
clear; clc; close all;

s = tf('s');

% System
G = 5/(s^2 - 11.5);

% Controllerdele
Cd = (0.16*s + 1)/(0.016*s + 1);
Ci = (0.2*s + 1)/(0.4*s + 1);

% Reduceret P-gain
Kp = 1;
```

```
% Open-loop
L = Kp * Cd * Ci * G;

% Closed-loop med negativ unity feedback
T = feedback(L, 1);

% Poler i closed-loop
p = pole(T)
```

```
p = 4×1
  -62.1523
    2.5388
   -3.5972
   -1.7893
```

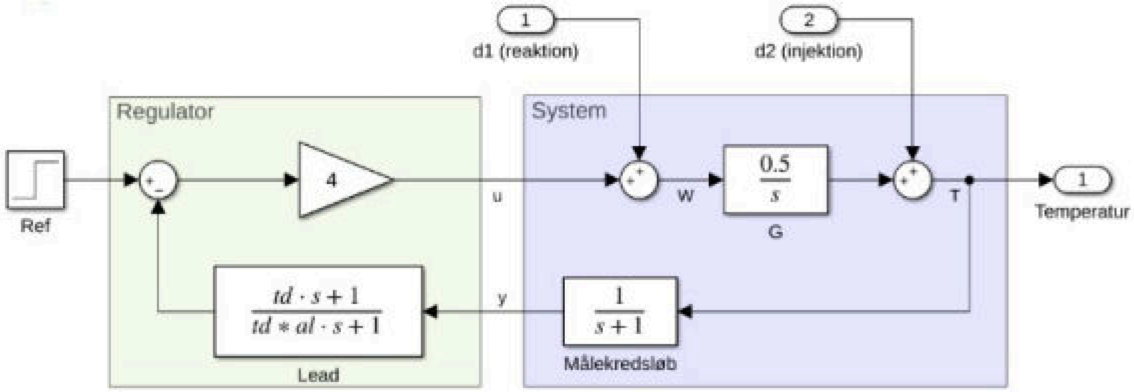
```
% Stabilitet
stable = isstable(T)
```

```
stable =
logical
   0
```

S20 Question 19 - Sensitivity

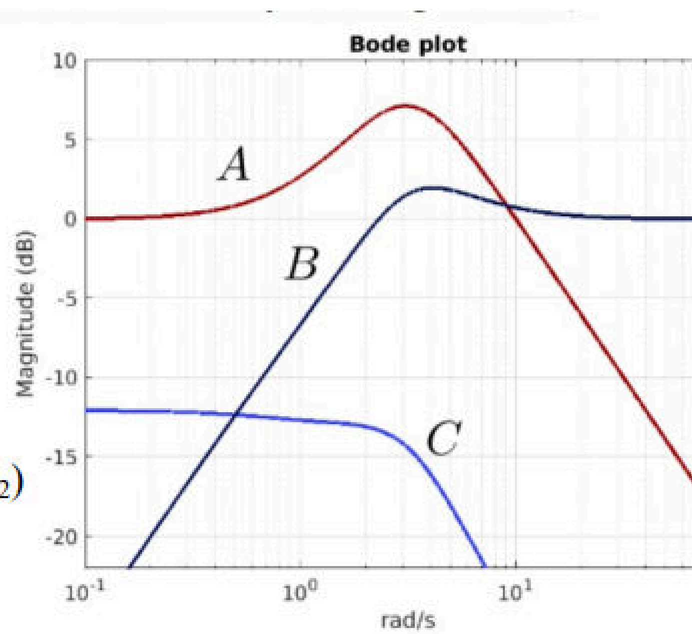
Which curve

An insulated tank (bio-reactor) with in-P-Lead temperature regulator is modelled as follows:



There are two disturbances, d_1 which is due to the development of chemical energy in the tank [Watt] and d_2 which is a change in temperature due to the replacement of part of the contents of the tank [°C].

The sensitivity to temperature change due to reaction (d_1) and due to replacement of content (d_2) is shown in the following amplitude Bode plot (together with the closed loop transfer function):




```
clear; clc; close all;

s = tf('s');

% System
G = 0.5/s;
H = 1/(s+1);

% Regulator
Kp = 4;

% Lead-led, indsæt dine egne td og alpha hvis de er givet
```

```
td = 1;
alpha = 0.1;
Lead = (td*s + 1)/(td*alpha*s + 1);

% Loop transfer
L = Kp * Lead * H * G;

% Transfer functions
T_ref = Kp*G/(1 + L);   % Ref -> Temperature, curve A
T_d1  = G/(1 + L);      % d1 -> Temperature, curve C
T_d2  = 1/(1 + L);      % d2 -> Temperature, curve B

% Bode plot
bode(T_ref, T_d2, T_d1)
grid on
legend('A: reference', 'B: d_2', 'C: d_1')
```